

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\=A150NRL31HN0030pe-Lug loc as per MDS.pvdb

Date of Analysis : Sep 18, 2026 4:02pm

PV Elite 28 U2, April 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 508.000 x 220.000 x 141.000 x 141.300 mm OD
 Vessel Design Length, Tangent to Tangent 1908.33 cm

Distance of Bottom Tangent above Grade 0.00 cm
 Specified Datum Line Distance 0.00 cm

Internal Design Temperature 85 °C
 Internal Design Pressure 25.492 kgf/cm²

External Design Temperature 0 °C

Maximum Allowable Working Pressure 39.3 kgf/cm²
 Shop Test Pressure 51.090 kgf/cm²

Required Minimum Design Metal Temperature -10.0 °C
 Warmest Computed Minimum Design Metal Temperature -17.0 °C

Wind Design Code ASCE/SEI 7-22
 Earthquake Design Code ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-312 TP304L	...	...	S30403	
Head	SA-403 304L	...	...	S30403	+IT
Cone	SA-312 TP304L	...	...	S30403	+IT
Flange	SA-182 F304L	...	> 130	S30403	
Nozzle	SA-182 F304L	...	<= 130	S30403	+IT
Nozzle Flg	SA-182 F304	...	> 130	S30400	
Nozzle Flg	SA-182 F304L	...	<= 130	S30403	
Sup. Lugs	SA-516 70	...	...	K02700	+IT
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Lug Bolting	SA-36	...	...	K02600	+IT

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Bottom head	25.590	0.00	45.300	0.0000	No	No
Shell-1	25.573	0.00	46.800	0.0000	N/A	No
conical-1	25.492	0.00	39.300	0.0000	N/A	No
Shell-2	25.492	0.00	94.000	0.0000	N/A	No
H6-2	25.492	0.00	69.500	0.0000	N/A	No
H6-1	25.492	0.00	69.700	0.0000	N/A	No
SHELL-3	25.492	0.00	74.700	0.0000	N/A	No
SHELL-4	25.492	0.00	74.700	0.0000	N/A	No
H5-2	25.492	0.00	67.800	0.0000	N/A	No
H5-1	25.492	0.00	67.800	0.0000	N/A	No
Shell-5	25.492	0.00	74.700	0.0000	N/A	No
Shell-6	25.492	0.00	74.700	0.0000	N/A	No
H4-2	25.492	0.00	67.800	0.0000	N/A	No
H4-1	25.492	0.00	67.800	0.0000	N/A	No
Shell-6	25.492	0.00	74.700	0.0000	N/A	No
conical -2	25.492	0.00	63.600	0.0000	N/A	No
shell-7	25.492	0.00	103.000	0.0000	N/A	No
H3-2	25.492	0.00	69.700	0.0000	N/A	No

H3-1	25.492	0.00	69.700	0.0000	N/A	No
shell-8	25.492	0.00	103.000	0.0000	N/A	No
H2-2	25.492	0.00	69.700	0.0000	N/A	No
H2-1	25.492	0.00	69.700	0.0000	N/A	No
SHELL9	25.492	0.00	103.000	0.0000	N/A	No
H1-2	25.492	0.00	69.700	0.0000	N/A	No
H1-1	25.492	0.00	55.100	0.0000	N/A	No

Liquid Level: 102.70 cm Dens.: 949.585 kg/m³ Sp. Gr.: 0.950

Element Types and Properties:

Element Type	"To" Elev cm	Element Length cm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	5.0	5.0	11.2	9.5	5.4	...	1.00	1.00
Cylinder	120.0	115.0	10.0	10.0	5.5	...	1.00	1.00
Conical	205.0	85.0	10.0	10.0	6.5	...	0.85	0.85
Cylinder	213.7	8.7	10.0	10.0	2.8	...	0.85	0.85
Body Flg	225.0	11.0	40.0	39.6	37.7	37.7	1.00	1.00
Body Flg	236.0	11.0	40.0	39.6	37.6	37.6	1.00	1.00
Cylinder	536.0	300.0	8.0	8.0	2.8	...	0.85	0.85
Cylinder	673.7	137.7	8.0	8.0	2.8	...	0.85	0.85
Body Flg	685.0	11.0	40.0	39.6	37.3	37.3	1.00	1.00
Body Flg	696.0	11.0	40.0	39.6	34.8	34.8	1.00	1.00
Cylinder	996.0	300.0	8.0	8.0	2.8	...	0.85	0.85
Cylinder	1103.7	107.7	8.0	8.0	2.8	...	0.85	0.85
Body Flg	1115.0	11.0	40.0	39.6	37.3	37.3	1.00	1.00
Body Flg	1126.0	11.0	40.0	39.6	37.3	37.3	1.00	1.00
Cylinder	1151.0	25.0	8.0	8.0	2.8	...	0.85	0.85
Conical	1166.2	15.2	7.0	7.0	2.9	...	0.85	0.85
Cylinder	1470.0	303.8	7.0	7.0	1.8	...	0.85	0.85
Body Flg	1480.0	9.7	34.0	33.3	30.7	30.7	1.00	1.00
Body Flg	1489.7	9.7	34.0	33.3	30.7	30.7	1.00	1.00
Cylinder	1785.0	295.3	7.0	7.0	1.8	...	0.85	0.85
Body Flg	1795.0	9.7	34.0	33.3	30.7	30.7	1.00	1.00
Body Flg	1804.7	9.7	34.0	33.3	30.7	30.7	1.00	1.00
Cylinder	1895.0	90.3	7.0	7.0	1.8	...	0.85	0.85
Body Flg	1905.0	9.7	34.0	33.3	30.7	30.7	1.00	1.00
Body Flg	1908.3	3.3	34.0	33.3	28.1	28.1	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
B2 2''	WRC-107/537	0.778	n/a	Passed
A1 2''	WRC-297	0.578	Shell	Passed
A3 2''	WRC-297	0.578	Shell	Passed
A2 2''	WRC-297	0.620	Shell	Passed

Loads for Foundation/Support Design:

Total Wind Shear on Support	2523. kgf
Total Earthquake Shear on Support	784. kgf
Wind Moment on Support	17940. kgf-m.
Earthquake Moment on Support	10900. kgf-m.
Maximum Initial Bolt Load	34083. kgf

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	1589.9 kg
Shop Test - Fabricated + water (full)	2368.7 kg
Shipping - Fab. + rem. intls.+ shipping app.	1589.9 kg
Erected - Fab. + rem. intls.+ insul. (etc)	1639.9 kg



Empty	- Fab. + intls. + details + wghts.	1711.1	kg
Operating	- Empty + operating liquid (no ca)	2021.1	kg

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Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
B2 2''	45.3	...	...	OK	4.80	OK	No Calc[*]
L2 2''	46.8	...	...	OK	4.80	OK	No Calc[*]
L4 2''	46.8	...	...	OK	4.80	OK	No Calc[*]
L6 2''	46.8	...	...	OK	4.80	OK	No Calc[*]
L1 2''	46.8	...	...	OK	4.80	OK	No Calc[*]
L3 2''	46.8	...	...	OK	4.80	OK	No Calc[*]
L5 2''	46.8	...	...	OK	4.80	OK	No Calc[*]
P3-2''	39.3	...	...	OK	4.80	OK	No Calc[*]
A4 2''	39.3	...	...	OK	4.80	OK	No Calc[*]
A1 2''	74.7	...	...	OK	3.49	OK	No Calc[*]
A3 2''	74.7	...	...	OK	3.49	OK	No Calc[*]
P2 2''	74.7	...	...	OK	3.49	OK	No Calc[*]
A2 2''	103	...	...	OK	2.18	OK	No Calc[*]
P1 2''	103	...	...	OK	2.18	OK	No Calc[*]
B1 2''	55.1	...	...	OK	4.80	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 39.3 Nozzle : A4 2''
 Minimum MAWP Shells/Flanges : 39.3 Element : conical-1
 Minimum MAPnc Shells/Flanges : 39.3 Element : conical-1

Computed Vessel M.A.W.P. : 39.3 kgf/cm²

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
10	B2 2''	0.000	0.000	103.328
20	L2 2''	500.000	0.000	104.328
20	L4 2''	500.000	45.000	104.328
20	L6 2''	500.000	90.000	104.328
20	L1 2''	950.000	0.000	104.328
20	L3 2''	950.000	45.000	104.328
20	L5 2''	950.000	90.000	104.328
30	P3-2''	1650.000	0.000	105.052
30	A4 2''	1350.000	180.000	105.052
70	A1 2''	2465.000	0.000	101.600
150	A3 2''	11400.003	0.000	101.600
150	P2 2''	11400.003	180.000	101.600
230	A2 2''	18150.000	180.000	101.600
230	P1 2''	18350.000	0.000	101.600
250	B1 2''	0.000	0.000	150.876

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

Checking Multiple Nozzles on Flat Head per ASME Sec. VIII Div. 1 UG-39

Comparing Nozzles on Element: H1-1

Note: No Nozzle pairs found on this element.

UG-39 Nozzle Diameter and Distance to Edge Checks:

Nozzle Description	Nozzle dia. mm	Head Dia. /2 mm	Distance from Edge mm	Nozzle dia./4 mm
B1 2''	50.8	83.375	97.536	12.7

No Multiple Nozzle spacing violations detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
B2 2''	2.000 in	Actual	LWN	3.320	16.764	...	...	336.49	600
L2 2''	2.000 in	Actual	LWN	3.320	16.764	...	...	189.10	600
L4 2''	2.000 in	Actual	LWN	3.320	16.764	...	...	189.10	600
L6 2''	2.000 in	Actual	LWN	3.320	16.764	...	...	189.10	600
L1 2''	2.000 in	Actual	LWN	3.320	16.764	...	...	189.10	600
L3 2''	2.000 in	Actual	LWN	3.320	16.764	...	...	189.10	600
L5 2''	2.000 in	Actual	LWN	3.320	16.764	...	...	189.10	600
P3-2''	2.000 in	Actual	LWN	3.320	16.764	...	...	165.39	600
A4 2''	2.000 in	Actual	LWN	3.320	16.764	...	...	164.11	600
A1 2''	2.000 in	Actual	LWN	3.320	16.764	...	...	186.55	600
A3 2''	2.000 in	Actual	LWN	3.320	16.764	...	...	186.55	600
P2 2''	2.000 in	Actual	LWN	3.320	16.764	...	...	186.55	600
A2 2''	2.000 in	Actual	LWN	3.320	16.764	...	...	201.74	600
P1 2''	2.000 in	Actual	LWN	3.320	16.764	...	...	201.74	600
B1 2''	2.000 in	Actual	LWN	3.320	16.764	...	...	208.32	600

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
B2 2''	SA-182 F304L	8.000	10.000	...	...	...
L2 2''	SA-182 F304L	10.000	10.000	...	...	...
L4 2''	SA-182 F304L	10.000	10.000	...	...	...
L6 2''	SA-182 F304L	10.000	10.000	...	...	...
L1 2''	SA-182 F304L	10.000	10.000	...	...	...
L3 2''	SA-182 F304L	10.000	10.000	...	...	...
L5 2''	SA-182 F304L	10.000	10.000	...	...	...
P3-2''	SA-182 F304L	8.000	10.000	...	...	...
A4 2''	SA-182 F304L	8.000	10.000	...	...	...
A1 2''	SA-182 F304L	8.000	10.000	...	...	...
A3 2''	SA-182 F304L	8.000	10.000	...	...	...
P2 2''	SA-182 F304L	8.000	10.000	...	...	...
A2 2''	SA-182 F304L	7.000	10.000	...	...	...
P1 2''	SA-182 F304L	7.000	10.000	...	...	...
B1 2''	SA-182 F304L	33.274	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
B2 2''	...	0.0	325.00	0.00	Bottom head
L2 2''	50.000	0.0	175.43	0.00	Shell-1
L4 2''	50.000	45.0	175.43	0.00	Shell-1
L6 2''	50.000	90.0	175.43	0.00	Shell-1
L1 2''	95.000	0.0	175.43	0.00	Shell-1
L3 2''	95.000	45.0	175.43	0.00	Shell-1
L5 2''	95.000	90.0	175.43	0.00	Shell-1



P3-2''	165.000	0.0	150.00	0.00	conical-1
A4 2''	135.000	180.0	150.00	0.00	conical-1
A1 2''	246.500	0.0	169.43	0.00	SHELL-3
A3 2''	1140.000	0.0	169.43	0.00	Shell-6
P2 2''	1140.000	180.0	169.43	0.00	Shell-6
A2 2''	1815.000	180.0	178.78	0.00	SHELL9
P1 2''	1835.000	0.0	178.78	0.00	SHELL9
B1 2''	...	0.0	175.00	0.00	H1-1

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	ELLIPTICAL HEAD: 2.0 X 1, 11.2mm THK X 489.0mm ID X 5.0cm	SA-403 304L
1	CYLINDER: 10.0mm THK X 488.0mm ID X 115.0cm	SA-312 TP304L
1	CONCENTRIC CONE: 10.0mm THK X 487.7mm X 199.7cm ID X 85.0 cm	SA-312 TP304L
1	CYLINDER: 10.0mm THK X 200.0mm ID X 8.7cm	SA-312 TP304L
6	WELD NECK FLANGE: 40.0mm THK X 381.0mm OD	SA-182 F304L
2	CYLINDER: 8.0mm THK X 204.0mm ID X 300.0cm	SA-312 TP304L
1	CYLINDER: 8.0mm THK X 204.0mm ID X 137.7cm	SA-312 TP304L
1	CYLINDER: 8.0mm THK X 204.0mm ID X 107.7cm	SA-312 TP304L
1	CYLINDER: 8.0mm THK X 204.0mm ID X 25.0cm	SA-312 TP304L
1	CONCENTRIC CONE: 7.0mm THK X 206.0mm X 127.0cm ID X 15.2 cm	SA-312 TP304L
1	CYLINDER: 7.0mm THK X 127.0mm ID X 303.8cm	SA-312 TP304L
5	WELD NECK FLANGE: 34.0mm THK X 279.4mm OD	SA-182 F304L
1	CYLINDER: 7.0mm THK X 127.3mm ID X 295.3cm	SA-312 TP304L
1	CYLINDER: 7.0mm THK X 127.3mm ID X 90.3cm	SA-312 TP304L
1	BLIND FLANGE: 34.0mm THK X 279.4mm OD	SA-182 F304L
1	PACKING: 2595.000mm	...
8	GUSSET PLATES: 250.0mm X 100.0mm X 220.0mm X 20.0mm	SA-516 70
1	TOP RING: 212.0mm X 412.0mm X 30.0mm	SA-516 70
1	BOTTOM RING: 212mm X 712mm X 44mm	SA-516 70
4	LUG BOLTS: 72.0mm TEMA Metric	SA-36
1	PACKING: 825.000mm	...
1	PACKING: 2870.000mm	...
1	PACKING: 550.000mm	...
2	PACKING: 2370.000mm	...
1	PACKING: 150.000mm	...
3	GASKET: 263.7mm OD X 233.4mm ID	...
36	BODY FLANGE BOLTS: 22.2mm DIA	SA-193 B7
72	NUTS FOR BODY FLANGE BOLTS: 22.2mm DIA	...
3	GASKET: 177.8mm OD X 155.7mm ID	...
24	BODY FLANGE BOLTS: 19.0mm DIA	SA-193 B7
48	NUTS FOR BODY FLANGE BOLTS: 19.0mm DIA	...
1	NAMEPLATE	...

PV Elite 28 U2 Licensee: SL User

FileName : =A150NRL31HN003_Ope-Lug loc as per MDS

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Bill of Material: Step: 70 4:02pm Sep 18, 2026



HEXAGON

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Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
	Notes							
H6-2	[11]	-196						
H6-1	[11]	-196						
H5-2	[11]	-196						
H5-1	[11]	-196						
H4-2	[11]	-196						
H4-1	[11]	-196						
H3-2	[11]	-196						
H3-1	[11]	-196						
H2-2	[11]	-196						
H2-1	[11]	-196						
H1-2	[11]	-196						
H1-1	[11]	-196						
Bottom head	[10] !	-10	-17		0.872	9.500	1.00	No
Bottom head	[7] !	-10	-24		0.753	11.177	1.00	No
Shell-1	[16]	-196						
conical-1	[8] !	-10	-18		0.850	10.000	0.85	No
Shell-2	[8] !	-10	-78		0.363	10.000	0.85	No
SHELL-3	[16]	-196						
SHELL-4	[16]	-196						
Shell-5	[16]	-196						
Shell-6	[16]	-196						
Shell-6	[16]	-196						
conical -2	[16]	-196						
shell-7	[16]	-196						
shell-8	[16]	-196						
SHELL9	[16]	-196						
B2 2''	[1] !	-10	-22		0.781	9.500	1.00	No
L2 2''	[1] !	-10	-19		0.835	10.000	1.00	No
L4 2''	[1] !	-10	-19		0.835	10.000	1.00	No
L6 2''	[1] !	-10	-19		0.835	10.000	1.00	No
L1 2''	[1] !	-10	-19		0.835	10.000	1.00	No
L3 2''	[1] !	-10	-19		0.835	10.000	1.00	No
L5 2''	[1] !	-10	-19		0.835	10.000	1.00	No
P3-2''	[1] !	-10	-34		0.581	10.000	1.00	No
A4 2''	[1] !	-10	-23		0.758	10.000	1.00	No
A1 2''	[15]	-196						
A3 2''	[1] !	-10	-53		0.436	8.000	1.00	No
P2 2''	[1] !	-10	-53		0.436	8.000	1.00	No
A2 2''	[1] !	-10	-104		0.311	7.000	1.00	No
P1 2''	[1] !	-10	-104		0.311	7.000	1.00	No
B1 2''	[1] !	-10	-104		0.052	16.764	1.00	No
Nozl. Bolting	[22]	-48			1.000	64.000	1.00	No
Bolting	[21]	-48						
Bolting	[21]	-48						
Bolting	[21]	-48						
Bolting	[21]	-48						
Bolting	[21]	-48						
Bolting	[21]	-48						

Warmest MDMT: -10 -17

Required Minimum Design Metal Temperature -10.0 °C
 Warmest Computed Minimum Design Metal Temperature -17.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.

- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note	10	20	The wind diameter multiplier is very big.
Note	20	30	The wind diameter multiplier is very big.
Note	30	40	The wind diameter multiplier is very big.
Note	40	50	The wind diameter multiplier is very big.
Note	50	60	The wind diameter multiplier is very big.
Note	60	70	The wind diameter multiplier is very big.
Note	70	80	The wind diameter multiplier is very big.
Note	80	90	The wind diameter multiplier is very big.
Note	90	100	The wind diameter multiplier is very big.
Note	100	110	The wind diameter multiplier is very big.
Note	110	120	The wind diameter multiplier is very big.
Note	120	130	The wind diameter multiplier is very big.
Note	130	140	The wind diameter multiplier is very big.
Note	140	150	The wind diameter multiplier is very big.
Note	150	160	The wind diameter multiplier is very big.
Note	160	170	The wind diameter multiplier is very big.
Note	170	180	The wind diameter multiplier is very big.
Note	180	190	The wind diameter multiplier is very big.
Note	190	200	The wind diameter multiplier is very big.
Note	200	210	The wind diameter multiplier is very big.
Note	210	220	The wind diameter multiplier is very big.
Note	220	230	The wind diameter multiplier is very big.
Note	230	240	The wind diameter multiplier is very big.
Note	240	250	The wind diameter multiplier is very big.
Note	250	260	The wind diameter multiplier is very big.

Class From To: Check of Additional Element Data

Note: Lug supports should be modeled on an element of their own whose length is just slightly greater than the height of the lug assembly. The vessel does not have to be constructed with this extra element, just modeled this way. This is to insure there is proper mass distribution for these types of models.

Warning:

This is a vertical vessel on support lugs and the earthquake type selected is a building code. Building codes assume all of the mass is located above the point of support. This makes their use on lug supported vessels questionable. It is strongly suggested that based on the seismic zone and applicable parameters, an appropriate lateral (Gx) and (+-) vertical (Gy) acceleration be applied using the G loading option.

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into



the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (Kgfc^{m2}):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in. ⁴	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in. ³	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	25.492	kgf/cm ²
Design Internal Temperature	85.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-10.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	Y	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW



Wind Design Code	ASCE/SEI 7-22	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 208.8	km/hr
Surface Roughness Category	D: Flat, unobstructed	
Type of Surface	Moderately Smooth	
Base Elevation	0	cm
Percent Wind for Hydrotest	33.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	H or Hh	0 cm
Distance Upwind of Crest	Lh	0 cm
Distance from Crest to the Vessel	x	0 cm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	cm

Seismic Design Code	ASCE/SEI 7-22	
Importance Factor	1.500	
Force Modification Factor R	1.000	
Site Class	D	
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Component resonance duct. fact.	[CAR]	0.000
Component strength factor	[Rpo]	0.000
Overstrength factor	[Omega0]	0.000
Response Acceleration Factor	[SMS]	0.326
Response Acceleration Factor	[SM1]	0.139
Building Importance Factor	[Ie]	1.000
Site Latitude		0.000 deg
Site Longitude		0.000 deg

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/25
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	Bottom head



Distance "FROM" to "TO"	5	cm
Element Outside Diameter	508	mm
Element Thickness	9.5	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	11.177	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	25.492	kgf/cm ²
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.6900001	
Material Name	SA-403 304L	+IT
Allowable Stress, Ambient	1172.6	kgf/cm ²
Allowable Stress, Operating	1172.6	kgf/cm ²
Allowable Stress, Hydrotest	1560.1	kgf/cm ²
Material Density	0.008027	kg/cm ³
P Number Thickness	0	mm
Yield Stress, Operating	1536.8	kgf/cm ²
UCS-66 Chart Curve Designation	Impact Tested	
External Pressure Chart Name	HA-3	
UNS Number	S30403	
Product Form	Smls. & wld. fittings	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid 10	
Dist. from "FROM" Node / Offset dist	-12.7	cm
Height/Length of Liquid	17.7	cm
Liquid Density	949.56	kg/m ³
Element From Node	10	
Detail Type	Nozzle	
Detail ID	B2 2''	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	14.736	kgf
Grade of Attached Flange	GR 2.1	
Nozzle Matl	SA-182 F304L	+IT
Element From Node	10	
Detail Type	Weight	
Detail ID	Vortex breaker	
Dist. from "FROM" Node / Offset dist	-12.7	cm
Miscellaneous Weight	5	kgf
Offset from Element Centerline	0	mm

Element#	2/25	
Element From Node	20	
Element To Node	30	
Element Type	Cylinder	
Description	Shell-1	
Distance "FROM" to "TO"	115	cm
Element Outside Diameter	508	mm
Element Thickness	10	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	10	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	25.492	kgf/cm ²
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	kgf/cm ²



Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.6900001
 Material Name SA-312 TP304L
 Allowable Stress, Ambient 1172.6 kgf/cm²
 Allowable Stress, Operating 1172.6 kgf/cm²
 Allowable Stress, Hydrotest 1560.1 kgf/cm²
 Material Density 0.008027 kg/cm³
 P Number Thickness 0 mm
 Yield Stress, Operating 1536.8 kgf/cm²
 External Pressure Chart Name HA-3
 UNS Number S30403
 Product Form Smls. & wld. pipe
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 20
 Detail Type Liquid
 Detail ID Liquid 20
 Dist. from "FROM" Node / Offset dist 0 cm
 Height/Length of Liquid 85 cm
 Liquid Density 949.58 kg/m³

Element From Node 20
 Detail Type Nozzle
 Detail ID L2 2''
 Dist. from "FROM" Node / Offset dist 45 cm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 600
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 9.7279 kgf
 Grade of Attached Flange GR 2.3
 Nozzle Matl SA-182 F304L +IT

Element From Node 20
 Detail Type Nozzle
 Detail ID L4 2''
 Dist. from "FROM" Node / Offset dist 45 cm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 600
 Layout Angle 45.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 9.7279 kgf
 Grade of Attached Flange GR 2.1
 Nozzle Matl SA-182 F304L +IT

Element From Node 20
 Detail Type Nozzle
 Detail ID L6 2''
 Dist. from "FROM" Node / Offset dist 45 cm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 600
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 9.7279 kgf
 Grade of Attached Flange GR 2.1
 Nozzle Matl SA-182 F304L +IT

Element From Node 20
 Detail Type Nozzle
 Detail ID L1 2''
 Dist. from "FROM" Node / Offset dist 90 cm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 600
 Layout Angle 0.0



Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	9.7279 kgf
Grade of Attached Flange	GR 2.1
Nozzle Matl	SA-182 F304L +IT

Element From Node	20
Detail Type	Nozzle
Detail ID	L3 2''
Dist. from "FROM" Node / Offset dist	90 cm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	45.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	9.7279 kgf
Grade of Attached Flange	GR 2.3
Nozzle Matl	SA-182 F304L +IT

Element From Node	20
Detail Type	Nozzle
Detail ID	L5 2''
Dist. from "FROM" Node / Offset dist	90 cm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	10.612 kgf
Grade of Attached Flange	GR 2.1
Nozzle Matl	SA-182 F304L +IT

Element#	3/25
Element From Node	30
Element To Node	40
Element Type	Conical
Description	conical-1
Distance "FROM" to "TO"	85 cm
Element Outside Diameter	508 mm
Element Thickness	10 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	10 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	25.492 kgf/cm ²
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.6900001
Material Name	SA-312 TP304L +IT
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Cone Diameter at "To" End	220 mm
Design Length of Cone	850 mm
Half Apex Angle of Cone	9.615284 degrees
Toriconical (Y/N)	N
Weld is pre-Heated	No

Element From Node	30
Detail Type	Nozzle
Detail ID	P3-2''
Dist. from "FROM" Node / Offset dist	45 cm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	9.0017 kgf
Grade of Attached Flange	GR 2.1
Nozzle Matl	SA-182 F304L +IT



Element From Node	30
Detail Type	Nozzle
Detail ID	A4 2''
Dist. from "FROM" Node / Offset dist	15 cm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	9.0017 kgf
Grade of Attached Flange	GR 2.1
Nozzle Matl	SA-182 F304L +IT

Element#	4/25
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell-2
Distance "FROM" to "TO"	8.71 cm
Element Outside Diameter	220 mm
Element Thickness	10 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	10 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	25.492 kgf/cm ²
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.6900001
Material Name	SA-312 TP304L +IT
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element#	5/25
Element From Node	50
Element To Node	60
Element Type	Flange
Description	H6-2
Distance "FROM" to "TO"	10.973 cm
Flange Inside Diameter	202.69 mm
Element Thickness	39.624 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	40 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	25.492 kgf/cm ²
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.6900001
Material Name	SA-182 F304L
Allowable Stress, Ambient	1172.6 kgf/cm ²
Allowable Stress, Operating	1172.6 kgf/cm ²
Allowable Stress, Hydrotest	1560.1 kgf/cm ²
Material Density	0.008027 kg/cm ³
P Number Thickness	0 mm
Yield Stress, Operating	1536.8 kgf/cm ²
External Pressure Chart Name	HA-3
UNS Number	S30403
Class / Thickness / Grade	:: > 130
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No



Element#	6/25
Element From Node	60
Element To Node	70
Element Type	Flange
Description	H6-1
Distance "FROM" to "TO"	10.973 cm
Flange Inside Diameter	202.69 mm
Element Thickness	39.624 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	40 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	25.492 kgf/cm ²
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.6900001
Material Name	SA-182 F304L
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	7/25
Element From Node	70
Element To Node	80
Element Type	Cylinder
Description	SHELL-3
Distance "FROM" to "TO"	300 cm
Element Outside Diameter	220 mm
Element Thickness	8 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	8 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	25.492 kgf/cm ²
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.6900001
Material Name	SA-312 TP304L
Allowable Stress, Ambient	1172.6 kgf/cm ²
Allowable Stress, Operating	1172.6 kgf/cm ²
Allowable Stress, Hydrotest	1560.1 kgf/cm ²
Material Density	0.008027 kg/cm ³
P Number Thickness	0 mm
Yield Stress, Operating	1536.8 kgf/cm ²
External Pressure Chart Name	HA-3
UNS Number	S30403
Product Form	Smls. & wld. pipe
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	70
Detail Type	Packing
Detail ID	BED 1
Dist. from "FROM" Node / Offset dist	40.5 cm
Height of Packed Section	259.5 cm
Density	0.00025 kg/cm ³
Percent Volume Holdup	50.0
Specific Gravity of Packing Liquid	0.94999999

Element From Node	70
Detail Type	Nozzle
Detail ID	A1 2''
Dist. from "FROM" Node / Offset dist	10.527 cm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0



Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 10.25 kgf
 Grade of Attached Flange GR 2.1
 Nozzle Matl SA-182 F304L

Element From Node 70
 Detail Type Lug
 Detail ID Support Lugs
 Dist. from "FROM" Node / Offset dist 60 cm
 Number of Lugs 4
 Dist. from OD to Lug Cntrline(dlug) 150 mm
 Height of Gusset Plates (hgp) 220 mm
 Force Bearing Width (wfb) 46 mm
 Weight of Lug 49.25 kgf
 Lug Start Angle (degrees) 0.0

Element From Node 70
 Detail Type Weight
 Detail ID Bed1 sup ring
 Dist. from "FROM" Node / Offset dist 40 cm
 Miscellaneous Weight 2 kgf
 Offset from Element Centerline 0 mm

Element From Node 70
 Detail Type Weight
 Detail ID Bed1 sup grid
 Dist. from "FROM" Node / Offset dist 40.5 cm
 Miscellaneous Weight 5 kgf
 Offset from Element Centerline 0 mm

Element# 8/25
 Element From Node 80
 Element To Node 90
 Element Type Cylinder
 Description SHELL-4
 Distance "FROM" to "TO" 137.74 cm
 Element Outside Diameter 220 mm
 Element Thickness 8 mm
 Internal Corrosion Allowance 0 mm
 Nominal Thickness 8 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 25.492 kgf/cm²
 Design Temperature Internal Pressure 85 °C
 Design External Pressure 0 kgf/cm²
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.6900001
 Material Name SA-312 TP304L
 Efficiency, Longitudinal Seam 0.85
 Efficiency, Circumferential Seam 0.85
 Weld is pre-Heated No

Element From Node 80
 Detail Type Packing
 Detail ID BED 1
 Dist. from "FROM" Node / Offset dist 0 cm
 Height of Packed Section 82.5 cm
 Density 0.00025 kg/cm³
 Percent Volume Holdup 50.0
 Specific Gravity of Packing Liquid 0.94999999

Element From Node 80
 Detail Type Weight
 Detail ID Cleat wt
 Dist. from "FROM" Node / Offset dist 100 cm
 Miscellaneous Weight 49.999 kgf
 Offset from Element Centerline 0 mm

Element From Node 80



Detail Type	Weight
Detail ID	Bed 1 Dist wt
Dist. from "FROM" Node / Offset dist	90 cm
Miscellaneous Weight	5 kgf
Offset from Element Centerline	0 mm

Element From Node	80
Detail Type	Weight
Detail ID	Bed1 Dist sup w
Dist. from "FROM" Node / Offset dist	88 cm
Miscellaneous Weight	2 kgf
Offset from Element Centerline	0 mm

Element#	9/25
Element From Node	90
Element To Node	100
Element Type	Flange
Description	H5-2
Distance "FROM" to "TO"	10.973 cm
Flange Inside Diameter	202.69 mm
Element Thickness	39.624 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	40 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	25.492 kgf/cm ²
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.6900001
Material Name	SA-182 F304L
Allowable Stress, Ambient	1172.6 kgf/cm ²
Allowable Stress, Operating	1172.6 kgf/cm ²
Allowable Stress, Hydrotest	1560.1 kgf/cm ²
Material Density	0.008027 kg/cm ³
P Number Thickness	0 mm
Yield Stress, Operating	1536.8 kgf/cm ²
External Pressure Chart Name	HA-3
UNS Number	S30403
Class / Thickness / Grade	:: > 130
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	10/25
Element From Node	100
Element To Node	110
Element Type	Flange
Description	H5-1
Distance "FROM" to "TO"	10.973 cm
Flange Inside Diameter	202.69 mm
Element Thickness	39.624 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	40 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	25.492 kgf/cm ²
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.6900001
Material Name	SA-182 F304L
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No



Element From Node	110	
Element To Node	120	
Element Type	Cylinder	
Description	Shell-5	
Distance "FROM" to "TO"	300	cm
Element Outside Diameter	220	mm
Element Thickness	8	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	8	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	25.492	kgf/cm ²
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.6900001	
Material Name	SA-312 TP304L	
Allowable Stress, Ambient	1172.6	kgf/cm ²
Allowable Stress, Operating	1172.6	kgf/cm ²
Allowable Stress, Hydrotest	1560.1	kgf/cm ²
Material Density	0.008027	kg/cm ³
P Number Thickness	0	mm
Yield Stress, Operating	1536.8	kgf/cm ²
External Pressure Chart Name	HA-3	
UNS Number	S30403	
Product Form	Smls. & wld. pipe	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	110	
Detail Type	Packing	
Detail ID	BED-2	
Dist. from "FROM" Node / Offset dist	13	cm
Height of Packed Section	287	cm
Density	0.00025	kg/cm ³
Percent Volume Holdup	50.0	
Specific Gravity of Packing Liquid	0.94999999	

Element From Node	110	
Detail Type	Weight	
Detail ID	Bed2 sup ring	
Dist. from "FROM" Node / Offset dist	12	cm
Miscellaneous Weight	2	kgf
Offset from Element Centerline	0	mm

Element From Node	110	
Detail Type	Weight	
Detail ID	Bed 2 Sup grid	
Dist. from "FROM" Node / Offset dist	12.5	cm
Miscellaneous Weight	5	kgf
Offset from Element Centerline	0	mm

Element#	12/25	
Element From Node	120	
Element To Node	130	
Element Type	Cylinder	
Description	Shell-6	
Distance "FROM" to "TO"	107.74	cm
Element Outside Diameter	220	mm
Element Thickness	8	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	8	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	25.492	kgf/cm ²
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.6900001	



Material Name	SA-312 TP304L	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	
Element From Node	120	
Detail Type	Packing	
Detail ID	Bed2	
Dist. from "FROM" Node / Offset dist	0	cm
Height of Packed Section	55	cm
Density	0.00025	kg/cm ³
Percent Volume Holdup	50.0	
Specific Gravity of Packing Liquid	0.94999999	
Element From Node	120	
Detail Type	Weight	
Detail ID	A3 dist wt	
Dist. from "FROM" Node / Offset dist	60	cm
Miscellaneous Weight	5	kgf
Offset from Element Centerline	0	mm
Element From Node	120	
Detail Type	Weight	
Detail ID	A3 dist sup wt	
Dist. from "FROM" Node / Offset dist	58	cm
Miscellaneous Weight	2	kgf
Offset from Element Centerline	0	mm

Element#	13/25	
Element From Node	130	
Element To Node	140	
Element Type	Flange	
Description	H4-2	
Distance "FROM" to "TO"	10.973	cm
Flange Inside Diameter	202.69	mm
Element Thickness	39.624	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	40	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	25.492	kgf/cm ²
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.6900001	
Material Name	SA-182 F304L	
Allowable Stress, Ambient	1172.6	kgf/cm ²
Allowable Stress, Operating	1172.6	kgf/cm ²
Allowable Stress, Hydrotest	1560.1	kgf/cm ²
Material Density	0.008027	kg/cm ³
P Number Thickness	0	mm
Yield Stress, Operating	1536.8	kgf/cm ²
External Pressure Chart Name	HA-3	
UNS Number	S30403	
Class / Thickness / Grade	:: > 130	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element#	14/25	
Element From Node	140	
Element To Node	150	
Element Type	Flange	
Description	H4-1	
Distance "FROM" to "TO"	10.973	cm
Flange Inside Diameter	202.69	mm
Element Thickness	39.624	mm



Internal Corrosion Allowance	0	mm
Nominal Thickness	40	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	25.492	kgf/cm ²
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.6900001	
Material Name	SA-182 F304L	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element#	15/25	
Element From Node	150	
Element To Node	160	
Element Type	Cylinder	
Description	Shell-6	
Distance "FROM" to "TO"	25	cm
Element Outside Diameter	220	mm
Element Thickness	8	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	8	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	25.492	kgf/cm ²
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.6900001	
Material Name	SA-312 TP304L	
Allowable Stress, Ambient	1172.6	kgf/cm ²
Allowable Stress, Operating	1172.6	kgf/cm ²
Allowable Stress, Hydrotest	1560.1	kgf/cm ²
Material Density	0.008027	kg/cm ³
P Number Thickness	0	mm
Yield Stress, Operating	1536.8	kgf/cm ²
External Pressure Chart Name	HA-3	
UNS Number	S30403	
Product Form	Smls. & wld. pipe	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	150	
Detail Type	Nozzle	
Detail ID	A3 2''	
Dist. from "FROM" Node / Offset dist	14.027	cm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	10.25	kgf
Grade of Attached Flange	GR 2.1	
Nozzle Matl	SA-182 F304L +IT	

Element From Node	150	
Detail Type	Nozzle	
Detail ID	P2 2''	
Dist. from "FROM" Node / Offset dist	14.027	cm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	9.4994	kgf
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F304L +IT	



Element From Node	150	
Detail Type	Weight	
Detail ID	A3 feed pipe	
Dist. from "FROM" Node / Offset dist	14.027	cm
Miscellaneous Weight	5	kgf
Offset from Element Centerline	0	mm

Element#	16/25	
Element From Node	160	
Element To Node	170	
Element Type	Conical	
Description	conical -2	
Distance "FROM" to "TO"	15.2	cm
Element Outside Diameter	220	mm
Element Thickness	7	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	7	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	25.492	kgf/cm ²
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.6900001	
Material Name	SA-312 TP304L	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Cone Diameter at "To" End	141	mm
Design Length of Cone	152	mm
Large knuckle radius	31	mm
Large knuckle thickness	8	mm
Small knuckle radius	12	mm
Small knuckle thickness	8	mm
Half Apex Angle of Cone	15.219764	degrees
Toriconical (Y/N)	Y	
Weld is pre-Heated	No	

Element#	17/25	
Element From Node	170	
Element To Node	180	
Element Type	Cylinder	
Description	shell-7	
Distance "FROM" to "TO"	303.81	cm
Element Outside Diameter	141	mm
Element Thickness	7	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	7	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	25.492	kgf/cm ²
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.6900001	
Material Name	SA-312 TP304L	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	170	
Detail Type	Packing	
Detail ID	BED 3	
Dist. from "FROM" Node / Offset dist	10	cm
Height of Packed Section	237	cm
Density	0.00025	kg/cm ³
Percent Volume Holdup	50.0	
Specific Gravity of Packing Liquid	0.94999999	



Element From Node	170	
Detail Type	Weight	
Detail ID	Bed3 sup ring	
Dist. from "FROM" Node / Offset dist	8	cm
Miscellaneous Weight	2	kgf
Offset from Element Centerline	0	mm

Element From Node	170	
Detail Type	Weight	
Detail ID	Bed 3 Sup grid	
Dist. from "FROM" Node / Offset dist	9.5	cm
Miscellaneous Weight	5	kgf
Offset from Element Centerline	0	mm

Element From Node	170	
Detail Type	Weight	
Detail ID	Bed3 dist wt	
Dist. from "FROM" Node / Offset dist	260	cm
Miscellaneous Weight	5	kgf
Offset from Element Centerline	0	mm

Element From Node	170	
Detail Type	Weight	
Detail ID	Bed3 dist sup w	
Dist. from "FROM" Node / Offset dist	258	cm
Miscellaneous Weight	2	kgf
Offset from Element Centerline	0	mm

Element#	18/25	
Element From Node	180	
Element To Node	190	
Element Type	Flange	
Description	H3-2	
Distance "FROM" to "TO"	9.7028	cm
Flange Inside Diameter	128.27	mm
Element Thickness	33.274	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	34	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	25.492	kgf/cm ²
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.6900001	
Material Name	SA-182 F304L	
Allowable Stress, Ambient	1172.6	kgf/cm ²
Allowable Stress, Operating	1172.6	kgf/cm ²
Allowable Stress, Hydrotest	1560.1	kgf/cm ²
Material Density	0.008027	kg/cm ³
P Number Thickness	0	mm
Yield Stress, Operating	1536.8	kgf/cm ²
External Pressure Chart Name	HA-3	
UNS Number	S30403	
Class / Thickness / Grade	:: > 130	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element#	19/25	
Element From Node	190	
Element To Node	200	
Element Type	Flange	
Description	H3-1	
Distance "FROM" to "TO"	9.7028	cm
Flange Inside Diameter	128.27	mm
Element Thickness	33.274	mm



Internal Corrosion Allowance	0	mm
Nominal Thickness	34	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	25.492	kgf/cm ²
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.6900001	
Material Name	SA-182 F304L	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element#	20/25	
Element From Node	200	
Element To Node	210	
Element Type	Cylinder	
Description	shell-8	
Distance "FROM" to "TO"	295.28	cm
Element Outside Diameter	141.3	mm
Element Thickness	7	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	7	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	25.492	kgf/cm ²
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.6900001	
Material Name	SA-312 TP304L	
Allowable Stress, Ambient	1172.6	kgf/cm ²
Allowable Stress, Operating	1172.6	kgf/cm ²
Allowable Stress, Hydrotest	1560.1	kgf/cm ²
Material Density	0.008027	kg/cm ³
P Number Thickness	0	mm
Yield Stress, Operating	1536.8	kgf/cm ²
External Pressure Chart Name	HA-3	
UNS Number	S30403	
Product Form	Smls. & wld. pipe	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	200	
Detail Type	Packing	
Detail ID	BED 4	
Dist. from "FROM" Node / Offset dist	11.2	cm
Height of Packed Section	237	cm
Density	0.00025	kg/cm ³
Percent Volume Holdup	50.0	
Specific Gravity of Packing Liquid	0.94999999	

Element From Node	200	
Detail Type	Weight	
Detail ID	Bed1 sup ring	
Dist. from "FROM" Node / Offset dist	8	cm
Miscellaneous Weight	2	kgf
Offset from Element Centerline	0	mm

Element From Node	200	
Detail Type	Weight	
Detail ID	Bed1 sup grid	
Dist. from "FROM" Node / Offset dist	10	cm
Miscellaneous Weight	5	kgf
Offset from Element Centerline	0	mm

Element From Node	200	
Detail Type	Weight	
Detail ID	Bed1 dist wt	



Dist. from "FROM" Node / Offset dist 260 cm
 Miscellaneous Weight 5 kgf
 Offset from Element Centerline 0 mm

Element From Node 200
 Detail Type Weight
 Detail ID Bed1 dist sup w
 Dist. from "FROM" Node / Offset dist 258 cm
 Miscellaneous Weight 2 kgf
 Offset from Element Centerline 0 mm

Element From Node 200
 Detail Type Weight
 Detail ID Lifting lug1
 Dist. from "FROM" Node / Offset dist 265.3 cm
 Miscellaneous Weight 29.999 kgf
 Offset from Element Centerline 70 mm

Element From Node 200
 Detail Type Weight
 Detail ID Lifting lug2
 Dist. from "FROM" Node / Offset dist 265.3 cm
 Miscellaneous Weight 29.999 kgf
 Offset from Element Centerline 70 mm

Element# 21/25
 Element From Node 210
 Element To Node 220
 Element Type Flange
 Description H2-2
 Distance "FROM" to "TO" 9.7028 cm
 Flange Inside Diameter 128.27 mm
 Element Thickness 33.274 mm
 Internal Corrosion Allowance 0 mm
 Nominal Thickness 34 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 25.492 kgf/cm²
 Design Temperature Internal Pressure 85 °C
 Design External Pressure 0 kgf/cm²
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.6900001
 Material Name SA-182 F304L
 Allowable Stress, Ambient 1172.6 kgf/cm²
 Allowable Stress, Operating 1172.6 kgf/cm²
 Allowable Stress, Hydrotest 1560.1 kgf/cm²
 Material Density 0.008027 kg/cm³
 P Number Thickness 0 mm
 Yield Stress, Operating 1536.8 kgf/cm²
 External Pressure Chart Name HA-3
 UNS Number S30403
 Class / Thickness / Grade :: > 130
 Product Form Forgings
 Perform Flange Stress Calculation (Y/N) Y
 Weld is pre-Heated No

Element# 22/25
 Element From Node 220
 Element To Node 230
 Element Type Flange
 Description H2-1
 Distance "FROM" to "TO" 9.7028 cm
 Flange Inside Diameter 128.27 mm
 Element Thickness 33.274 mm
 Internal Corrosion Allowance 0 mm
 Nominal Thickness 34 mm
 External Corrosion Allowance 0 mm



Design Internal Pressure	25.492	kgf/cm ²
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.6900001	
Material Name	SA-182 F304L	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element#	23/25	
Element From Node	230	
Element To Node	240	
Element Type	Cylinder	
Description	SHELL9	
Distance "FROM" to "TO"	90.277	cm
Element Outside Diameter	141.3	mm
Element Thickness	7	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	7	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	25.492	kgf/cm ²
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.6900001	
Material Name	SA-312 TP304L	
Allowable Stress, Ambient	1172.6	kgf/cm ²
Allowable Stress, Operating	1172.6	kgf/cm ²
Allowable Stress, Hydrotest	1560.1	kgf/cm ²
Material Density	0.008027	kg/cm ³
P Number Thickness	0	mm
Yield Stress, Operating	1536.8	kgf/cm ²
External Pressure Chart Name	HA-3	
UNS Number	S30403	
Product Form	Smls. & wld. pipe	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	230	
Detail Type	Packing	
Detail ID	Demister	
Dist. from "FROM" Node / Offset dist	70.277	cm
Height of Packed Section	15	cm
Density	0.60000E-04	kg/cm ³
Percent Volume Holdup	50.0	
Specific Gravity of Packing Liquid	0.94999999	

Element From Node	230	
Detail Type	Nozzle	
Detail ID	A2 2''	
Dist. from "FROM" Node / Offset dist	10.297	cm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	10.488	kgf
Grade of Attached Flange	GR 2.1	
Nozzle Matl	SA-182 F304L +IT	

Element From Node	230	
Detail Type	Nozzle	
Detail ID	P1 2''	
Dist. from "FROM" Node / Offset dist	30.297	cm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	



Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	10.526	kgf
Grade of Attached Flange	GR 2.1	
Nozzle Matl	SA-182 F304L	+IT

Element From Node	230	
Detail Type	Weight	
Detail ID	A2 feed pipe	
Dist. from "FROM" Node / Offset dist	10.297	cm
Miscellaneous Weight	5	kgf
Offset from Element Centerline	0	mm

Element From Node	230	
Detail Type	Weight	
Detail ID	Demister sup	
Dist. from "FROM" Node / Offset dist	70	cm
Miscellaneous Weight	5	kgf
Offset from Element Centerline	0	mm

Element#	24/25	
Element From Node	240	
Element To Node	250	
Element Type	Flange	
Description	H1-2	
Distance "FROM" to "TO"	9.7028	cm
Flange Inside Diameter	128.27	mm
Element Thickness	33.274	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	34	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	25.492	kgf/cm ²
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.6900001	
Material Name	SA-182 F304L	
Allowable Stress, Ambient	1172.6	kgf/cm ²
Allowable Stress, Operating	1172.6	kgf/cm ²
Allowable Stress, Hydrotest	1560.1	kgf/cm ²
Material Density	0.008027	kg/cm ³
P Number Thickness	0	mm
Yield Stress, Operating	1536.8	kgf/cm ²
External Pressure Chart Name	HA-3	
UNS Number	S30403	
Class / Thickness / Grade	:: > 130	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element#	25/25	
Element From Node	250	
Element To Node	260	
Element Type	Flange	
Description	H1-1	
Distance "FROM" to "TO"	3.3274	cm
Flange Outside Diameter	279.4	mm
Element Thickness	33.274	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	34	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	25.492	kgf/cm ²
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.6900001	



Material Name	SA-182 F304L
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	250
Detail Type	Nozzle
Detail ID	B1 2''
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	10.381 kgf
Grade of Attached Flange	GR 2.1
Nozzle Matl	SA-182 F304L +IT
Element From Node	250
Detail Type	For./Mom.
Detail ID	B1 F/M
Dist. from "FROM" Node / Offset dist	3 cm
Force in X Direction	317.25 kgf
Force in Y Direction	-358.96 kgf
Force in Z Direction	317.25 kgf
Moment about X Axis	41.6 kgf-m.
Moment about Y Axis	0 kgf-m.
Moment about Z Axis	41.6 kgf-m.
Force/Moment Combination Method	SRSS



XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
Bottom head	...	...	5	...	5
Shell-1	...	...	120	...	115
conical-1	...	...	205	...	85
Shell-2	...	...	213.71	...	8.71
H6-2	...	...	225	...	10.9728
H6-1	...	...	235.973	...	10.9728
SHELL-3	...	...	535.973	...	300
SHELL-4	...	...	673.71	...	137.737
H5-2	...	...	685	...	10.9728
H5-1	...	...	695.973	...	10.9728
Shell-5	...	...	995.973	...	300
Shell-6	...	...	1103.71	...	107.737
H4-2	...	...	1115	...	10.9728
H4-1	...	...	1125.97	...	10.9728
Shell-6	...	...	1150.97	...	25
conical -2	...	...	1166.17	...	15.2
shell-7	...	...	1469.98	...	303.806
H3-2	...	...	1480	...	9.7028
H3-1	...	...	1489.7	...	9.7028
shell-8	...	...	1784.98	...	295.277
H2-2	...	...	1795	...	9.7028
H2-1	...	...	1804.7	...	9.7028
SHELL9	...	...	1894.98	...	90.277
H1-2	...	...	1905	...	9.7028
H1-1	...	...	1908.33	...	3.3274

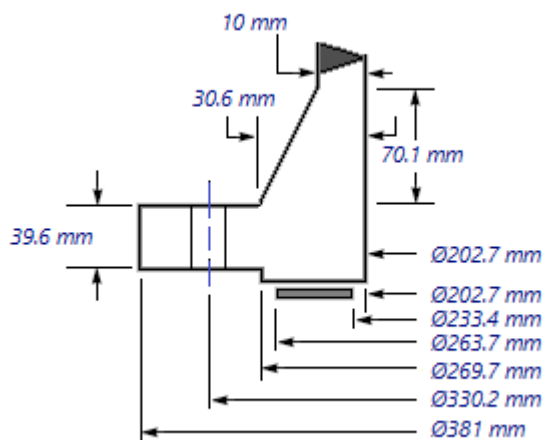
The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flange Input Data Values Description: H6-2 :

H6-2

Flange Type	Integral Weld Neck		
Design Pressure	P	25.49	kgf/cm ²
Design Temperature		85	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	202.692	mm
Flange Outside Diameter	A	381.000	mm
Flange Thickness	t	39.6240	mm
Thickness of Hub at Small End	go	10.0000	mm
Thickness of Hub at Large End	gl	30.5740	mm
Length of Hub	h	70.1040	mm
Flange Material	SA-182 F304L		
Flange Material UNS number		S30403	
Flange Allowable Stress At Temperature	Sfo	1172.61	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1172.61	kgf/cm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	330.200	mm
Nominal Bolt Diameter	a	22.2250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		12	
Flange Face Outside Diameter	Fod	269.748	mm
Flange Face Inside Diameter	Fid	202.692	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	263.652	mm
Gasket Inside Diameter	Gi	233.426	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Hub Small End Required Thickness due to Internal Pressure:
 $= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (25.5 \cdot (203/2 + 0)) / (1173 \cdot 1 - 0.6 \cdot 25.5) + Ca$
 $= 2.2323 \text{ mm}$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1173 \cdot 1 \cdot 10) / (101 + 0.6 \cdot 10)$$

$$= 109.237 \text{ kgf/cm}^2$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1173 \cdot 1 \cdot 10) / (101 + 0.6 \cdot 10)$$

$$= 109.237 \text{ kgf/cm}^2$$

Corroded Flange ID,	Bcor = B+2*ci	202.692 mm
Corroded Large Hub,	g1Cor = g1-ci	30.574 mm
Corroded Small Hub,	g0Cor = go-ci	10.000 mm
Code R Dimension,	R = (C-B)/2 - g1	33.180 mm

Gasket Contact Width,	N = (Go - Gi)/2	15.113 mm
Basic Gasket Width,	bo = N/2	7.557 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	6.927 mm
Gasket Reaction Diameter,	G = Go - 2 * b	249.798 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 250^2 \cdot 25.5$$

$$= 12493.130 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.93 \cdot \pi \cdot 250 \cdot 3 \cdot 25.5$$

$$= 8314.592 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 203^2 \cdot 25.5$$

$$= 8225.584 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 12493 - 8226$$

$$= 4267.545 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(12493 + 8315 + 0, 0)$$

$$= 20807.721 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 703 \cdot 6.93 \cdot \pi \cdot 250 + 0 \cdot 0$$

$$= 38219.406 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((20808 + 0 + 4 \cdot 0 / 250) / 1754, 38219 / 1754)$$

$$= 21.792 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 22.2 + 6 \cdot 39.6 / (3 + 0.5)$$

$$= 112.377 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 330 \cdot \sin(\pi / 12)$$

$$= 85.462 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{(85.5/(2 \cdot 22.2 + 39.6))}, 1)$$

$$= 1.0082$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	21.792	32.439	
Radial Distance between Hub and Bolts:	31.750	33.180	
Radial Distance between Bolts and Edge:	23.812	25.400	
Circ. Spacing between the Bolts:	52.400	85.462	112.377

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi(G_o + G_i))$$

$$= 32.4 \cdot 1754 / (703 \cdot \pi(264 + 233))$$

$$= 5.182 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 1754 \cdot 32.4$$

$$= 56891.45 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 20808 - 12493$$

$$= 8314.59 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (330 - 250) / 2$$

$$= 40.2010 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + h_g)$$

$$= 0.5((330 - 203) / 2 + 40.2)$$

$$= 51.9775 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g_1) / 2$$

$$= (330 - 203 - 30.6) / 2$$

$$= 48.4670 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	8226.	48.4670	1.0082	402.
Face Pressure, Mt	4268.	51.9775	1.0082	224.
Gasket Load, Mg	8315.	40.2010	1.0082	337.
Gasket Seating, Matm	56891.	40.2010	1.0082	2306.

Total Moment for Operation, Mo 963. kgf-m.
 Total Moment for Gasket seating, Mg 2306. kgf-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 45.021 mm
 Hub Ratio, $h/h_o = H_L / H_o$ 1.557
 Thickness Ratio, $g_1/g_o = (g_{1Cor}/g_{oCor})$ 3.057

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.618
 Factor V per 4.16.5 0.057
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 381 / 203$$

$$= 1.880$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.88^2 \cdot \log_{10}(1.88))) / (1.88^2 - 1)) / (1.88 - 1)$$

$$= 3.244$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.88^2(1 + 8.55246 \cdot \log_{10}(1.88)) - 1) / ((1.0472 + 1.9448 \cdot 1.88^2) \cdot (1.88 - 1))$$

$$= 1.553$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.88^2(1 + 8.55246 \cdot \log_{10}(1.88)) - 1) / (1.36136(1.88^2 - 1) \cdot (1.88 - 1))$$

$$= 3.565$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.88^2 + 1) / (1.88^2 - 1)$$

$$= 1.789$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 3.57 \cdot 10^2 \cdot 45 / 0.057$$

$$= 283935.031 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.62 / 45$$

$$= 0.014 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3 / d$$

$$= (39.6 \cdot 0.014 + 1) / 1.55 + 39.6^3 / 283935$$

$$= 1.213$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 963 / 203) / (1.21 \cdot 30.6^2)$$

$$= 418.69 \text{ kgf/cm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 2306 / 203) / (1.21 \cdot 30.6^2)$$

$$= 1002.99 \text{ kgf/cm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.73 \cdot 963 / 203) / (1.21 \cdot 39.6^2)$$

$$= 430.01 \text{ kgf/cm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.73 \cdot 2306 / 203) / (1.21 \cdot 39.6^2)$$

$$= 1030.10 \text{ kgf/cm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (3.24 \cdot 963 / (39.6^2 \cdot 203)) - 1.79 \cdot 430$$

$$= 211.79 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (3.24 \cdot 2306 / (39.6^2 \cdot 203)) - 1.79 \cdot 1030$$

$$= 507.34 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (419 + \max(430, 212)) / 2$$

$$= 424.35 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (1003 + \max(1030, 507)) / 2$$

$$= 1016.54 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 20808 / 32.4$$

$$= 641.45 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (38219 / 32.4)$$

$$= 1178.21 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	418.69	1758.92	1002.99 1758.92
Radial Flange	430.01	1172.61	1030.10 1172.61
Tangential Flange	211.79	1172.61	507.34 1172.61
Maximum Average	424.35	1172.61	1016.54 1172.61
Bolting	641.45	1753.82	1178.21 1753.82

Minimum Required Flange Thickness 37.719 mm

Estimated M.A.W.P. (Operating) 69.5 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 34.0 kg
 Estimated Unfinished Weight of Forging at given Thk 72.0 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 2332 / 1.02 \cdot 100000 \cdot 0.057 / (1.17 \cdot 1989688 \cdot 10^2 \cdot 45 \cdot 0.3)$$

$$= 0.215 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 974 / 1.02 \cdot 100000 \cdot 0.057 / (1.17 \cdot 1939938 \cdot 10^2 \cdot 45 \cdot 0.3)$$

$$= 0.092 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 2306 / 1.01 \cdot 100000 \cdot 0.057 / (1.21 \cdot 1989688 \cdot 10^2 \cdot 45 \cdot 0.3)$$

$$= 0.206 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 963 / 1.01 \cdot 100000 \cdot 0.057 / (1.21 \cdot 1939938 \cdot 10^2 \cdot 45 \cdot 0.3)$$

$$= 0.088 \text{ (should be } \leq 1)$$

SA-182 F304L, Min Metal Temp without impact per UHA-51: -196 °C

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	418.69	1758.92	1002.99 1758.92
Radial Flange	430.01	1172.61	1030.10 1172.61



Tangential Flange	211.79	1172.61	507.34	1172.61
Maximum Average	424.35	1172.61	1016.54	1172.61
Bolting	641.45	1753.82	1178.21	1753.82

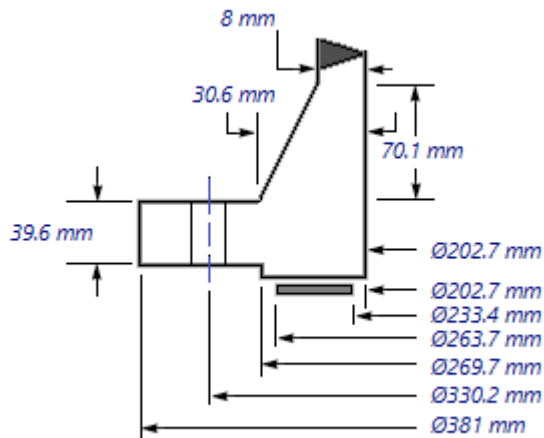
Minimum Req'd Thickness, ambient, no Corrosion: 37.719 mm

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Flange Input Data Values Description: H6-1 :

H6-1

Flange Type	Integral Weld Neck		
Design Pressure	P	25.49	kgf/cm ²
Design Temperature		85	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	202.692	mm
Flange Outside Diameter	A	381.000	mm
Flange Thickness	t	39.6240	mm
Thickness of Hub at Small End	go	8.0000	mm
Thickness of Hub at Large End	gl	30.5740	mm
Length of Hub	h	70.1040	mm
Flange Material	SA-182 F304L		
Flange Material UNS number	S30403		
Flange Allowable Stress At Temperature	Sfo	1172.61	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1172.61	kgf/cm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	330.200	mm
Nominal Bolt Diameter	a	22.2250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	12		
Flange Face Outside Diameter	Fod	269.748	mm
Flange Face Inside Diameter	Fid	202.692	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	263.652	mm
Gasket Inside Diameter	Gi	233.426	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Hub Small End Required Thickness due to Internal Pressure:
 $= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (25.5 \cdot (203/2 + 0)) / (1173 \cdot 1 - 0.6 \cdot 25.5) + Ca$
 $= 2.2323 \text{ mm}$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1173 \cdot 1 \cdot 8) / (101 + 0.6 \cdot 8)$$

$$= 88.377 \text{ kgf/cm}^2$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1173 \cdot 1 \cdot 8) / (101 + 0.6 \cdot 8)$$

$$= 88.377 \text{ kgf/cm}^2$$

Corroded Flange ID,	Bcor = B+2*ci	202.692 mm
Corroded Large Hub,	g1Cor = g1-ci	30.574 mm
Corroded Small Hub,	g0Cor = go-ci	8.000 mm
Code R Dimension,	R = (C-B)/2 - g1	33.180 mm

Gasket Contact Width,	N = (Go - Gi)/2	15.113 mm
Basic Gasket Width,	bo = N/2	7.557 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	6.927 mm
Gasket Reaction Diameter,	G = Go - 2 * b	249.798 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 250^2 \cdot 25.5$$

$$= 12493.130 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.93 \cdot \pi \cdot 250 \cdot 3 \cdot 25.5$$

$$= 8314.592 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 203^2 \cdot 25.5$$

$$= 8225.584 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 12493 - 8226$$

$$= 4267.545 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(12493 + 8315 + 0, 0)$$

$$= 20807.721 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 703 \cdot 6.93 \cdot \pi \cdot 250 + 0 \cdot 0 \cdot 0$$

$$= 38219.406 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((20808 + 0 + 4 \cdot 0 / 250) / 1754, 38219 / 1754)$$

$$= 21.792 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 22.2 + 6 \cdot 39.6 / (3 + 0.5)$$

$$= 112.377 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 330 \cdot \sin(\pi / 12)$$

$$= 85.462 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{(85.5/(2 \cdot 22.2 + 39.6))}, 1)$$

$$= 1.0082$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	21.792	32.439	
Radial Distance between Hub and Bolts:	31.750	33.180	
Radial Distance between Bolts and Edge:	23.812	25.400	
Circ. Spacing between the Bolts:	52.400	85.462	112.377

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi(G_o + G_i))$$

$$= 32.4 \cdot 1754 / (703 \cdot \pi(264 + 233))$$

$$= 5.182 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 1754 \cdot 32.4$$

$$= 56891.45 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 20808 - 12493$$

$$= 8314.59 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (330 - 250) / 2$$

$$= 40.2010 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + h_g)$$

$$= 0.5((330 - 203) / 2 + 40.2)$$

$$= 51.9775 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g_1) / 2$$

$$= (330 - 203 - 30.6) / 2$$

$$= 48.4670 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	8226.	48.4670	1.0082	402.
Face Pressure, Mt	4268.	51.9775	1.0082	224.
Gasket Load, Mg	8315.	40.2010	1.0082	337.
Gasket Seating, Matm	56891.	40.2010	1.0082	2306.

Total Moment for Operation, Mo 963. kgf-m.
 Total Moment for Gasket seating, Mg 2306. kgf-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 40.268 mm
 Hub Ratio, $h/h_o = H_L / H_o$ 1.741
 Thickness Ratio, $g_1/g_o = (g_{1Cor}/g_{oCor})$ 3.822

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.564
 Factor V per 4.16.5 0.034
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 381 / 203$$

$$= 1.880$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.88^2 \cdot \log_{10}(1.88))) / (1.88^2 - 1)) / (1.88 - 1)$$

$$= 3.244$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.88^2(1 + 8.55246 \cdot \log_{10}(1.88)) - 1) / ((1.0472 + 1.9448 \cdot 1.88^2) \cdot (1.88 - 1))$$

$$= 1.553$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.88^2(1 + 8.55246 \cdot \log_{10}(1.88)) - 1) / (1.36136(1.88^2 - 1) \cdot (1.88 - 1))$$

$$= 3.565$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.88^2 + 1) / (1.88^2 - 1)$$

$$= 1.789$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 3.57 \cdot 8^2 \cdot 40.3 / 0.034$$

$$= 266976.188 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.56 / 40.3$$

$$= 0.014 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (39.6 \cdot 0.014 + 1) / 1.55 + 39.6^3 / 266976$$

$$= 1.234$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 963 / 203) / (1.23 \cdot 30.6^2)$$

$$= 411.60 \text{ kgf/cm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 2306 / 203) / (1.23 \cdot 30.6^2)$$

$$= 985.98 \text{ kgf/cm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.74 \cdot 963 / 203) / (1.23 \cdot 39.6^2)$$

$$= 426.28 \text{ kgf/cm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.74 \cdot 2306 / 203) / (1.23 \cdot 39.6^2)$$

$$= 1021.15 \text{ kgf/cm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (3.24 \cdot 963 / (39.6^2 \cdot 203)) - 1.79 \cdot 426$$

$$= 218.47 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (3.24 \cdot 2306 / (39.6^2 \cdot 203)) - 1.79 \cdot 1021$$

$$= 523.36 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (412 + \max(426, 218)) / 2$$

$$= 418.94 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (986 + \max(1021, 523)) / 2$$

$$= 1003.57 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 20808 / 32.4$$

$$= 641.45 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (38219 / 32.4)$$

$$= 1178.21 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	411.60	1758.92	985.98 1758.92
Radial Flange	426.28	1172.61	1021.15 1172.61
Tangential Flange	218.47	1172.61	523.36 1172.61
Maximum Average	418.94	1172.61	1003.57 1172.61
Bolting	641.45	1753.82	1178.21 1753.82

Minimum Required Flange Thickness 37.617 mm

Estimated M.A.W.P. (Operating) 69.7 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 33.6 kg
 Estimated Unfinished Weight of Forging at given Thk 72.0 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 2334 / 1.02 \cdot 100000 \cdot 0.034 / (1.18 \cdot 1989688 \cdot 8^2 \cdot 40.3 \cdot 0.3)$
 $= 0.225$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 974 / 1.02 \cdot 100000 \cdot 0.034 / (1.18 \cdot 1939938 \cdot 8^2 \cdot 40.3 \cdot 0.3)$
 $= 0.096$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 2306 / 1.01 \cdot 100000 \cdot 0.034 / (1.23 \cdot 1989688 \cdot 8^2 \cdot 40.3 \cdot 0.3)$
 $= 0.216$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 963 / 1.01 \cdot 100000 \cdot 0.034 / (1.23 \cdot 1939938 \cdot 8^2 \cdot 40.3 \cdot 0.3)$
 $= 0.092$ (should be ≤ 1)

SA-182 F304L, Min Metal Temp without impact per UHA-51: -196 °C

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	411.60	1758.92	985.98 1758.92
Radial Flange	426.28	1172.61	1021.15 1172.61



Tangential Flange	218.47	1172.61	523.36	1172.61
Maximum Average	418.94	1172.61	1003.57	1172.61
Bolting	641.45	1753.82	1178.21	1753.82

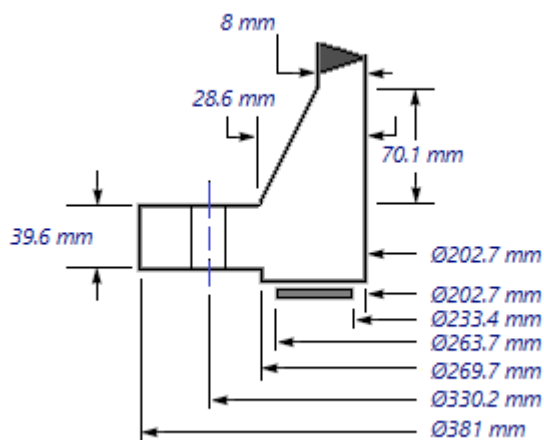
Minimum Req'd Thickness, ambient, no Corrosion: 37.617 mm

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Flange Input Data Values **Description: H5-2** :

H5-2

Flange Type	Integral Weld Neck		
Design Pressure	P	25.49	kgf/cm ²
Design Temperature		85	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	202.692	mm
Flange Outside Diameter	A	381.000	mm
Flange Thickness	t	39.6240	mm
Thickness of Hub at Small End	go	8.0000	mm
Thickness of Hub at Large End	gl	28.5740	mm
Length of Hub	h	70.1040	mm
Flange Material	SA-182 F304L		
Flange Material UNS number		S30403	
Flange Allowable Stress At Temperature	Sfo	1172.61	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1172.61	kgf/cm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	330.200	mm
Nominal Bolt Diameter	a	22.2250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		12	
Flange Face Outside Diameter	Fod	269.748	mm
Flange Face Inside Diameter	Fid	202.692	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	263.652	mm
Gasket Inside Diameter	Gi	233.426	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Hub Small End Required Thickness due to Internal Pressure:
 $= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (25.5 \cdot (203/2 + 0)) / (1173 \cdot 1 - 0.6 \cdot 25.5) + Ca$
 $= 2.2323 \text{ mm}$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1173 \cdot 1 \cdot 8) / (101 + 0.6 \cdot 8)$$

$$= 88.377 \text{ kgf/cm}^2$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1173 \cdot 1 \cdot 8) / (101 + 0.6 \cdot 8)$$

$$= 88.377 \text{ kgf/cm}^2$$

Corroded Flange ID,	Bcor = B+2*ci	202.692 mm
Corroded Large Hub,	g1Cor = g1-ci	28.574 mm
Corroded Small Hub,	g0Cor = go-ci	8.000 mm
Code R Dimension,	R = (C-B)/2 - g1	35.180 mm

Gasket Contact Width,	N = (Go - Gi)/2	15.113 mm
Basic Gasket Width,	bo = N/2	7.557 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	6.927 mm
Gasket Reaction Diameter,	G = Go - 2 * b	249.798 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 250^2 \cdot 25.5$$

$$= 12493.130 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.93 \cdot \pi \cdot 250 \cdot 3 \cdot 25.5$$

$$= 8314.592 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 203^2 \cdot 25.5$$

$$= 8225.584 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 12493 - 8226$$

$$= 4267.545 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(12493 + 8315 + 0, 0)$$

$$= 20807.721 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 703 \cdot 6.93 \cdot \pi \cdot 250 + 0 \cdot 0$$

$$= 38219.406 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((20808 + 0 + 4 \cdot 0 / 250) / 1754, 38219 / 1754)$$

$$= 21.792 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 22.2 + 6 \cdot 39.6 / (3 + 0.5)$$

$$= 112.377 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 330 \cdot \sin(\pi / 12)$$

$$= 85.462 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{(85.5/(2 \cdot 22.2 + 39.6))}, 1)$$

$$= 1.0082$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	21.792	32.439	
Radial Distance between Hub and Bolts:	31.750	35.180	
Radial Distance between Bolts and Edge:	23.812	25.400	
Circ. Spacing between the Bolts:	52.400	85.462	112.377

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi(G_o + G_i))$$

$$= 32.4 \cdot 1754 / (703 \cdot \pi(264 + 233))$$

$$= 5.182 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 1754 \cdot 32.4$$

$$= 56891.45 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 20808 - 12493$$

$$= 8314.59 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (330 - 250) / 2$$

$$= 40.2010 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + h_g)$$

$$= 0.5((330 - 203) / 2 + 40.2)$$

$$= 51.9775 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g_1) / 2$$

$$= (330 - 203 - 28.6) / 2$$

$$= 49.4670 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	8226.	49.4670	1.0082	410.
Face Pressure, Mt	4268.	51.9775	1.0082	224.
Gasket Load, Mg	8315.	40.2010	1.0082	337.
Gasket Seating, Matm	56891.	40.2010	1.0082	2306.

Total Moment for Operation, Mo 971. kgf-m.
 Total Moment for Gasket seating, Mg 2306. kgf-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 40.268 mm
 Hub Ratio, $h/h_o = H_L / H_o$ 1.741
 Thickness Ratio, $g_1/g_o = (g_{1Cor}/g_{oCor})$ 3.572

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.574
 Factor V per 4.16.5 0.039
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 381 / 203$$

$$= 1.880$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.88^2 \cdot \log_{10}(1.88))) / (1.88^2 - 1)) / (1.88 - 1)$$

$$= 3.244$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.88^2(1 + 8.55246 \cdot \log_{10}(1.88)) - 1) / ((1.0472 + 1.9448 \cdot 1.88^2) \cdot (1.88 - 1))$$

$$= 1.553$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.88^2(1 + 8.55246 \cdot \log_{10}(1.88)) - 1) / (1.36136(1.88^2 - 1) \cdot (1.88 - 1))$$

$$= 3.565$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.88^2 + 1) / (1.88^2 - 1)$$

$$= 1.789$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 3.57 \cdot 8^2 \cdot 40.3 / 0.039$$

$$= 234872.734 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.57 / 40.3$$

$$= 0.014 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (39.6 \cdot 0.014 + 1) / 1.55 + 39.6^3 / 234873$$

$$= 1.272$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 971 / 203) / (1.27 \cdot 28.6^2)$$

$$= 461.05 \text{ kgf/cm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 2306 / 203) / (1.27 \cdot 28.6^2)$$

$$= 1095.01 \text{ kgf/cm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.75 \cdot 971 / 203) / (1.27 \cdot 39.6^2)$$

$$= 420.18 \text{ kgf/cm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.75 \cdot 2306 / 203) / (1.27 \cdot 39.6^2)$$

$$= 997.95 \text{ kgf/cm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (3.24 \cdot 971 / (39.6^2 \cdot 203)) - 1.79 \cdot 420$$

$$= 237.83 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (3.24 \cdot 2306 / (39.6^2 \cdot 203)) - 1.79 \cdot 998$$

$$= 564.86 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (461 + \max(420, 238)) / 2$$

$$= 440.61 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (1095 + \max(998, 565)) / 2$$

$$= 1046.48 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 20808 / 32.4$$

$$= 641.45 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (38219 / 32.4)$$

$$= 1178.21 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	461.05	1758.92	1095.01 1758.92
Radial Flange	420.18	1172.61	997.95 1172.61
Tangential Flange	237.83	1172.61	564.86 1172.61
Maximum Average	440.61	1172.61	1046.48 1172.61
Bolting	641.45	1753.82	1178.21 1753.82

Minimum Required Flange Thickness 37.313 mm

Estimated M.A.W.P. (Operating) 67.8 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 33.1 kg
 Estimated Unfinished Weight of Forging at given Thk 72.0 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 2338 / 1.02 \cdot 100000 \cdot 0.039 / (1.21 \cdot 1989688 \cdot 8^2 \cdot 40.3 \cdot 0.3)$$

$$= 0.251 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 985 / 1.02 \cdot 100000 \cdot 0.039 / (1.21 \cdot 1939938 \cdot 8^2 \cdot 40.3 \cdot 0.3)$$

$$= 0.108 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 2306 / 1.01 \cdot 100000 \cdot 0.039 / (1.27 \cdot 1989688 \cdot 8^2 \cdot 40.3 \cdot 0.3)$$

$$= 0.238 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 971 / 1.01 \cdot 100000 \cdot 0.039 / (1.27 \cdot 1939938 \cdot 8^2 \cdot 40.3 \cdot 0.3)$$

$$= 0.102 \text{ (should be } \leq 1)$$

SA-182 F304L, Min Metal Temp without impact per UHA-51: -196 °C

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	461.05	1758.92	1095.01 1758.92
Radial Flange	420.18	1172.61	997.95 1172.61



Tangential Flange	237.83	1172.61	564.86	1172.61
Maximum Average	440.61	1172.61	1046.48	1172.61
Bolting	641.45	1753.82	1178.21	1753.82

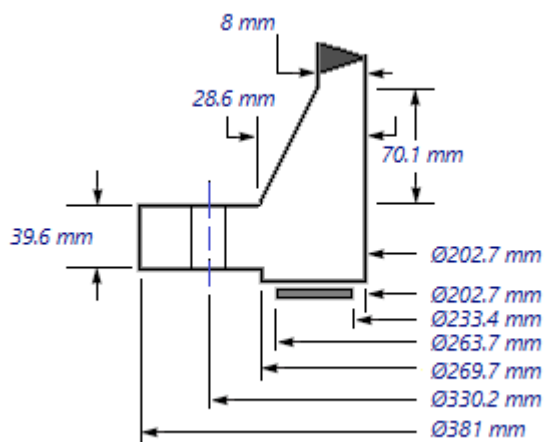
Minimum Req'd Thickness, ambient, no Corrosion: 37.313 mm

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Flange Input Data Values Description: H5-1 :

H5-1

Flange Type	Integral Weld Neck		
Design Pressure	P	25.49	kgf/cm ²
Design Temperature		85	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	202.692	mm
Flange Outside Diameter	A	381.000	mm
Flange Thickness	t	39.6240	mm
Thickness of Hub at Small End	go	8.0000	mm
Thickness of Hub at Large End	gl	28.5740	mm
Length of Hub	h	70.1040	mm
Flange Material	SA-182 F304L		
Flange Material UNS number	S30403		
Flange Allowable Stress At Temperature	Sfo	1172.61	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1172.61	kgf/cm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	330.200	mm
Nominal Bolt Diameter	a	22.2250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	12		
Flange Face Outside Diameter	Fod	269.748	mm
Flange Face Inside Diameter	Fid	202.692	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	263.652	mm
Gasket Inside Diameter	Gi	233.426	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Hub Small End Required Thickness due to Internal Pressure:
 $= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (25.5 \cdot (203/2 + 0)) / (1173 \cdot 1 - 0.6 \cdot 25.5) + Ca$
 $= 2.2323 \text{ mm}$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1173 \cdot 1 \cdot 8) / (101 + 0.6 \cdot 8)$$

$$= 88.377 \text{ kgf/cm}^2$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1173 \cdot 1 \cdot 8) / (101 + 0.6 \cdot 8)$$

$$= 88.377 \text{ kgf/cm}^2$$

Corroded Flange ID,	Bcor = B+2*ci	202.692 mm
Corroded Large Hub,	g1Cor = g1-ci	28.574 mm
Corroded Small Hub,	g0Cor = go-ci	8.000 mm
Code R Dimension,	R = (C-B)/2 - g1	35.180 mm

Gasket Contact Width,	N = (Go - Gi)/2	15.113 mm
Basic Gasket Width,	bo = N/2	7.557 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	6.927 mm
Gasket Reaction Diameter,	G = Go - 2 * b	249.798 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 250^2 \cdot 25.5$$

$$= 12493.130 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.93 \cdot \pi \cdot 250 \cdot 3 \cdot 25.5$$

$$= 8314.592 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 203^2 \cdot 25.5$$

$$= 8225.584 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 12493 - 8226$$

$$= 4267.545 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(12493 + 8315 + 0, 0)$$

$$= 20807.721 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 703 \cdot 6.93 \cdot \pi \cdot 250 + 0 \cdot 0$$

$$= 38219.406 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((20808 + 0 + 4 \cdot 0 / 250) / 1754, 38219 / 1754)$$

$$= 21.792 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 22.2 + 6 \cdot 39.6 / (3 + 0.5)$$

$$= 112.377 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 330 \cdot \sin(\pi / 12)$$

$$= 85.462 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{(85.5/(2 \cdot 22.2 + 39.6))}, 1)$$

$$= 1.0082$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	21.792	32.439	
Radial Distance between Hub and Bolts:	31.750	35.180	
Radial Distance between Bolts and Edge:	23.812	25.400	
Circ. Spacing between the Bolts:	52.400	85.462	112.377

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi(G_o + G_i))$$

$$= 32.4 \cdot 1754 / (703 \cdot \pi(264 + 233))$$

$$= 5.182 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a (A_m + A_b) / 2$$

$$= 1754 (21.8 + 32.4) / 2$$

$$= 47555.43 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 20808 - 12493$$

$$= 8314.59 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (330 - 250) / 2$$

$$= 40.2010 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + h_g)$$

$$= 0.5((330 - 203) / 2 + 40.2)$$

$$= 51.9775 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g_1) / 2$$

$$= (330 - 203 - 28.6) / 2$$

$$= 49.4670 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	8226.	49.4670	1.0082	410.
Face Pressure, Mt	4268.	51.9775	1.0082	224.
Gasket Load, Mg	8315.	40.2010	1.0082	337.
Gasket Seating, Matm	47555.	40.2010	1.0082	1927.

Total Moment for Operation, Mo 971. kgf-m.
 Total Moment for Gasket seating, Mg 1927. kgf-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_o \cdot C_{cor}}$ 40.268 mm
 Hub Ratio, $h/h_o = H_L / H_o$ 1.741
 Thickness Ratio, $g_1/g_o = (g_1 \cdot C_{cor}) / (g_o \cdot C_{cor})$ 3.572

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.574
 Factor V per 4.16.5 0.039
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 381 / 203$$

$$= 1.880$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.88^2 \cdot \log_{10}(1.88))) / (1.88^2 - 1)) / (1.88 - 1)$$

$$= 3.244$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.88^2(1 + 8.55246 \cdot \log_{10}(1.88)) - 1) / ((1.0472 + 1.9448 \cdot 1.88^2) \cdot (1.88 - 1))$$

$$= 1.553$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.88^2(1 + 8.55246 \cdot \log_{10}(1.88)) - 1) / (1.36136(1.88^2 - 1) \cdot (1.88 - 1))$$

$$= 3.565$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.88^2 + 1) / (1.88^2 - 1)$$

$$= 1.789$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 3.57 \cdot 8^2 \cdot 40.3 / 0.039$$

$$= 234872.734 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.57 / 40.3$$

$$= 0.014 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (39.6 \cdot 0.014 + 1) / 1.55 + 39.6^3 / 234873$$

$$= 1.272$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 971 / 203) / (1.27 \cdot 28.6^2)$$

$$= 461.05 \text{ kgf/cm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 1927 / 203) / (1.27 \cdot 28.6^2)$$

$$= 915.31 \text{ kgf/cm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.75 \cdot 971 / 203) / (1.27 \cdot 39.6^2)$$

$$= 420.18 \text{ kgf/cm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.75 \cdot 1927 / 203) / (1.27 \cdot 39.6^2)$$

$$= 834.19 \text{ kgf/cm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (3.24 \cdot 971 / (39.6^2 \cdot 203)) - 1.79 \cdot 420$$

$$= 237.83 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (3.24 \cdot 1927 / (39.6^2 \cdot 203)) - 1.79 \cdot 834$$

$$= 472.17 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (461 + \max(420, 238)) / 2$$

$$= 440.61 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (915 + \max(834, 472)) / 2$$

$$= 874.75 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 20808 / 32.4$$

$$= 641.45 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (38219 / 32.4)$$

$$= 1178.21 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	461.05	1758.92	915.31 1758.92
Radial Flange	420.18	1172.61	834.19 1172.61
Tangential Flange	237.83	1172.61	472.17 1172.61
Maximum Average	440.61	1172.61	874.75 1172.61
Bolting	641.45	1753.82	1178.21 1753.82

Minimum Required Flange Thickness 34.849 mm

Estimated M.A.W.P. (Operating) 67.8 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 33.1 kg
 Estimated Unfinished Weight of Forging at given Thk 72.0 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 1985 / 1.04 \cdot 100000 \cdot 0.039 / (1.14 \cdot 1989688 \cdot 8^2 \cdot 40.3 \cdot 0.3)$$

$$= 0.221 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 1000 / 1.04 \cdot 100000 \cdot 0.039 / (1.14 \cdot 1939938 \cdot 8^2 \cdot 40.3 \cdot 0.3)$$

$$= 0.114 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 1927 / 1.01 \cdot 100000 \cdot 0.039 / (1.27 \cdot 1989688 \cdot 8^2 \cdot 40.3 \cdot 0.3)$$

$$= 0.199 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 971 / 1.01 \cdot 100000 \cdot 0.039 / (1.27 \cdot 1939938 \cdot 8^2 \cdot 40.3 \cdot 0.3)$$

$$= 0.102 \text{ (should be } \leq 1)$$

SA-182 F304L, Min Metal Temp without impact per UHA-51: -196 °C

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	461.05	1758.92	915.31 1758.92
Radial Flange	420.18	1172.61	834.19 1172.61



Tangential Flange	237.83	1172.61	472.17	1172.61
Maximum Average	440.61	1172.61	874.75	1172.61
Bolting	641.45	1753.82	1178.21	1753.82

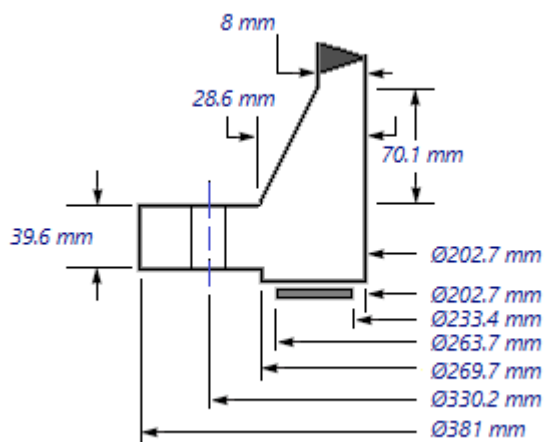
Minimum Req'd Thickness, ambient, no Corrosion: 34.849 mm

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Flange Input Data Values Description: H4-2 :

H4-2

Flange Type	Integral Weld Neck		
Design Pressure	P	25.49	kgf/cm ²
Design Temperature		85	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	202.692	mm
Flange Outside Diameter	A	381.000	mm
Flange Thickness	t	39.6240	mm
Thickness of Hub at Small End	go	8.0000	mm
Thickness of Hub at Large End	gl	28.5740	mm
Length of Hub	h	70.1040	mm
Flange Material	SA-182 F304L		
Flange Material UNS number		S30403	
Flange Allowable Stress At Temperature	Sfo	1172.61	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1172.61	kgf/cm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	330.200	mm
Nominal Bolt Diameter	a	22.2250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		12	
Flange Face Outside Diameter	Fod	269.748	mm
Flange Face Inside Diameter	Fid	202.692	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	263.652	mm
Gasket Inside Diameter	Gi	233.426	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:
 $= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (25.5 \cdot (203/2 + 0)) / (1173 \cdot 1 - 0.6 \cdot 25.5) + Ca$
 $= 2.2323 \text{ mm}$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1173 \cdot 1 \cdot 8) / (101 + 0.6 \cdot 8)$$

$$= 88.377 \text{ kgf/cm}^2$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1173 \cdot 1 \cdot 8) / (101 + 0.6 \cdot 8)$$

$$= 88.377 \text{ kgf/cm}^2$$

Corroded Flange ID,	Bcor = B+2*ci	202.692 mm
Corroded Large Hub,	g1Cor = g1-ci	28.574 mm
Corroded Small Hub,	g0Cor = go-ci	8.000 mm
Code R Dimension,	R = (C-B)/2 - g1	35.180 mm

Gasket Contact Width,	N = (Go - Gi)/2	15.113 mm
Basic Gasket Width,	bo = N/2	7.557 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	6.927 mm
Gasket Reaction Diameter,	G = Go - 2 * b	249.798 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 250^2 \cdot 25.5$$

$$= 12493.130 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.93 \cdot \pi \cdot 250 \cdot 3 \cdot 25.5$$

$$= 8314.592 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 203^2 \cdot 25.5$$

$$= 8225.584 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 12493 - 8226$$

$$= 4267.545 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(12493 + 8315 + 0, 0)$$

$$= 20807.721 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 703 \cdot 6.93 \cdot \pi \cdot 250 + 0 \cdot 0$$

$$= 38219.406 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((20808 + 0 + 4 \cdot 0 / 250) / 1754, 38219 / 1754)$$

$$= 21.792 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 22.2 + 6 \cdot 39.6 / (3 + 0.5)$$

$$= 112.377 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 330 \cdot \sin(\pi / 12)$$

$$= 85.462 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{(85.5/(2 \cdot 22.2 + 39.6))}, 1)$$

$$= 1.0082$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	21.792	32.439	
Radial Distance between Hub and Bolts:	31.750	35.180	
Radial Distance between Bolts and Edge:	23.812	25.400	
Circ. Spacing between the Bolts:	52.400	85.462	112.377

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi(G_o + G_i))$$

$$= 32.4 \cdot 1754 / (703 \cdot \pi(264 + 233))$$

$$= 5.182 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 1754 \cdot 32.4$$

$$= 56891.45 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 20808 - 12493$$

$$= 8314.59 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (330 - 250) / 2$$

$$= 40.2010 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + h_g)$$

$$= 0.5((330 - 203) / 2 + 40.2)$$

$$= 51.9775 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g_1) / 2$$

$$= (330 - 203 - 28.6) / 2$$

$$= 49.4670 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	8226.	49.4670	1.0082	410.
Face Pressure, Mt	4268.	51.9775	1.0082	224.
Gasket Load, Mg	8315.	40.2010	1.0082	337.
Gasket Seating, Matm	56891.	40.2010	1.0082	2306.

Total Moment for Operation, Mo 971. kgf-m.
 Total Moment for Gasket seating, Mg 2306. kgf-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 40.268 mm
 Hub Ratio, $h/h_o = H_L / H_o$ 1.741
 Thickness Ratio, $g_1/g_o = (g_{1Cor}/g_{oCor})$ 3.572

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.574
 Factor V per 4.16.5 0.039
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 381/203$$

$$= 1.880$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.88^2 \cdot \log_{10}(1.88))) / (1.88^2 - 1)) / (1.88 - 1)$$

$$= 3.244$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.88^2(1 + 8.55246 \cdot \log_{10}(1.88)) - 1) / ((1.0472 + 1.9448 \cdot 1.88^2) \cdot (1.88 - 1))$$

$$= 1.553$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.88^2(1 + 8.55246 \cdot \log_{10}(1.88)) - 1) / (1.36136(1.88^2 - 1) \cdot (1.88 - 1))$$

$$= 3.565$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.88^2 + 1) / (1.88^2 - 1)$$

$$= 1.789$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 3.57 \cdot 8^2 \cdot 40.3 / 0.039$$

$$= 234872.734 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.57 / 40.3$$

$$= 0.014 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (39.6 \cdot 0.014 + 1) / 1.55 + 39.6^3 / 234873$$

$$= 1.272$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 971 / 203) / (1.27 \cdot 28.6^2)$$

$$= 461.05 \text{ kgf/cm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 2306 / 203) / (1.27 \cdot 28.6^2)$$

$$= 1095.01 \text{ kgf/cm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.75 \cdot 971 / 203) / (1.27 \cdot 39.6^2)$$

$$= 420.18 \text{ kgf/cm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.75 \cdot 2306 / 203) / (1.27 \cdot 39.6^2)$$

$$= 997.95 \text{ kgf/cm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (3.24 \cdot 971 / (39.6^2 \cdot 203)) - 1.79 \cdot 420$$

$$= 237.83 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (3.24 \cdot 2306 / (39.6^2 \cdot 203)) - 1.79 \cdot 998$$

$$= 564.86 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (461 + \max(420, 238)) / 2$$

$$= 440.61 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (1095 + \max(998, 565)) / 2$$

$$= 1046.48 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 20808 / 32.4$$

$$= 641.45 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (38219 / 32.4)$$

$$= 1178.21 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	461.05	1758.92	1095.01 1758.92
Radial Flange	420.18	1172.61	997.95 1172.61
Tangential Flange	237.83	1172.61	564.86 1172.61
Maximum Average	440.61	1172.61	1046.48 1172.61
Bolting	641.45	1753.82	1178.21 1753.82

Minimum Required Flange Thickness 37.313 mm

Estimated M.A.W.P. (Operating) 67.8 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 33.1 kg
 Estimated Unfinished Weight of Forging at given Thk 72.0 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 2338 / 1.02 \cdot 100000 \cdot 0.039 / (1.21 \cdot 1989688 \cdot 8^2 \cdot 40.3 \cdot 0.3)$
 $= 0.251$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 985 / 1.02 \cdot 100000 \cdot 0.039 / (1.21 \cdot 1939938 \cdot 8^2 \cdot 40.3 \cdot 0.3)$
 $= 0.108$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 2306 / 1.01 \cdot 100000 \cdot 0.039 / (1.27 \cdot 1989688 \cdot 8^2 \cdot 40.3 \cdot 0.3)$
 $= 0.238$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 971 / 1.01 \cdot 100000 \cdot 0.039 / (1.27 \cdot 1939938 \cdot 8^2 \cdot 40.3 \cdot 0.3)$
 $= 0.102$ (should be ≤ 1)

SA-182 F304L, Min Metal Temp without impact per UHA-51: -196 °C

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	461.05	1758.92	1095.01 1758.92
Radial Flange	420.18	1172.61	997.95 1172.61



Tangential Flange	237.83	1172.61	564.86	1172.61
Maximum Average	440.61	1172.61	1046.48	1172.61
Bolting	641.45	1753.82	1178.21	1753.82

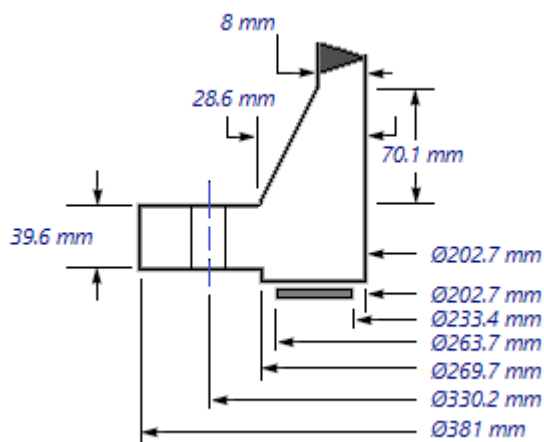
Minimum Req'd Thickness, ambient, no Corrosion: 37.313 mm

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Flange Input Data Values Description: H4-1 :

H4-1

Flange Type	Integral Weld Neck		
Design Pressure	P	25.49	kgf/cm ²
Design Temperature		85	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	202.692	mm
Flange Outside Diameter	A	381.000	mm
Flange Thickness	t	39.6240	mm
Thickness of Hub at Small End	go	8.0000	mm
Thickness of Hub at Large End	gl	28.5740	mm
Length of Hub	h	70.1040	mm
Flange Material	SA-182 F304L		
Flange Material UNS number	S30403		
Flange Allowable Stress At Temperature	Sfo	1172.61	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1172.61	kgf/cm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	330.200	mm
Nominal Bolt Diameter	a	22.2250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	12		
Flange Face Outside Diameter	Fod	269.748	mm
Flange Face Inside Diameter	Fid	202.692	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	263.652	mm
Gasket Inside Diameter	Gi	233.426	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Hub Small End Required Thickness due to Internal Pressure:
 $= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (25.5 \cdot (203/2 + 0)) / (1173 \cdot 1 - 0.6 \cdot 25.5) + Ca$
 $= 2.2323 \text{ mm}$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1173 \cdot 1 \cdot 8) / (101 + 0.6 \cdot 8)$$

$$= 88.377 \text{ kgf/cm}^2$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1173 \cdot 1 \cdot 8) / (101 + 0.6 \cdot 8)$$

$$= 88.377 \text{ kgf/cm}^2$$

Corroded Flange ID,	Bcor = B+2*ci	202.692 mm
Corroded Large Hub,	g1Cor = g1-ci	28.574 mm
Corroded Small Hub,	g0Cor = go-ci	8.000 mm
Code R Dimension,	R = (C-B)/2 - g1	35.180 mm

Gasket Contact Width,	N = (Go - Gi)/2	15.113 mm
Basic Gasket Width,	bo = N/2	7.557 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	6.927 mm
Gasket Reaction Diameter,	G = Go - 2 * b	249.798 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 250^2 \cdot 25.5$$

$$= 12493.130 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.93 \cdot \pi \cdot 250 \cdot 3 \cdot 25.5$$

$$= 8314.592 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 203^2 \cdot 25.5$$

$$= 8225.584 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 12493 - 8226$$

$$= 4267.545 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(12493 + 8315 + 0, 0)$$

$$= 20807.721 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 703 \cdot 6.93 \cdot \pi \cdot 250 + 0 \cdot 0$$

$$= 38219.406 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((20808 + 0 + 4 \cdot 0 / 250) / 1754, 38219 / 1754)$$

$$= 21.792 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 22.2 + 6 \cdot 39.6 / (3 + 0.5)$$

$$= 112.377 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 330 \cdot \sin(\pi / 12)$$

$$= 85.462 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{(85.5/(2 \cdot 22.2 + 39.6))}, 1)$$

$$= 1.0082$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	21.792	32.439	
Radial Distance between Hub and Bolts:	31.750	35.180	
Radial Distance between Bolts and Edge:	23.812	25.400	
Circ. Spacing between the Bolts:	52.400	85.462	112.377

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi(G_o + G_i))$$

$$= 32.4 \cdot 1754 / (703 \cdot \pi(264 + 233))$$

$$= 5.182 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 1754 \cdot 32.4$$

$$= 56891.45 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 20808 - 12493$$

$$= 8314.59 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (330 - 250) / 2$$

$$= 40.2010 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + h_g)$$

$$= 0.5((330 - 203) / 2 + 40.2)$$

$$= 51.9775 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g_1) / 2$$

$$= (330 - 203 - 28.6) / 2$$

$$= 49.4670 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	8226.	49.4670	1.0082	410.
Face Pressure, Mt	4268.	51.9775	1.0082	224.
Gasket Load, Mg	8315.	40.2010	1.0082	337.
Gasket Seating, Matm	56891.	40.2010	1.0082	2306.

Total Moment for Operation, Mo 971. kgf-m.
 Total Moment for Gasket seating, Mg 2306. kgf-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 40.268 mm
 Hub Ratio, $h/h_o = H_L / H_o$ 1.741
 Thickness Ratio, $g_1/g_o = (g_{1Cor}/g_{oCor})$ 3.572

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.574
 Factor V per 4.16.5 0.039
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 381 / 203$$

$$= 1.880$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.88^2 \cdot \log_{10}(1.88))) / (1.88^2 - 1)) / (1.88 - 1)$$

$$= 3.244$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.88^2(1 + 8.55246 \cdot \log_{10}(1.88)) - 1) / ((1.0472 + 1.9448 \cdot 1.88^2) \cdot (1.88 - 1))$$

$$= 1.553$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.88^2(1 + 8.55246 \cdot \log_{10}(1.88)) - 1) / (1.36136(1.88^2 - 1) \cdot (1.88 - 1))$$

$$= 3.565$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.88^2 + 1) / (1.88^2 - 1)$$

$$= 1.789$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 3.57 \cdot 8^2 \cdot 40.3 / 0.039$$

$$= 234872.734 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.57 / 40.3$$

$$= 0.014 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (39.6 \cdot 0.014 + 1) / 1.55 + 39.6^3 / 234873$$

$$= 1.272$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 971 / 203) / (1.27 \cdot 28.6^2)$$

$$= 461.05 \text{ kgf/cm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 2306 / 203) / (1.27 \cdot 28.6^2)$$

$$= 1095.01 \text{ kgf/cm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.75 \cdot 971 / 203) / (1.27 \cdot 39.6^2)$$

$$= 420.18 \text{ kgf/cm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.75 \cdot 2306 / 203) / (1.27 \cdot 39.6^2)$$

$$= 997.95 \text{ kgf/cm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (3.24 \cdot 971 / (39.6^2 \cdot 203)) - 1.79 \cdot 420$$

$$= 237.83 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (3.24 \cdot 2306 / (39.6^2 \cdot 203)) - 1.79 \cdot 998$$

$$= 564.86 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (461 + \max(420, 238)) / 2$$

$$= 440.61 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (1095 + \max(998, 565)) / 2$$

$$= 1046.48 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 20808 / 32.4$$

$$= 641.45 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (38219 / 32.4)$$

$$= 1178.21 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	461.05	1758.92	1095.01 1758.92
Radial Flange	420.18	1172.61	997.95 1172.61
Tangential Flange	237.83	1172.61	564.86 1172.61
Maximum Average	440.61	1172.61	1046.48 1172.61
Bolting	641.45	1753.82	1178.21 1753.82

Minimum Required Flange Thickness 37.313 mm

Estimated M.A.W.P. (Operating) 67.8 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 33.1 kg
 Estimated Unfinished Weight of Forging at given Thk 72.0 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 2338 / 1.02 \cdot 100000 \cdot 0.039 / (1.21 \cdot 1989688 \cdot 8^2 \cdot 40.3 \cdot 0.3)$
 $= 0.251$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 985 / 1.02 \cdot 100000 \cdot 0.039 / (1.21 \cdot 1939938 \cdot 8^2 \cdot 40.3 \cdot 0.3)$
 $= 0.108$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 2306 / 1.01 \cdot 100000 \cdot 0.039 / (1.27 \cdot 1989688 \cdot 8^2 \cdot 40.3 \cdot 0.3)$
 $= 0.238$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 971 / 1.01 \cdot 100000 \cdot 0.039 / (1.27 \cdot 1939938 \cdot 8^2 \cdot 40.3 \cdot 0.3)$
 $= 0.102$ (should be ≤ 1)

SA-182 F304L, Min Metal Temp without impact per UHA-51: -196 °C

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	461.05	1758.92	1095.01 1758.92
Radial Flange	420.18	1172.61	997.95 1172.61



Tangential Flange	237.83	1172.61	564.86	1172.61
Maximum Average	440.61	1172.61	1046.48	1172.61
Bolting	641.45	1753.82	1178.21	1753.82

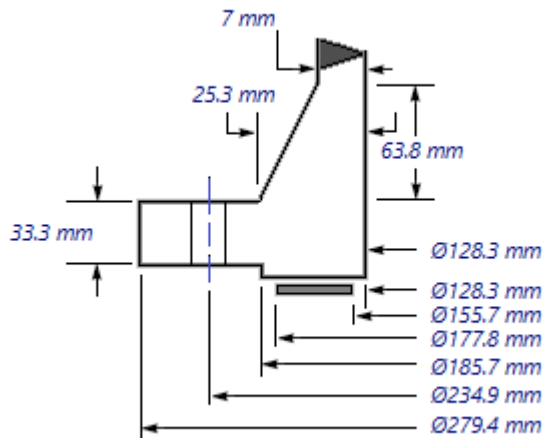
Minimum Req'd Thickness, ambient, no Corrosion: 37.313 mm

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Flange Input Data Values **Description: H3-2** :

H3-2

Flange Type	Integral Weld Neck		
Design Pressure	P	25.49	kgf/cm ²
Design Temperature		85	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	128.270	mm
Flange Outside Diameter	A	279.400	mm
Flange Thickness	t	33.2740	mm
Thickness of Hub at Small End	go	7.0000	mm
Thickness of Hub at Large End	gl	25.2880	mm
Length of Hub	h	63.7540	mm
Flange Material	SA-182 F304L		
Flange Material UNS number	S30403		
Flange Allowable Stress At Temperature	Sfo	1172.61	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1172.61	kgf/cm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	234.950	mm
Nominal Bolt Diameter	a	19.0500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	8		
Flange Face Outside Diameter	Fod	185.674	mm
Flange Face Inside Diameter	Fid	128.270	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	177.800	mm
Gasket Inside Diameter	Gi	155.702	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Hub Small End Required Thickness due to Internal Pressure:
 $= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (25.5 \cdot (128/2 + 0)) / (1173 \cdot 1 - 0.6 \cdot 25.5) + Ca$
 $= 1.4127 \text{ mm}$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1173 \cdot 1 \cdot 7) / (64.1 + 0.6 \cdot 7)$$

$$= 120.118 \text{ kgf/cm}^2$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1173 \cdot 1 \cdot 7) / (64.1 + 0.6 \cdot 7)$$

$$= 120.118 \text{ kgf/cm}^2$$

Corroded Flange ID,	Bcor = B+2*ci	128.270 mm
Corroded Large Hub,	g1Cor = g1-ci	25.288 mm
Corroded Small Hub,	g0Cor = go-ci	7.000 mm
Code R Dimension,	R = (C-B)/2 - g1	28.052 mm

Gasket Contact Width,	N = (Go - Gi)/2	11.049 mm
Basic Gasket Width,	bo = N/2	5.524 mm
Effective Gasket Width,	b = bo	5.524 mm
Gasket Reaction Diameter,	G = (Go + Gi)/2	166.751 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 167^2 \cdot 25.5$$

$$= 5567.119 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 5.52 \cdot \pi \cdot 167 \cdot 3 \cdot 25.5$$

$$= 4426.558 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 128^2 \cdot 25.5$$

$$= 3294.154 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 5567 - 3294$$

$$= 2272.965 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(5567 + 4427 + 0, 0)$$

$$= 9993.677 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 703 \cdot 5.52 \cdot \pi \cdot 167 + 0 \cdot 0$$

$$= 20347.412 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((9994 + 0 + 4 \cdot 0 / 167) / 1754, 20347 / 1754)$$

$$= 11.602 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 19 + 6 \cdot 33.3 / (3 + 0.5)$$

$$= 95.141 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 235 \cdot \sin(\pi / 8)$$

$$= 89.911 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$



$$= \max(\sqrt{(89.9 / (2 \cdot 19 + 33.3))}, 1)$$

$$= 1.1224$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	11.602	15.587	
Radial Distance between Hub and Bolts:	28.575	28.052	
Radial Distance between Bolts and Edge:	20.637	22.225	
Circ. Spacing between the Bolts:	44.450	89.911	95.141

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi(G_o + G_i))$$

$$= 15.6 \cdot 1754 / (703 \cdot \pi(178 + 156))$$

$$= 3.711 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 1754 \cdot 15.6$$

$$= 27336.86 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 9994 - 5567$$

$$= 4426.56 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (235 - 167) / 2$$

$$= 34.0995 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((235 - 128) / 2 + 34.1)$$

$$= 43.7197 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g_1) / 2$$

$$= (235 - 128 - 25.3) / 2$$

$$= 40.6960 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	3294.	40.6960	1.1224	150.
Face Pressure, Mt	2273.	43.7197	1.1224	112.
Gasket Load, Mg	4427.	34.0995	1.1224	169.
Gasket Seating, Matm	27337.	34.0995	1.1224	1046.

Total Moment for Operation, Mo 431. kgf-m.
 Total Moment for Gasket seating, Mg 1046. kgf-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 29.965 mm
 Hub Ratio, $h/h_o = H_L / H_o$ 2.128
 Thickness Ratio, $g_1/g_o = (g_{1Cor}/g_{oCor})$ 3.613

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.545
 Factor V per 4.16.5 0.035
 Factor f per 4.16.5 1.000

Warning: The hub ratio Xh was greater than 2 and has been set to 2.

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 279/128$$

$$= 2.178$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(2.18^2 \cdot \log_{10}(2.18))) / (2.18^2 - 1)) / (2.18 - 1)$$

$$= 2.646$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (2.18^2(1 + 8.55246 \cdot \log_{10}(2.18)) - 1) / ((1.0472 + 1.9448 \cdot 2.18^2) \cdot (2.18 - 1))$$

$$= 1.443$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (2.18^2(1 + 8.55246 \cdot \log_{10}(2.18)) - 1) / (1.36136(2.18^2 - 1) \cdot (2.18 - 1))$$

$$= 2.908$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (2.18^2 + 1) / (2.18^2 - 1)$$

$$= 1.534$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 2.91 \cdot 7^2 \cdot 30 / 0.035$$

$$= 122415.383 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.54/30$$

$$= 0.018 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (33.3 \cdot 0.018 + 1) / 1.44 + 33.3^3/122415$$

$$= 1.413$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 431/128) / (1.41 \cdot 25.3^2)$$

$$= 372.11 \text{ kgf/cm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 1046/128) / (1.41 \cdot 25.3^2)$$

$$= 902.43 \text{ kgf/cm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.81 \cdot 431/128) / (1.41 \cdot 33.3^2)$$

$$= 388.27 \text{ kgf/cm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.81 \cdot 1046/128) / (1.41 \cdot 33.3^2)$$

$$= 941.62 \text{ kgf/cm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (2.65 \cdot 431 / (33.3^2 \cdot 128)) - 1.53 \cdot 388$$

$$= 208.14 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (2.65 \cdot 1046 / (33.3^2 \cdot 128)) - 1.53 \cdot 942$$

$$= 504.78 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (372 + \max(388, 208)) / 2$$

$$= 380.19 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (902 + \max(942, 505)) / 2$$

$$= 922.03 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 9994 / 15.6$$

$$= 641.15 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (20347 / 15.6)$$

$$= 1305.41 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	372.11	1758.92	902.43	1758.92
Radial Flange	388.27	1172.61	941.62	1172.61
Tangential Flange	208.14	1172.61	504.78	1172.61
Maximum Average	380.19	1172.61	922.03	1172.61
Bolting	641.15	1753.82	1305.41	1753.82

Minimum Required Flange Thickness 30.658 mm

Estimated M.A.W.P. (Operating) 69.7 kgf/cm²

Estimated Finished Weight of Flange at given Thk. 16.7 kg

Estimated Unfinished Weight of Forging at given Thk 37.7 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 1066 / 1.14 \cdot 100000 \cdot 0.035 / (1.31 \cdot 1989688 \cdot 7^2 \cdot 30 \cdot 0.3)$$

$$= 0.147 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 440 / 1.14 \cdot 100000 \cdot 0.035 / (1.31 \cdot 1939938 \cdot 7^2 \cdot 30 \cdot 0.3)$$

$$= 0.062 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 1046 / 1.12 \cdot 100000 \cdot 0.035 / (1.41 \cdot 1989688 \cdot 7^2 \cdot 30 \cdot 0.3)$$

$$= 0.136 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 431 / 1.12 \cdot 100000 \cdot 0.035 / (1.41 \cdot 1939938 \cdot 7^2 \cdot 30 \cdot 0.3)$$

$$= 0.057 \text{ (should be } \leq 1)$$

SA-182 F304L, Min Metal Temp without impact per UHA-51: -196 °C

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
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Longitudinal Hub	372.11	1758.92	902.43	1758.92
Radial Flange	388.27	1172.61	941.62	1172.61
Tangential Flange	208.14	1172.61	504.78	1172.61
Maximum Average	380.19	1172.61	922.03	1172.61
Bolting	641.15	1753.82	1305.41	1753.82

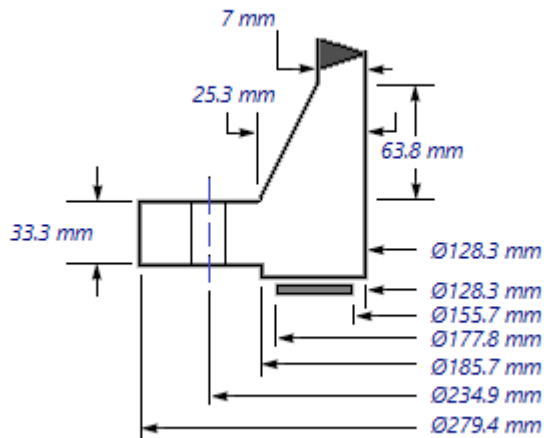
Minimum Req'd Thickness, ambient, no Corrosion: 30.658 mm

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Flange Input Data Values Description: H3-1 :

H3-1

Flange Type	Integral Weld Neck		
Design Pressure	P	25.49	kgf/cm ²
Design Temperature		85	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	128.270	mm
Flange Outside Diameter	A	279.400	mm
Flange Thickness	t	33.2740	mm
Thickness of Hub at Small End	go	7.0000	mm
Thickness of Hub at Large End	gl	25.2880	mm
Length of Hub	h	63.7540	mm
Flange Material	SA-182 F304L		
Flange Material UNS number	S30403		
Flange Allowable Stress At Temperature	Sfo	1172.61	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1172.61	kgf/cm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	234.950	mm
Nominal Bolt Diameter	a	19.0500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	8		
Flange Face Outside Diameter	Fod	185.674	mm
Flange Face Inside Diameter	Fid	128.270	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	177.800	mm
Gasket Inside Diameter	Gi	155.702	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (25.5 \cdot (128/2 + 0)) / (1173 \cdot 1 - 0.6 \cdot 25.5) + Ca$$

$$= 1.4127 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1173 \cdot 1 \cdot 7) / (64.1 + 0.6 \cdot 7)$$

$$= 120.118 \text{ kgf/cm}^2$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1173 \cdot 1 \cdot 7) / (64.1 + 0.6 \cdot 7)$$

$$= 120.118 \text{ kgf/cm}^2$$

Corroded Flange ID,	Bcor = B+2*ci	128.270 mm
Corroded Large Hub,	g1Cor = g1-ci	25.288 mm
Corroded Small Hub,	g0Cor = go-ci	7.000 mm
Code R Dimension,	R = (C-B)/2 - g1	28.052 mm

Gasket Contact Width,	N = (Go - Gi)/2	11.049 mm
Basic Gasket Width,	bo = N/2	5.524 mm
Effective Gasket Width,	b = bo	5.524 mm
Gasket Reaction Diameter,	G = (Go + Gi)/2	166.751 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 167^2 \cdot 25.5$$

$$= 5567.119 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 5.52 \cdot \pi \cdot 167 \cdot 3 \cdot 25.5$$

$$= 4426.558 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 128^2 \cdot 25.5$$

$$= 3294.154 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 5567 - 3294$$

$$= 2272.965 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(5567 + 4427 + 0, 0)$$

$$= 9993.677 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 703 \cdot 5.52 \cdot \pi \cdot 167 + 0 \cdot 0$$

$$= 20347.412 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((9994 + 0 + 4 \cdot 0 / 167) / 1754, 20347 / 1754)$$

$$= 11.602 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 19 + 6 \cdot 33.3 / (3 + 0.5)$$

$$= 95.141 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 235 \cdot \sin(\pi / 8)$$

$$= 89.911 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$



$$= \max(\sqrt{(89.9/(2 \cdot 19 + 33.3))}, 1)$$

$$= 1.1224$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	11.602	15.587	
Radial Distance between Hub and Bolts:	28.575	28.052	
Radial Distance between Bolts and Edge:	20.637	22.225	
Circ. Spacing between the Bolts:	44.450	89.911	95.141

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 15.6 \cdot 1754 / (703 \cdot \pi(178 + 156))$$

$$= 3.711 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 1754 \cdot 15.6$$

$$= 27336.86 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 9994 - 5567$$

$$= 4426.56 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (235 - 167) / 2$$

$$= 34.0995 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((235 - 128) / 2 + 34.1)$$

$$= 43.7197 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (235 - 128 - 25.3) / 2$$

$$= 40.6960 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	3294.	40.6960	1.1224	150.
Face Pressure, Mt	2273.	43.7197	1.1224	112.
Gasket Load, Mg	4427.	34.0995	1.1224	169.
Gasket Seating, Matm	27337.	34.0995	1.1224	1046.

Total Moment for Operation, Mo 431. kgf-m.
 Total Moment for Gasket seating, Mg 1046. kgf-m.

Effective Hub Length, $h_o = \sqrt{Bcor \cdot goCor}$ 29.965 mm
 Hub Ratio, $h/h_o = HL / H_o$ 2.128
 Thickness Ratio, $g1/g_o = (g1Cor/goCor)$ 3.613

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.545
 Factor V per 4.16.5 0.035
 Factor f per 4.16.5 1.000

Warning: The hub ratio Xh was greater than 2 and has been set to 2.

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 279/128$$

$$= 2.178$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(2.18^2 \cdot \log_{10}(2.18))) / (2.18^2 - 1)) / (2.18 - 1)$$

$$= 2.646$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (2.18^2(1 + 8.55246 \cdot \log_{10}(2.18)) - 1) / ((1.0472 + 1.9448 \cdot 2.18^2) \cdot (2.18 - 1))$$

$$= 1.443$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (2.18^2(1 + 8.55246 \cdot \log_{10}(2.18)) - 1) / (1.36136(2.18^2 - 1) \cdot (2.18 - 1))$$

$$= 2.908$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (2.18^2 + 1) / (2.18^2 - 1)$$

$$= 1.534$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 2.91 \cdot 7^2 \cdot 30 / 0.035$$

$$= 122415.383 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.54/30$$

$$= 0.018 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (33.3 \cdot 0.018 + 1) / 1.44 + 33.3^3/122415$$

$$= 1.413$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 431/128) / (1.41 \cdot 25.3^2)$$

$$= 372.11 \text{ kgf/cm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 1046/128) / (1.41 \cdot 25.3^2)$$

$$= 902.43 \text{ kgf/cm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.81 \cdot 431/128) / (1.41 \cdot 33.3^2)$$

$$= 388.27 \text{ kgf/cm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.81 \cdot 1046/128) / (1.41 \cdot 33.3^2)$$

$$= 941.62 \text{ kgf/cm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (2.65 \cdot 431 / (33.3^2 \cdot 128)) - 1.53 \cdot 388$$

$$= 208.14 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (2.65 \cdot 1046 / (33.3^2 \cdot 128)) - 1.53 \cdot 942$$

$$= 504.78 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (372 + \max(388, 208)) / 2$$

$$= 380.19 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (902 + \max(942, 505)) / 2$$

$$= 922.03 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 9994 / 15.6$$

$$= 641.15 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (20347 / 15.6)$$

$$= 1305.41 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	372.11	1758.92	902.43	1758.92
Radial Flange	388.27	1172.61	941.62	1172.61
Tangential Flange	208.14	1172.61	504.78	1172.61
Maximum Average	380.19	1172.61	922.03	1172.61
Bolting	641.15	1753.82	1305.41	1753.82

Minimum Required Flange Thickness 30.658 mm

Estimated M.A.W.P. (Operating) 69.7 kgf/cm²

Estimated Finished Weight of Flange at given Thk. 16.7 kg

Estimated Unfinished Weight of Forging at given Thk 37.7 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 1066 / 1.14 \cdot 100000 \cdot 0.035 / (1.31 \cdot 1989688 \cdot 7^2 \cdot 30 \cdot 0.3)$$

$$= 0.147 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 440 / 1.14 \cdot 100000 \cdot 0.035 / (1.31 \cdot 1939938 \cdot 7^2 \cdot 30 \cdot 0.3)$$

$$= 0.062 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 1046 / 1.12 \cdot 100000 \cdot 0.035 / (1.41 \cdot 1989688 \cdot 7^2 \cdot 30 \cdot 0.3)$$

$$= 0.136 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 431 / 1.12 \cdot 100000 \cdot 0.035 / (1.41 \cdot 1939938 \cdot 7^2 \cdot 30 \cdot 0.3)$$

$$= 0.057 \text{ (should be } \leq 1)$$

SA-182 F304L, Min Metal Temp without impact per UHA-51: -196 °C

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
--	--------	-------------------	-----------------------	---------



Longitudinal Hub	372.11	1758.92	902.43	1758.92
Radial Flange	388.27	1172.61	941.62	1172.61
Tangential Flange	208.14	1172.61	504.78	1172.61
Maximum Average	380.19	1172.61	922.03	1172.61
Bolting	641.15	1753.82	1305.41	1753.82

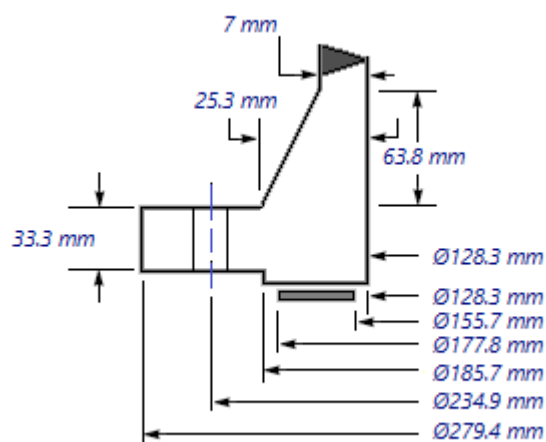
Minimum Req'd Thickness, ambient, no Corrosion: 30.658 mm

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Flange Input Data Values **Description: H2-2** :

H2-2

Flange Type	Integral Weld Neck		
Design Pressure	P	25.49	kgf/cm ²
Design Temperature		85	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	128.270	mm
Flange Outside Diameter	A	279.400	mm
Flange Thickness	t	33.2740	mm
Thickness of Hub at Small End	go	7.0000	mm
Thickness of Hub at Large End	gl	25.2880	mm
Length of Hub	h	63.7540	mm
Flange Material	SA-182 F304L		
Flange Material UNS number		S30403	
Flange Allowable Stress At Temperature	Sfo	1172.61	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1172.61	kgf/cm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	234.950	mm
Nominal Bolt Diameter	a	19.0500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		8	
Flange Face Outside Diameter	Fod	185.674	mm
Flange Face Inside Diameter	Fid	128.270	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	177.800	mm
Gasket Inside Diameter	Gi	155.702	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (25.5 \cdot (128/2 + 0)) / (1173 \cdot 1 - 0.6 \cdot 25.5) + Ca$$

$$= 1.4127 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1173 \cdot 1 \cdot 7) / (64.1 + 0.6 \cdot 7)$$

$$= 120.118 \text{ kgf/cm}^2$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1173 \cdot 1 \cdot 7) / (64.1 + 0.6 \cdot 7)$$

$$= 120.118 \text{ kgf/cm}^2$$

Corroded Flange ID,	Bcor = B+2*ci	128.270 mm
Corroded Large Hub,	g1Cor = g1-ci	25.288 mm
Corroded Small Hub,	g0Cor = go-ci	7.000 mm
Code R Dimension,	R = (C-B)/2 - g1	28.052 mm

Gasket Contact Width,	N = (Go - Gi)/2	11.049 mm
Basic Gasket Width,	bo = N/2	5.524 mm
Effective Gasket Width,	b = bo	5.524 mm
Gasket Reaction Diameter,	G = (Go + Gi)/2	166.751 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 167^2 \cdot 25.5$$

$$= 5567.119 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 5.52 \cdot \pi \cdot 167 \cdot 3 \cdot 25.5$$

$$= 4426.558 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 128^2 \cdot 25.5$$

$$= 3294.154 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 5567 - 3294$$

$$= 2272.965 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(5567 + 4427 + 0, 0)$$

$$= 9993.677 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 703 \cdot 5.52 \cdot \pi \cdot 167 + 0 \cdot 0$$

$$= 20347.412 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((9994 + 0 + 4 \cdot 0 / 167) / 1754, 20347 / 1754)$$

$$= 11.602 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 19 + 6 \cdot 33.3 / (3 + 0.5)$$

$$= 95.141 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 235 \cdot \sin(\pi / 8)$$

$$= 89.911 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{(89.9 / (2 \cdot 19 + 33.3))}, 1)$$

$$= 1.1224$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	11.602	15.587	
Radial Distance between Hub and Bolts:	28.575	28.052	
Radial Distance between Bolts and Edge:	20.637	22.225	
Circ. Spacing between the Bolts:	44.450	89.911	95.141

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi(G_o + G_i))$$

$$= 15.6 \cdot 1754 / (703 \cdot \pi(178 + 156))$$

$$= 3.711 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 1754 \cdot 15.6$$

$$= 27336.86 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 9994 - 5567$$

$$= 4426.56 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (235 - 167) / 2$$

$$= 34.0995 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + h_g)$$

$$= 0.5((235 - 128) / 2 + 34.1)$$

$$= 43.7197 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g_1) / 2$$

$$= (235 - 128 - 25.3) / 2$$

$$= 40.6960 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	3294.	40.6960	1.1224	150.
Face Pressure, Mt	2273.	43.7197	1.1224	112.
Gasket Load, Mg	4427.	34.0995	1.1224	169.
Gasket Seating, Matm	27337.	34.0995	1.1224	1046.

Total Moment for Operation, Mo 431. kgf-m.
 Total Moment for Gasket seating, Mg 1046. kgf-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 29.965 mm
 Hub Ratio, $h/h_o = H_L / H_o$ 2.128
 Thickness Ratio, $g_1/g_o = (g_{1Cor}/g_{oCor})$ 3.613

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.545
 Factor V per 4.16.5 0.035
 Factor f per 4.16.5 1.000

Warning: The hub ratio Xh was greater than 2 and has been set to 2.

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 279/128$$

$$= 2.178$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(2.18^2 \cdot \log_{10}(2.18))) / (2.18^2 - 1)) / (2.18 - 1)$$

$$= 2.646$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (2.18^2(1 + 8.55246 \cdot \log_{10}(2.18)) - 1) / ((1.0472 + 1.9448 \cdot 2.18^2) \cdot (2.18 - 1))$$

$$= 1.443$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (2.18^2(1 + 8.55246 \cdot \log_{10}(2.18)) - 1) / (1.36136(2.18^2 - 1) \cdot (2.18 - 1))$$

$$= 2.908$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (2.18^2 + 1) / (2.18^2 - 1)$$

$$= 1.534$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 2.91 \cdot 7^2 \cdot 30 / 0.035$$

$$= 122415.383 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.54/30$$

$$= 0.018 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (33.3 \cdot 0.018 + 1) / 1.44 + 33.3^3/122415$$

$$= 1.413$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 431/128) / (1.41 \cdot 25.3^2)$$

$$= 372.11 \text{ kgf/cm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 1046/128) / (1.41 \cdot 25.3^2)$$

$$= 902.43 \text{ kgf/cm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.81 \cdot 431/128) / (1.41 \cdot 33.3^2)$$

$$= 388.27 \text{ kgf/cm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.81 \cdot 1046/128) / (1.41 \cdot 33.3^2)$$

$$= 941.62 \text{ kgf/cm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (2.65 \cdot 431 / (33.3^2 \cdot 128)) - 1.53 \cdot 388$$

$$= 208.14 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (2.65 \cdot 1046 / (33.3^2 \cdot 128)) - 1.53 \cdot 942$$

$$= 504.78 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (372 + \max(388, 208)) / 2$$

$$= 380.19 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (902 + \max(942, 505)) / 2$$

$$= 922.03 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 9994 / 15.6$$

$$= 641.15 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (20347 / 15.6)$$

$$= 1305.41 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	372.11	1758.92	902.43	1758.92
Radial Flange	388.27	1172.61	941.62	1172.61
Tangential Flange	208.14	1172.61	504.78	1172.61
Maximum Average	380.19	1172.61	922.03	1172.61
Bolting	641.15	1753.82	1305.41	1753.82

Minimum Required Flange Thickness 30.658 mm

Estimated M.A.W.P. (Operating) 69.7 kgf/cm²

Estimated Finished Weight of Flange at given Thk. 16.7 kg

Estimated Unfinished Weight of Forging at given Thk 37.7 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 1066 / 1.14 \cdot 100000 \cdot 0.035 / (1.31 \cdot 1989688 \cdot 7^2 \cdot 30 \cdot 0.3)$$

$$= 0.147 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 440 / 1.14 \cdot 100000 \cdot 0.035 / (1.31 \cdot 1939938 \cdot 7^2 \cdot 30 \cdot 0.3)$$

$$= 0.062 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 1046 / 1.12 \cdot 100000 \cdot 0.035 / (1.41 \cdot 1989688 \cdot 7^2 \cdot 30 \cdot 0.3)$$

$$= 0.136 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 431 / 1.12 \cdot 100000 \cdot 0.035 / (1.41 \cdot 1939938 \cdot 7^2 \cdot 30 \cdot 0.3)$$

$$= 0.057 \text{ (should be } \leq 1)$$

SA-182 F304L, Min Metal Temp without impact per UHA-51: -196 °C

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
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Longitudinal Hub	372.11	1758.92	902.43	1758.92
Radial Flange	388.27	1172.61	941.62	1172.61
Tangential Flange	208.14	1172.61	504.78	1172.61
Maximum Average	380.19	1172.61	922.03	1172.61
Bolting	641.15	1753.82	1305.41	1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 30.658 mm

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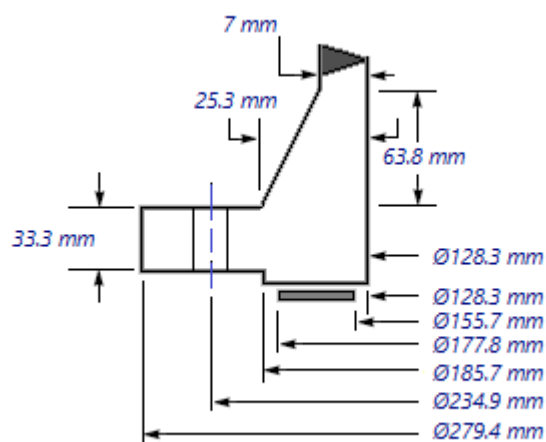


Flange Input Data Values

Description: H2-1 :

H2-1

Flange Type	Integral Weld Neck		
Design Pressure	P	25.49	kgf/cm ²
Design Temperature		85	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	128.270	mm
Flange Outside Diameter	A	279.400	mm
Flange Thickness	t	33.2740	mm
Thickness of Hub at Small End	go	7.0000	mm
Thickness of Hub at Large End	gl	25.2880	mm
Length of Hub	h	63.7540	mm
Flange Material	SA-182 F304L		
Flange Material UNS number		S30403	
Flange Allowable Stress At Temperature	Sfo	1172.61	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1172.61	kgf/cm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	234.950	mm
Nominal Bolt Diameter	a	19.0500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		8	
Flange Face Outside Diameter	Fod	185.674	mm
Flange Face Inside Diameter	Fid	128.270	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	177.800	mm
Gasket Inside Diameter	Gi	155.702	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (25.5 \cdot (128/2 + 0)) / (1173 \cdot 1 - 0.6 \cdot 25.5) + Ca$$

$$= 1.4127 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1173 \cdot 1 \cdot 7) / (64.1 + 0.6 \cdot 7)$$

$$= 120.118 \text{ kgf/cm}^2$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1173 \cdot 1 \cdot 7) / (64.1 + 0.6 \cdot 7)$$

$$= 120.118 \text{ kgf/cm}^2$$

Corroded Flange ID,	Bcor = B+2*ci	128.270 mm
Corroded Large Hub,	g1Cor = g1-ci	25.288 mm
Corroded Small Hub,	g0Cor = go-ci	7.000 mm
Code R Dimension,	R = (C-B)/2 - g1	28.052 mm

Gasket Contact Width,	N = (Go - Gi)/2	11.049 mm
Basic Gasket Width,	bo = N/2	5.524 mm
Effective Gasket Width,	b = bo	5.524 mm
Gasket Reaction Diameter,	G = (Go + Gi)/2	166.751 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 167^2 \cdot 25.5$$

$$= 5567.119 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 5.52 \cdot \pi \cdot 167 \cdot 3 \cdot 25.5$$

$$= 4426.558 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 128^2 \cdot 25.5$$

$$= 3294.154 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 5567 - 3294$$

$$= 2272.965 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(5567 + 4427 + 0, 0)$$

$$= 9993.677 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 703 \cdot 5.52 \cdot \pi \cdot 167 + 0 \cdot 0 \cdot 0$$

$$= 20347.412 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((9994 + 0 + 4 \cdot 0 / 167) / 1754, 20347 / 1754)$$

$$= 11.602 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 19 + 6 \cdot 33.3 / (3 + 0.5)$$

$$= 95.141 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 235 \cdot \sin(\pi / 8)$$

$$= 89.911 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$



$$= \max(\sqrt{(89.9/(2 \cdot 19 + 33.3))}, 1)$$

$$= 1.1224$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	11.602	15.587	
Radial Distance between Hub and Bolts:	28.575	28.052	
Radial Distance between Bolts and Edge:	20.637	22.225	
Circ. Spacing between the Bolts:	44.450	89.911	95.141

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 15.6 \cdot 1754 / (703 \cdot \pi(178 + 156))$$

$$= 3.711 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 1754 \cdot 15.6$$

$$= 27336.86 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 9994 - 5567$$

$$= 4426.56 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (235 - 167) / 2$$

$$= 34.0995 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((235 - 128) / 2 + 34.1)$$

$$= 43.7197 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (235 - 128 - 25.3) / 2$$

$$= 40.6960 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	3294.	40.6960	1.1224	150.
Face Pressure, Mt	2273.	43.7197	1.1224	112.
Gasket Load, Mg	4427.	34.0995	1.1224	169.
Gasket Seating, Matm	27337.	34.0995	1.1224	1046.

Total Moment for Operation, Mo 431. kgf-m.
 Total Moment for Gasket seating, Mg 1046. kgf-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 29.965 mm
 Hub Ratio, $h/h_o = HL / H_o$ 2.128
 Thickness Ratio, $g1/g_o = (g1_{cor}/g_{oCor})$ 3.613

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.545
 Factor V per 4.16.5 0.035
 Factor f per 4.16.5 1.000

Warning: The hub ratio Xh was greater than 2 and has been set to 2.

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 279/128$$

$$= 2.178$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(2.18^2 \cdot \log_{10}(2.18))) / (2.18^2 - 1)) / (2.18 - 1)$$

$$= 2.646$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (2.18^2(1 + 8.55246 \cdot \log_{10}(2.18)) - 1) / ((1.0472 + 1.9448 \cdot 2.18^2) \cdot (2.18 - 1))$$

$$= 1.443$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (2.18^2(1 + 8.55246 \cdot \log_{10}(2.18)) - 1) / (1.36136(2.18^2 - 1) \cdot (2.18 - 1))$$

$$= 2.908$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (2.18^2 + 1) / (2.18^2 - 1)$$

$$= 1.534$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 2.91 \cdot 7^2 \cdot 30 / 0.035$$

$$= 122415.383 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.54/30$$

$$= 0.018 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (33.3 \cdot 0.018 + 1) / 1.44 + 33.3^3/122415$$

$$= 1.413$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 431/128) / (1.41 \cdot 25.3^2)$$

$$= 372.11 \text{ kgf/cm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 1046/128) / (1.41 \cdot 25.3^2)$$

$$= 902.43 \text{ kgf/cm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.81 \cdot 431/128) / (1.41 \cdot 33.3^2)$$

$$= 388.27 \text{ kgf/cm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.81 \cdot 1046/128) / (1.41 \cdot 33.3^2)$$

$$= 941.62 \text{ kgf/cm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (2.65 \cdot 431 / (33.3^2 \cdot 128)) - 1.53 \cdot 388$$

$$= 208.14 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (2.65 \cdot 1046 / (33.3^2 \cdot 128)) - 1.53 \cdot 942$$

$$= 504.78 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:
 $= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$
 $= (372 + \max(388, 208)) / 2$
 $= 380.19 \text{ kgf/cm}^2$

Average Flange Stress, Seating [SAa]:
 $= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$
 $= (902 + \max(942, 505)) / 2$
 $= 922.03 \text{ kgf/cm}^2$

Bolt Stress, Operating [BSo]:
 $= W_{m1} / A_b$
 $= 9994 / 15.6$
 $= 641.15 \text{ kgf/cm}^2$

Bolt Stress, Seating [BSa]:
 $= (W_{m2} / A_b)$
 $= (20347 / 15.6)$
 $= 1305.41 \text{ kgf/cm}^2$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	372.11	1758.92	902.43	1758.92
Radial Flange	388.27	1172.61	941.62	1172.61
Tangential Flange	208.14	1172.61	504.78	1172.61
Maximum Average	380.19	1172.61	922.03	1172.61
Bolting	641.15	1753.82	1305.41	1753.82

Minimum Required Flange Thickness 30.658 mm

Estimated M.A.W.P. (Operating) 69.7 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 16.7 kg
 Estimated Unfinished Weight of Forging at given Thk 37.7 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 1066 / 1.14 \cdot 100000 \cdot 0.035 / (1.31 \cdot 1989688 \cdot 7^2 \cdot 30 \cdot 0.3)$
 $= 0.147 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 440 / 1.14 \cdot 100000 \cdot 0.035 / (1.31 \cdot 1939938 \cdot 7^2 \cdot 30 \cdot 0.3)$
 $= 0.062 \text{ (should be } \leq 1)$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 1046 / 1.12 \cdot 100000 \cdot 0.035 / (1.41 \cdot 1989688 \cdot 7^2 \cdot 30 \cdot 0.3)$
 $= 0.136 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 431 / 1.12 \cdot 100000 \cdot 0.035 / (1.41 \cdot 1939938 \cdot 7^2 \cdot 30 \cdot 0.3)$
 $= 0.057 \text{ (should be } \leq 1)$

SA-182 F304L, Min Metal Temp without impact per UHA-51: -196 °C

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
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Longitudinal Hub	372.11	1758.92	902.43	1758.92
Radial Flange	388.27	1172.61	941.62	1172.61
Tangential Flange	208.14	1172.61	504.78	1172.61
Maximum Average	380.19	1172.61	922.03	1172.61
Bolting	641.15	1753.82	1305.41	1753.82

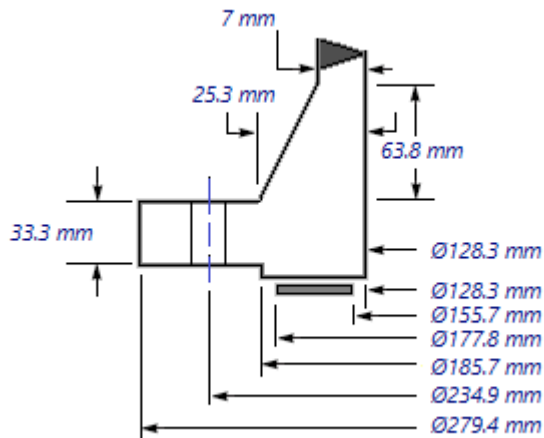
Minimum Req'd Thickness, ambient, no Corrosion: 30.658 mm

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Flange Input Data Values Description: H1-2 :

H1-2

Flange Type	Integral Weld Neck		
Design Pressure	P	25.49	kgf/cm ²
Design Temperature		85	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	128.270	mm
Flange Outside Diameter	A	279.400	mm
Flange Thickness	t	33.2740	mm
Thickness of Hub at Small End	go	7.0000	mm
Thickness of Hub at Large End	gl	25.2880	mm
Length of Hub	h	63.7540	mm
Flange Material	SA-182 F304L		
Flange Material UNS number		S30403	
Flange Allowable Stress At Temperature	Sfo	1172.61	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1172.61	kgf/cm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	234.950	mm
Nominal Bolt Diameter	a	19.0500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		8	
Flange Face Outside Diameter	Fod	185.674	mm
Flange Face Inside Diameter	Fid	128.270	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	177.800	mm
Gasket Inside Diameter	Gi	155.702	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Hub Small End Required Thickness due to Internal Pressure:
 $= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (25.5 \cdot (128/2 + 0)) / (1173 \cdot 1 - 0.6 \cdot 25.5) + Ca$
 $= 1.4127 \text{ mm}$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1173 \cdot 1 \cdot 7) / (64.1 + 0.6 \cdot 7)$$

$$= 120.118 \text{ kgf/cm}^2$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1173 \cdot 1 \cdot 7) / (64.1 + 0.6 \cdot 7)$$

$$= 120.118 \text{ kgf/cm}^2$$

Corroded Flange ID,	Bcor = B+2*ci	128.270 mm
Corroded Large Hub,	g1Cor = g1-ci	25.288 mm
Corroded Small Hub,	g0Cor = go-ci	7.000 mm
Code R Dimension,	R = (C-B)/2 - g1	28.052 mm

Gasket Contact Width,	N = (Go - Gi)/2	11.049 mm
Basic Gasket Width,	bo = N/2	5.524 mm
Effective Gasket Width,	b = bo	5.524 mm
Gasket Reaction Diameter,	G = (Go + Gi)/2	166.751 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 167^2 \cdot 25.5$$

$$= 5567.119 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 5.52 \cdot \pi \cdot 167 \cdot 3 \cdot 25.5$$

$$= 4426.558 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 128^2 \cdot 25.5$$

$$= 3294.154 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 5567 - 3294$$

$$= 2272.965 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(5567 + 4427 + 0, 0)$$

$$= 9993.677 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 703 \cdot 5.52 \cdot \pi \cdot 167 + 0 \cdot 0$$

$$= 20347.412 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((9994 + 0 + 4 \cdot 0 / 167) / 1754, 20347 / 1754)$$

$$= 11.602 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 19 + 6 \cdot 33.3 / (3 + 0.5)$$

$$= 95.141 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 235 \cdot \sin(\pi / 8)$$

$$= 89.911 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{(89.9 / (2 \cdot 19 + 33.3))}, 1)$$

$$= 1.1224$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	11.602	15.587	
Radial Distance between Hub and Bolts:	28.575	28.052	
Radial Distance between Bolts and Edge:	20.637	22.225	
Circ. Spacing between the Bolts:	44.450	89.911	95.141

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi(G_o + G_i))$$

$$= 15.6 \cdot 1754 / (703 \cdot \pi(178 + 156))$$

$$= 3.711 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 1754 \cdot 15.6$$

$$= 27336.86 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 9994 - 5567$$

$$= 4426.56 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (235 - 167) / 2$$

$$= 34.0995 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + h_g)$$

$$= 0.5((235 - 128) / 2 + 34.1)$$

$$= 43.7197 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g_1) / 2$$

$$= (235 - 128 - 25.3) / 2$$

$$= 40.6960 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	3294.	40.6960	1.1224	150.
Face Pressure, Mt	2273.	43.7197	1.1224	112.
Gasket Load, Mg	4427.	34.0995	1.1224	169.
Gasket Seating, Matm	27337.	34.0995	1.1224	1046.

Total Moment for Operation, Mo 431. kgf-m.
 Total Moment for Gasket seating, Mg 1046. kgf-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_o \cdot C_{cor}}$ 29.965 mm
 Hub Ratio, $h/h_o = H_L / H_o$ 2.128
 Thickness Ratio, $g_1/g_o = (g_1 \cdot C_{cor} / g_o \cdot C_{cor})$ 3.613

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.545
 Factor V per 4.16.5 0.035
 Factor f per 4.16.5 1.000

Warning: The hub ratio Xh was greater than 2 and has been set to 2.

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 279/128$$

$$= 2.178$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(2.18^2 \cdot \log_{10}(2.18))) / (2.18^2 - 1)) / (2.18 - 1)$$

$$= 2.646$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (2.18^2(1 + 8.55246 \cdot \log_{10}(2.18)) - 1) / ((1.0472 + 1.9448 \cdot 2.18^2) \cdot (2.18 - 1))$$

$$= 1.443$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (2.18^2(1 + 8.55246 \cdot \log_{10}(2.18)) - 1) / (1.36136(2.18^2 - 1) \cdot (2.18 - 1))$$

$$= 2.908$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (2.18^2 + 1) / (2.18^2 - 1)$$

$$= 1.534$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 2.91 \cdot 7^2 \cdot 30 / 0.035$$

$$= 122415.383 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.54/30$$

$$= 0.018 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (33.3 \cdot 0.018 + 1) / 1.44 + 33.3^3/122415$$

$$= 1.413$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 431/128) / (1.41 \cdot 25.3^2)$$

$$= 372.11 \text{ kgf/cm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 1046/128) / (1.41 \cdot 25.3^2)$$

$$= 902.43 \text{ kgf/cm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.81 \cdot 431/128) / (1.41 \cdot 33.3^2)$$

$$= 388.27 \text{ kgf/cm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.81 \cdot 1046/128) / (1.41 \cdot 33.3^2)$$

$$= 941.62 \text{ kgf/cm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (2.65 \cdot 431 / (33.3^2 \cdot 128)) - 1.53 \cdot 388$$

$$= 208.14 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (2.65 \cdot 1046 / (33.3^2 \cdot 128)) - 1.53 \cdot 942$$

$$= 504.78 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:
 $= (SHo + \max(SRo, STo)) / 2$
 $= (372 + \max(388, 208)) / 2$
 $= 380.19 \text{ kgf/cm}^2$

Average Flange Stress, Seating [SAa]:
 $= (SHa + \max(SRa, STa)) / 2$
 $= (902 + \max(942, 505)) / 2$
 $= 922.03 \text{ kgf/cm}^2$

Bolt Stress, Operating [BSo]:
 $= Wm1 / Ab$
 $= 9994 / 15.6$
 $= 641.15 \text{ kgf/cm}^2$

Bolt Stress, Seating [BSa]:
 $= (Wm2 / Ab)$
 $= (20347 / 15.6)$
 $= 1305.41 \text{ kgf/cm}^2$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	372.11	1758.92	902.43	1758.92
Radial Flange	388.27	1172.61	941.62	1172.61
Tangential Flange	208.14	1172.61	504.78	1172.61
Maximum Average	380.19	1172.61	922.03	1172.61
Bolting	641.15	1753.82	1305.41	1753.82

Minimum Required Flange Thickness 30.658 mm

Estimated M.A.W.P. (Operating) 69.7 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 16.7 kg
 Estimated Unfinished Weight of Forging at given Thk 37.7 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 1066 / 1.14 \cdot 100000 \cdot 0.035 / (1.31 \cdot 1989688 \cdot 7^2 \cdot 30 \cdot 0.3)$
 $= 0.147 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 440 / 1.14 \cdot 100000 \cdot 0.035 / (1.31 \cdot 1939938 \cdot 7^2 \cdot 30 \cdot 0.3)$
 $= 0.062 \text{ (should be } \leq 1)$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 1046 / 1.12 \cdot 100000 \cdot 0.035 / (1.41 \cdot 1989688 \cdot 7^2 \cdot 30 \cdot 0.3)$
 $= 0.136 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 431 / 1.12 \cdot 100000 \cdot 0.035 / (1.41 \cdot 1939938 \cdot 7^2 \cdot 30 \cdot 0.3)$
 $= 0.057 \text{ (should be } \leq 1)$

SA-182 F304L, Min Metal Temp without impact per UHA-51: -196 °C

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
--	--------	-------------------	-----------------------	---------



Longitudinal Hub	372.11	1758.92	902.43	1758.92
Radial Flange	388.27	1172.61	941.62	1172.61
Tangential Flange	208.14	1172.61	504.78	1172.61
Maximum Average	380.19	1172.61	922.03	1172.61
Bolting	641.15	1753.82	1305.41	1753.82

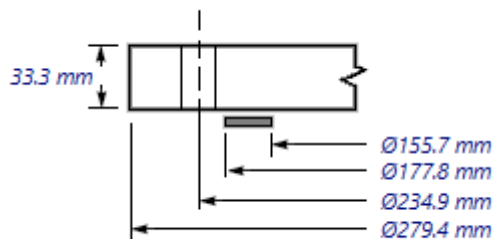
Minimum Req'd Thickness, ambient, no Corrosion: 30.658 mm

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Flange Input Data Values Description: H1-1 :

H1-1

Flange Type		Blind	
Design Pressure	P	25.49	kgf/cm ²
Design Temperature		85	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	279.400	mm
Flange Thickness	t	33.2740	mm
Flange Material		SA-182 F304L	
Flange Material UNS number		S30403	
Flange Allowable Stress At Temperature	Sfo	1172.61	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1172.61	kgf/cm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Perimeter along the Center of the Bolts	L	738.117	mm
Diameter of Bolt Circle	C	234.950	mm
Nominal Bolt Diameter	a	19.0500	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		8	
Flange Face Outside Diameter	Fod	185.674	mm
Flange Face Inside Diameter	Fid	128.270	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	177.800	mm
Gasket Inside Diameter	Gi	155.702	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Gasket Contact Width,	$N = (G_o - G_i)/2$	11.049 mm
Basic Gasket Width,	$b_o = N/2$	5.524 mm
Effective Gasket Width,	$b = b_o$	5.524 mm
Gasket Reaction Diameter,	$G = (G_o + G_i)/2$	166.751 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 167^2 \cdot 25.5$$

$$= 5567.119 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:



$$\begin{aligned}
 &= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P \\
 &= 2 \cdot 5.52 \cdot \pi \cdot 167 \cdot 3 \cdot 25.5 \\
 &= 4426.558 \text{ kgf}
 \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned}
 &= \max(W_{o,1} + W_{o,2} + H'p, 0) \\
 &= \max(5567 + 4427 + 0, 0) \\
 &= 9993.677 \text{ kgf}
 \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned}
 &= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p \\
 &= 703 \cdot 5.52 \cdot \pi \cdot 167 + 0 \cdot 0 \cdot 0 \\
 &= 20347.412 \text{ kgf}
 \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned}
 &= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a) \\
 &= \max((9994 + 0 + 4 \cdot 0 / 167) / 1754, 20347 / 1754) \\
 &= 11.602 \text{ cm}^2
 \end{aligned}$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	11.602	15.587	
Radial Distance between Bolts and Edge:	20.637	22.225	
Circ. Spacing between the Bolts:	44.450	89.911	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned}
 &= A_b \cdot S_a / (y \cdot \pi (G_o + G_i)) \\
 &= 15.6 \cdot 1754 / (703 \cdot \pi (178 + 156)) \\
 &= 3.711 \text{ mm}
 \end{aligned}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned}
 &= S_a (A_m + A_b) / 2 \\
 &= 1754 (11.6 + 15.6) / 2 \\
 &= 23842.14 \text{ kgf}
 \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned}
 &= W_o \\
 &= 9993.68 \text{ kgf}
 \end{aligned}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$\begin{aligned}
 &= (C - G) / 2 \\
 &= (235 - 167) / 2 \\
 &= 34.0995 \text{ mm}
 \end{aligned}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$\begin{aligned}
 &= 1.9 \cdot W_{m1} \cdot h_G \cdot B_{sc} / (t^2 \cdot G) + C \cdot Z \cdot P_{eq} \cdot G^2 / t^2 \\
 &= 1.9 \cdot 9994 \cdot 34.1 \cdot 1 / (33.3^2 \cdot 167) + 0.3 \cdot 1 \cdot 25.5 \cdot 167^2 / 33.3^2 \\
 &= 542.78 \text{ kgf/cm}^2
 \end{aligned}$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$\begin{aligned}
 &= 1.9 \cdot W \cdot h_G \cdot B_{sc} / (t^2 \cdot G) \\
 &= 1.9 \cdot 23842 \cdot 34.1 \cdot 1 / (33.3^2 \cdot 167) \\
 &= 836.70 \text{ kgf/cm}^2
 \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned}
 &= W_{m1} / A_b \\
 &= 9994 / 15.6 \\
 &= 641.15 \text{ kgf/cm}^2
 \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned}
 &= (W_{m2} / A_b) \\
 &= (20347 / 15.6) \\
 &= 1305.41 \text{ kgf/cm}^2
 \end{aligned}$$



Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	542.78	1172.61	836.70 1172.61
Bolting	641.15	1753.82	1305.41 1753.82

Reqd. Blind Flange Thickness at Center 28.107 mm
 Reqd. Blind Flange Thickness at Gasket 28.107 mm

Estimated M.A.W.P. (Operating) 55.1 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 16.4 kg
 Estimated Unfinished Weight of Forging at given Thk 16.4 kg

SA-182 F304L, Min Metal Temp without impact per UHA-51: -196 °C

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	542.78	1172.61	836.70 1172.61
Bolting	641.15	1753.82	1305.41 1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 28.118 mm

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**Internal Pressure Results Summary:****Element Thickness, Pressure, Diameter and Allowable Stress :**

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
Bottom head		25.59	11.177	...	508	1172.6	1
Shell-1		25.573	10	...	508	1172.6	2
conical-1		25.492	10	...	508	996.72	3
Shell-2		25.492	10	...	220	996.72	4
H6-2		25.492	40	...	202.69	1172.6	5
H6-1		25.492	40	...	202.69	1172.6	6
SHELL-3		25.492	8	...	220	996.72	7
SHELL-4		25.492	8	...	220	996.72	8
H5-2		25.492	40	...	202.69	1172.6	9
H5-1		25.492	40	...	202.69	1172.6	10
Shell-5		25.492	8	...	220	996.72	11
Shell-6		25.492	8	...	220	996.72	12
H4-2		25.492	40	...	202.69	1172.6	13
H4-1		25.492	40	...	202.69	1172.6	14
Shell-6		25.492	8	...	220	996.72	15
conical -2		25.492	7	...	217.26	996.72	16
shell-7		25.492	7	...	141	996.72	17
H3-2		25.492	34	...	128.27	1172.6	18
H3-1		25.492	34	...	128.27	1172.6	19
shell-8		25.492	7	...	141.3	996.72	20
H2-2		25.492	34	...	128.27	1172.6	21
H2-1		25.492	34	...	128.27	1172.6	22
SHELL9		25.492	7	...	141.3	996.72	23
H1-2		25.492	34	...	128.27	1172.6	24
H1-1		25.492	34	...	279.4	1172.6	25

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
Bottom head		25.492	45.3	45.4	9.5	5.43621	1
Shell-1		25.492	46.8	46.9	10	5.49142	2
conical-1		25.492	39.3	39.3	10	6.52212	3
Shell-2		25.492	94	94	10	2.78486	4
H6-2		25.492	69.5	69.5	39.624	37.719	5
H6-1		25.492	69.7	69.7	39.624	37.6174	6
SHELL-3		25.492	74.7	74.7	8	2.78486	7
SHELL-4		25.492	74.7	74.7	8	2.78486	8
H5-2		25.492	67.8	67.8	39.624	37.3126	9
H5-1		25.492	67.8	67.8	39.624	34.8488	10
Shell-5		25.492	74.7	74.7	8	2.78486	11
Shell-6		25.492	74.7	74.7	8	2.78486	12
H4-2		25.492	67.8	67.8	39.624	37.3126	13
H4-1		25.492	67.8	67.8	39.624	37.3126	14
Shell-6		25.492	74.7	74.7	8	2.78486	15
conical -2		25.492	63.6	63.6	7	2.8502	16
shell-7		25.492	103	103	7	1.78484	17
H3-2		25.492	69.7	69.7	33.274	30.6578	18
H3-1		25.492	69.7	69.7	33.274	30.6578	19
shell-8		25.492	103	103	7	1.78864	20
H2-2		25.492	69.7	69.7	33.274	30.6578	21
H2-1		25.492	69.7	69.7	33.274	30.6578	22
SHELL9		25.492	103	103	7	1.78864	23
H1-2		25.492	69.7	69.7	33.274	30.6578	24
H1-1		25.492	55.1	55.1	33.274	28.1069	25

Minimum

39.3

39.3



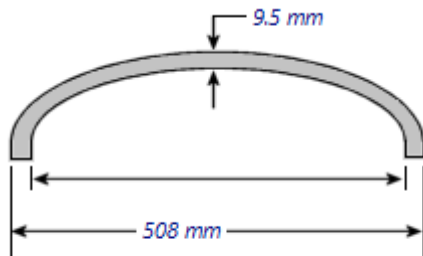
MAWP: 39.3 kgf/cm², limited by: conical-1.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Bottom head

Elliptical Head Design Information:							
Design Pressure	P	25.590	kgf/cm ²	Int. Design Temperature		85	°C
Material		SA-403	304L	Ext. Design Temperature		0	°C
Length		5.000	cm	External Pressure Chart		HA-3	
				UNS Number		S30403	
Int. Corr. All.	c	0.0000	mm	Allowable Stress (ope)	S	1172.6	kgf/cm ²
Ext. Corr. All.	ce	0.0000	mm	Allowable Stress (amb)	Sa	1172.6	kgf/cm ²
Outside Diameter	Do	508.002	mm	Actual Stress		661.2	kgf/cm ²
Spec. Min. Thk t _e , or sa		9.5000	mm	Spec. Nominal Thk.		11.177	mm
Surface Area		3637.7	cm ²	SG of Contents		0.950	
Element Volume		24696.7	cm ³	Weight of Contents		23.5	kgf
Empty Weight		33.9	kgf	Operating Weight		57.4	kgf
				Comp. All. Ope. Stress		890.1	kgf/cm ²
				Aspect Ratio		2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.210 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.105 mm

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D_o \cdot K_{cor}) / (2 \cdot S \cdot E + 2 \cdot P \cdot (K_{cor} - 0.1)) \text{ per Appendix 1-4 (c)}$$

$$= (25.6 \cdot 508 \cdot 1) / (2 \cdot 1173 \cdot 1 + 2 \cdot 25.6 \cdot (1 - 0.1))$$

$$= 5.4362 + 0.0000 = 5.4362 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.098 kgf/cm²

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D_o - 2 \cdot t \cdot (K_{cor} - 0.1)) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1173 \cdot 1 \cdot 9.5) / (1 \cdot 508 - 2 \cdot 9.5 \cdot (1 - 0.1))$$

$$= 45.4 - 0.098 = 45.3 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D_o - 2 \cdot t \cdot (K - 0.1)) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1173 \cdot 1 \cdot 9.5) / (1 \cdot 508 - 2 \cdot 9.5 \cdot (1 - 0.1))$$

$$= 45.4 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D_o - 2 \cdot t \cdot (K_{cor} - 0.1))) / (2 \cdot E \cdot t)$$

$$= (25.6 \cdot (1 \cdot 508 - 2 \cdot 9.5 \cdot (1 - 0.1))) / (2 \cdot 1 \cdot 9.5)$$

= 661.156 kgf/cm²

Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot Ro) / (S \cdot E + 0.4 \cdot P) + ca \text{ per Appendix 1-1 (a)(1)} \\ &= (25.6 \cdot 254) / (1173 \cdot 1 + 0.4 \cdot 25.6) + 0 \\ &= 5.495 \text{ mm} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.086 kgf/cm²

$$\begin{aligned} &= (S \cdot E \cdot t) / (Ro - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)} \\ &= (1173 \cdot 1 \cdot 11.2) / (254 - 0.4 \cdot 11.2) \\ &= 52.5 - 0.086 = 52.4 \text{ kgf/cm}^2 \end{aligned}$$

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per UG-81(a):

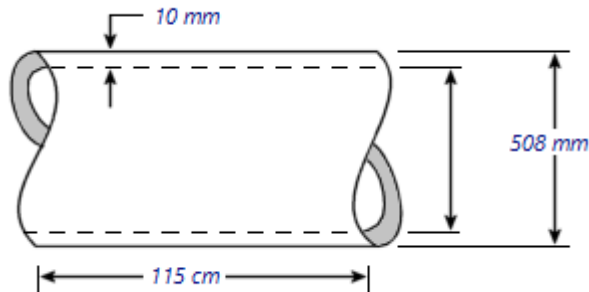
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.210 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.105 mm

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	25.573 kgf/cm²	Int. Design Temperature	85 °C	
Material	SA-312 TP304L		Ext. Design Temperature	0 °C	
Length	115.000 cm		External Pressure Chart	HA-3	
			UNS Number	S30403	
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S	1172.6 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1172.6 kgf/cm²
Outside Diameter	Do	508.000 mm	Actual Stress	639.3 kgf/cm²	
Spec. Min. Thk t _e , or s _a	10.0000 mm		Spec. Nominal Thk.	10.000 mm	
Surface Area	18353.2 cm²		SG of Contents	0.950	
Element Volume	215094.0 cm³		Weight of Contents	151.0 kgf	
Empty Weight	144.4 kgf		Operating Weight	295.4 kgf	
			Comp. All. Ope. Stress	897.1 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R_o) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (25.6 \cdot 254) / (1173 \cdot 1 + 0.4 \cdot 25.6)$$

$$= 5.4914 + 0.0000 = 5.4914 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.081 kgf/cm²

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1173 \cdot 1 \cdot 10) / (254 - 0.4 \cdot 10)$$

$$= 46.9 - 0.081 = 46.8 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1173 \cdot 1 \cdot 10) / (254 - 0.4 \cdot 10)$$

$$= 46.9 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R_o - 0.4 \cdot t)) / (E \cdot t)$$

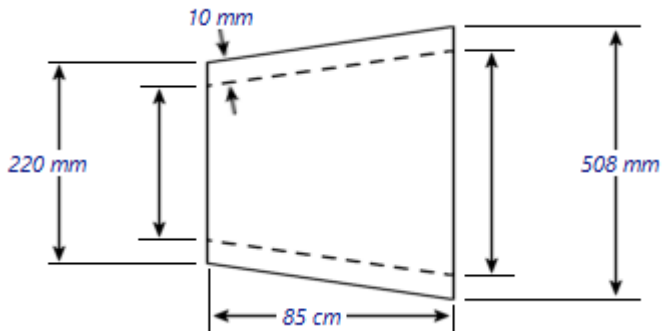
$$= (25.6 \cdot (254 - 0.4 \cdot 10)) / (1 \cdot 10)$$

$$= 639.318 \text{ kgf/cm}^2$$

SA-312 TP304L, Min Metal Temp without impact per UHA-51: -196 °C

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: conical-1

Conical Head/Section Design Information:							
Design Pressure	P	25.492	kgf/cm²	Int. Design Temperature		85	°C
Material	SA-312	TP304L		Ext. Design Temperature		0	°C
Length		85.000	cm	External Pressure Chart		HA-3	
				UNS Number		S30403	
Int. Corr. All.	c	0.0000	mm	Allowable Stress (ope)	S	1172.6	kgf/cm²
Ext. Corr. All.	ce	0.0000	mm	Allowable Stress (amb)	Sa	1172.6	kgf/cm²
Outside Diameter	Do	508.000	mm	Actual Stress		760.6	kgf/cm²
Spec. Min. Thk t,e, or sa		10.0000	mm	Spec. Nominal Thk.		10.000	mm
Surface Area		9858.6	cm²	SG of Contents		0.000	
Element Volume		83483.5	cm³	Weight of Contents		0.0	kgf
Empty Weight		76.9	kgf	Operating Weight		76.9	kgf
				Comp. All. Ope. Stress		899.0	kgf/cm²
				To End Diameter		220.000	mm
Line of Support	Both Ends						

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot Do) / (2 \cdot \cos(a) \cdot (S \cdot E + 0.4 \cdot P))$ per Appendix 1-4 (e)
 $= (25.5 \cdot 508) / (2 \cdot 0.99 \cdot (1173 \cdot 0.85 + 0.4 \cdot 25.5))$
 $= 6.5221 + 0.0000 = 6.5221 \text{ mm}$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
 $= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D - 0.8 \cdot t \cdot \cos(a))$ per Appendix 1-4 (e)(5)
 $= (2 \cdot 1173 \cdot 0.85 \cdot (10) \cdot 0.99) / (508 - 0.8 \cdot (10) \cdot 0.99)$
 $= 39.3 \text{ kgf/cm}^2$

Maximum Allowable Pressure, New and Cold [MAPNC]:
 $= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D - 0.8 \cdot t \cdot \cos(a))$ per Appendix 1-4(e)
 $= (2 \cdot 1173 \cdot 0.85 \cdot 10 \cdot 0.99) / (508 - 0.8 \cdot 10 \cdot 0.99)$
 $= 39.3 \text{ kgf/cm}^2$

Actual stress at given pressure and thickness, corroded [Sact]:
 $= (P \cdot (D - 0.8 \cdot t \cdot \cos(a))) / (2 \cdot E \cdot t \cdot \cos(a))$
 $= (25.5 \cdot (508 - 0.8 \cdot (10) \cdot 0.99)) / (2 \cdot 0.85 \cdot (10) \cdot 0.99)$
 $= 760.620 \text{ kgf/cm}^2$

Minimum Design Metal Temperature Results:



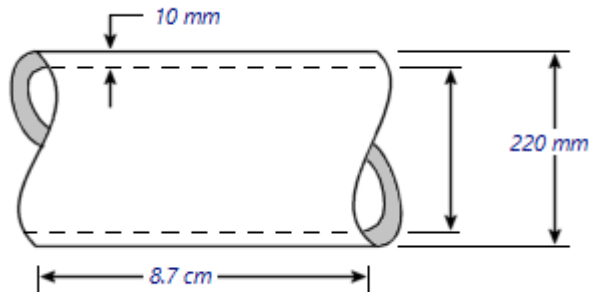
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	25.492 kgf/cm²	Int. Design Temperature	85 °C	
Material	SA-312 TP304L		Ext. Design Temperature	0 °C	
Length	8.710 cm		External Pressure Chart	HA-3	
			UNS Number	S30403	
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S	1172.6 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1172.6 kgf/cm²
Outside Diameter	Do	220.000 mm	Actual Stress	317.9 kgf/cm²	
Spec. Min. Thk t _e , or sa		10.0000 mm	Spec. Nominal Thk.	10.000 mm	
Surface Area		602.0 cm²	SG of Contents	0.000	
Element Volume		2736.3 cm³	Weight of Contents	0.0 kgf	
Empty Weight		4.6 kgf	Operating Weight	4.6 kgf	
			Comp. All. Ope. Stress	1007.5 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot Ro) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (25.5 \cdot 110) / (1173 \cdot 0.85 + 0.4 \cdot 25.5)$$

$$= 2.7849 + 0.0000 = 2.7849 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (Ro - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1173 \cdot 0.85 \cdot 10) / (110 - 0.4 \cdot 10)$$

$$= 94 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (Ro - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1173 \cdot 0.85 \cdot 10) / (110 - 0.4 \cdot 10)$$

$$= 94 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (Ro - 0.4 \cdot t)) / (E \cdot t)$$

$$= (25.5 \cdot (110 - 0.4 \cdot 10)) / (0.85 \cdot 10)$$

$$= 317.901 \text{ kgf/cm}^2$$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

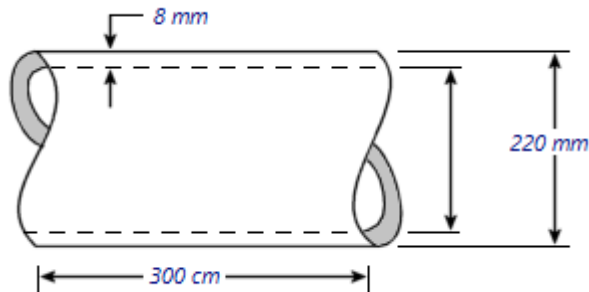


Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: SHELL-3

Cylindrical Shell Design Information:					
Design Pressure	P	25.492 kgf/cm²	Int. Design Temperature	85 °C	
Material	SA-312 TP304L		Ext. Design Temperature	0 °C	
Length	300.000 cm		External Pressure Chart	HA-3	
			UNS Number	S30403	
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S	1172.6 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1172.6 kgf/cm²
Outside Diameter	Do	220.000 mm	Actual Stress	400.4 kgf/cm²	
Spec. Min. Thk t _e , or s _a		8.0000 mm	Spec. Nominal Thk.	8.000 mm	
Surface Area		20734.5 cm²	SG of Contents	0.000	
Element Volume		98055.6 cm³	Weight of Contents	0.0 kgf	
Empty Weight		128.3 kgf	Operating Weight	128.3 kgf	
			Comp. All. Ope. Stress	978.8 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R_o) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (25.5 \cdot 110) / (1173 \cdot 0.85 + 0.4 \cdot 25.5)$$

$$= 2.7849 + 0.0000 = 2.7849 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1173 \cdot 0.85 \cdot 8) / (110 - 0.4 \cdot 8)$$

$$= 74.7 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1173 \cdot 0.85 \cdot 8) / (110 - 0.4 \cdot 8)$$

$$= 74.7 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R_o - 0.4 \cdot t)) / (E \cdot t)$$

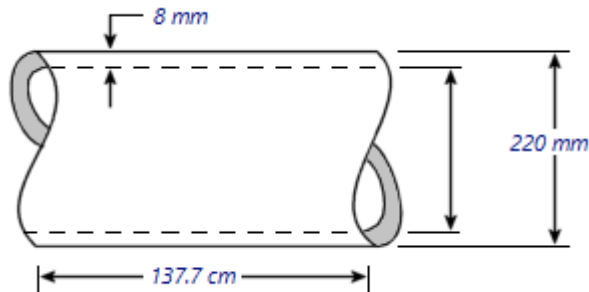
$$= (25.5 \cdot (110 - 0.4 \cdot 8)) / (0.85 \cdot 8)$$

$$= 400.375 \text{ kgf/cm}^2$$

SA-312 TP304L, Min Metal Temp without impact per UHA-51: -196 °C

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: SHELL-4

Cylindrical Shell Design Information:					
Design Pressure	P	25.492 kgf/cm²	Int. Design Temperature	85 °C	
Material	SA-312 TP304L		Ext. Design Temperature	0 °C	
Length	137.737 cm		External Pressure Chart	HA-3	
			UNS Number	S30403	
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S	1172.6 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1172.6 kgf/cm²
Outside Diameter	Do	220.000 mm	Actual Stress	400.4 kgf/cm²	
Spec. Min. Thk t _e , or s _a	8.0000 mm		Spec. Nominal Thk.	8.000 mm	
Surface Area	9519.7 cm²		SG of Contents	0.000	
Element Volume	45019.6 cm³		Weight of Contents	0.0 kgf	
Empty Weight	58.9 kgf		Operating Weight	58.9 kgf	
			Comp. All. Ope. Stress	978.8 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R_o) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (25.5 \cdot 110) / (1173 \cdot 0.85 + 0.4 \cdot 25.5)$$

$$= 2.7849 + 0.0000 = 2.7849 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1173 \cdot 0.85 \cdot 8) / (110 - 0.4 \cdot 8)$$

$$= 74.7 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1173 \cdot 0.85 \cdot 8) / (110 - 0.4 \cdot 8)$$

$$= 74.7 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R_o - 0.4 \cdot t)) / (E \cdot t)$$

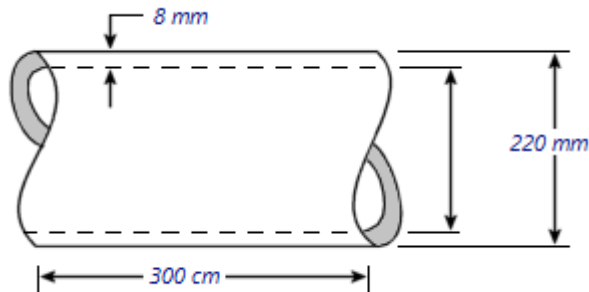
$$= (25.5 \cdot (110 - 0.4 \cdot 8)) / (0.85 \cdot 8)$$

$$= 400.375 \text{ kgf/cm}^2$$

SA-312 TP304L, Min Metal Temp without impact per UHA-51: -196 °C

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-5

Cylindrical Shell Design Information:							
Design Pressure	P	25.492	kgf/cm²	Int. Design Temperature		85	°C
Material		SA-312	TP304L	Ext. Design Temperature		0	°C
Length		300.000	cm	External Pressure Chart		HA-3	
				UNS Number		S30403	
Int. Corr. All.	c	0.0000	mm	Allowable Stress (ope)	S	1172.6	kgf/cm²
Ext. Corr. All.	ce	0.0000	mm	Allowable Stress (amb)	Sa	1172.6	kgf/cm²
Outside Diameter	Do	220.000	mm	Actual Stress		400.4	kgf/cm²
Spec. Min. Thk t _e , or s _a		8.0000	mm	Spec. Nominal Thk.		8.000	mm
Surface Area		20734.5	cm²	SG of Contents		0.000	
Element Volume		98055.6	cm³	Weight of Contents		0.0	kgf
Empty Weight		128.3	kgf	Operating Weight		128.3	kgf
				Comp. All. Ope. Stress		978.8	kgf/cm²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R_o) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (25.5 \cdot 110) / (1173 \cdot 0.85 + 0.4 \cdot 25.5)$$

$$= 2.7849 + 0.0000 = 2.7849 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1173 \cdot 0.85 \cdot 8) / (110 - 0.4 \cdot 8)$$

$$= 74.7 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1173 \cdot 0.85 \cdot 8) / (110 - 0.4 \cdot 8)$$

$$= 74.7 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R_o - 0.4 \cdot t)) / (E \cdot t)$$

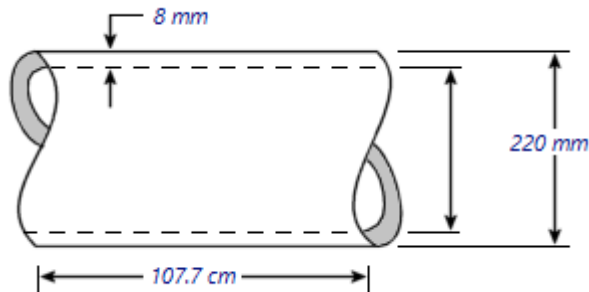
$$= (25.5 \cdot (110 - 0.4 \cdot 8)) / (0.85 \cdot 8)$$

$$= 400.375 \text{ kgf/cm}^2$$

SA-312 TP304L, Min Metal Temp without impact per UHA-51: -196 °C

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-6

Cylindrical Shell Design Information:					
Design Pressure	P	25.492 kgf/cm²	Int. Design Temperature		85 °C
Material		SA-312 TP304L	Ext. Design Temperature		0 °C
Length		107.737 cm	External Pressure Chart		HA-3
			UNS Number		S30403
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S	1172.6 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1172.6 kgf/cm²
Outside Diameter	Do	220.000 mm	Actual Stress		400.4 kgf/cm²
Spec. Min. Thk t _e , or s _a		8.0000 mm	Spec. Nominal Thk.		8.000 mm
Surface Area		7446.2 cm²	SG of Contents		0.000
Element Volume		35214.1 cm³	Weight of Contents		0.0 kgf
Empty Weight		46.1 kgf	Operating Weight		46.1 kgf
			Comp. All. Ope. Stress		978.8 kgf/cm²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R_o) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (25.5 \cdot 110) / (1173 \cdot 0.85 + 0.4 \cdot 25.5)$$

$$= 2.7849 + 0.0000 = 2.7849 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1173 \cdot 0.85 \cdot 8) / (110 - 0.4 \cdot 8)$$

$$= 74.7 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1173 \cdot 0.85 \cdot 8) / (110 - 0.4 \cdot 8)$$

$$= 74.7 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R_o - 0.4 \cdot t)) / (E \cdot t)$$

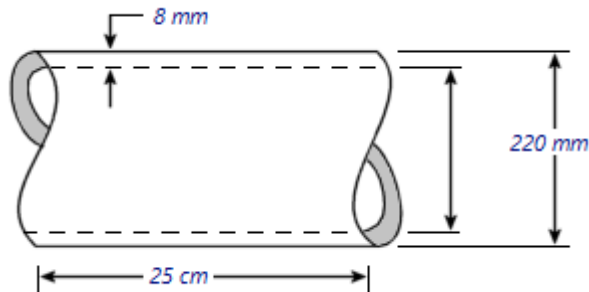
$$= (25.5 \cdot (110 - 0.4 \cdot 8)) / (0.85 \cdot 8)$$

$$= 400.375 \text{ kgf/cm}^2$$

SA-312 TP304L, Min Metal Temp without impact per UHA-51: -196 °C

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-6

Cylindrical Shell Design Information:					
Design Pressure	P	25.492 kgf/cm²	Int. Design Temperature	85 °C	
Material	SA-312 TP304L		Ext. Design Temperature	0 °C	
Length		25.000 cm	External Pressure Chart	HA-3	
			UNS Number	S30403	
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S	1172.6 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1172.6 kgf/cm²
Outside Diameter	Do	220.000 mm	Actual Stress	400.4 kgf/cm²	
Spec. Min. Thk t _e , or s _a		8.0000 mm	Spec. Nominal Thk.	8.000 mm	
Surface Area		1727.9 cm²	SG of Contents	0.000	
Element Volume		8171.3 cm³	Weight of Contents	0.0 kgf	
Empty Weight		10.7 kgf	Operating Weight	10.7 kgf	
			Comp. All. Ope. Stress	978.8 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R_o) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (25.5 \cdot 110) / (1173 \cdot 0.85 + 0.4 \cdot 25.5)$$

$$= 2.7849 + 0.0000 = 2.7849 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1173 \cdot 0.85 \cdot 8) / (110 - 0.4 \cdot 8)$$

$$= 74.7 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1173 \cdot 0.85 \cdot 8) / (110 - 0.4 \cdot 8)$$

$$= 74.7 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R_o - 0.4 \cdot t)) / (E \cdot t)$$

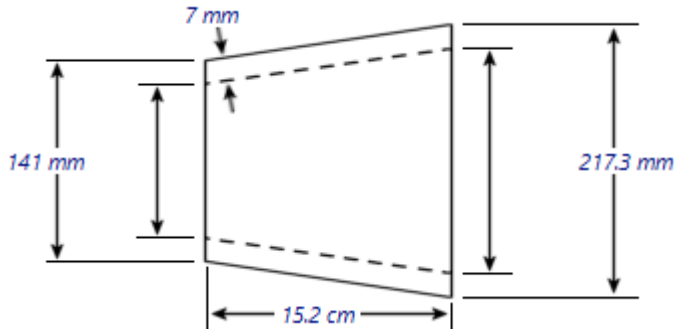
$$= (25.5 \cdot (110 - 0.4 \cdot 8)) / (0.85 \cdot 8)$$

$$= 400.375 \text{ kgf/cm}^2$$

SA-312 TP304L, Min Metal Temp without impact per UHA-51: -196 °C

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: conical -2

Conical Head/Section Design Information:					
Design Pressure	P	25.492 kgf/cm²	Int. Design Temperature		85 °C
Material	SA-312 TP304L		Ext. Design Temperature		0 °C
Length		15.200 cm	External Pressure Chart		HA-3
			UNS Number		S30403
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S	1172.6 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1172.6 kgf/cm²
Outside Diameter	Do	217.264 mm	Actual Stress		470.3 kgf/cm²
Spec. Min. Thk t _e , or s _a		7.0000 mm	Spec. Nominal Thk.		7.000 mm
Surface Area		825.8 cm²	SG of Contents		0.000
Element Volume		3401.8 cm³	Weight of Contents		0.0 kgf
Empty Weight		5.0 kgf	Operating Weight		5.0 kgf
			Comp. All. Ope. Stress		977.6 kgf/cm²
			To End Diameter		141.000 mm
Ins. Knuckle Radius L		31.0000 mm	Small End Flare Radius r		12.000 mm
Knuckle Thickness		8.0000 mm	Flare Thickness		8.000 mm
Line of Support	Both Ends				

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot Do) / (2 \cdot \cos(a) \cdot (S \cdot E + 0.4 \cdot P))$$
 per Appendix 1-4 (e)

$$= (25.5 \cdot 217) / (2 \cdot 0.96 \cdot (1173 \cdot 0.85 + 0.4 \cdot 25.5))$$

$$= 2.8502 + 0.0000 = 2.8502 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D - 0.8 \cdot t \cdot \cos(a))$$
 per Appendix 1-4 (e)(5)

$$= (2 \cdot 1173 \cdot 0.85 \cdot (7) \cdot 0.96) / (217 - 0.8 \cdot (7) \cdot 0.96)$$

$$= 63.6 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D - 0.8 \cdot t \cdot \cos(a))$$
 per Appendix 1-4(e)

$$= (2 \cdot 1173 \cdot 0.85 \cdot 7 \cdot 0.96) / (217 - 0.8 \cdot 7 \cdot 0.96)$$

$$= 63.6 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (D - 0.8 \cdot t \cdot \cos(a))) / (2 \cdot E \cdot t \cdot \cos(a))$$

$$= (25.5 \cdot (217 - 0.8 \cdot (7) \cdot 0.96)) / (2 \cdot 0.85 \cdot (7) \cdot 0.96)$$

$$= 470.342 \text{ kgf/cm}^2$$

Thickness and Maximum Pressure for Large Knuckle:



Equivalent Crown Radius for the Cone Large End Knuckle [L]:

$$\begin{aligned}
 &= Di/(2(\cos(\alpha)) + ca \text{ per UG-32(h)} \\
 &= 217/(2(\cos(15.2)) + 0 \\
 &= 112.581 \text{ mm}
 \end{aligned}$$

Stress Concentration Factor [M]:

$$\begin{aligned}
 &= (3 + \sqrt{L/(r + ca)})/4 \\
 &= (3 + \sqrt{31/(31 + 0)})/4 \\
 &= 1.226
 \end{aligned}$$

Required Knuckle Thickness due to Internal Pressure [TrKnuckle]:

$$\begin{aligned}
 &= (P(L+ca)M)/(2 \cdot S \cdot E - 0.2 \cdot P) + ca \text{ per App. 1-4(d)} \\
 &= (25.5 (113) 1.23)/(2 \cdot 1173 \cdot 0.85 - 0.2 \cdot 25.5) + 0 \\
 &= 1.770 \text{ mm}
 \end{aligned}$$

Large End Knuckle Maximum Allowable Working Pressure [KnuckleMAWP]:

$$\begin{aligned}
 &= (2 \cdot S \cdot E (t - ca))/(M (L + ca) + 0.2 (t - ca)) \text{ per Appendix 1-4 (d)} \\
 &= (2 \cdot 1173 \cdot 0.85 (8 - 0))/(1.23 (113 + 0) + 0.2 (8 - 0)) \\
 &= 114 \text{ kgf/cm}^2
 \end{aligned}$$

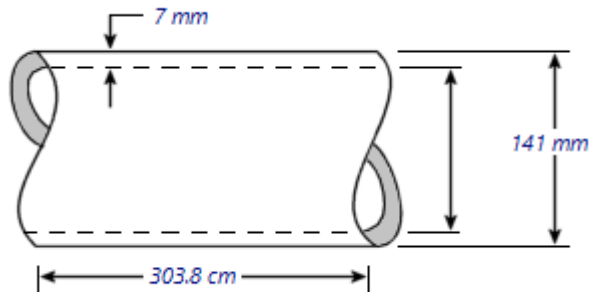
Minimum Knuckle Radius check per Figure UG-36 [rL]:

$$\begin{aligned}
 &= \max(3 \cdot t, 0.12(RL + t)) \\
 &= \max(3 \cdot 7, 0.12(103 + 7)) \\
 &= 21 \text{ mm}
 \end{aligned}$$

SA-312 TP304L, Min Metal Temp without impact per UHA-51: -196 °C

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: shell-7

Cylindrical Shell Design Information:					
Design Pressure	P	25.492 kgf/cm²	Int. Design Temperature	85 °C	
Material	SA-312 TP304L		Ext. Design Temperature	0 °C	
Length	303.806 cm		External Pressure Chart	HA-3	
			UNS Number	S30403	
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S	1172.6 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1172.6 kgf/cm²
Outside Diameter	Do	141.000 mm	Actual Stress	290.1 kgf/cm²	
Spec. Min. Thk t _e , or s _a	7.0000 mm		Spec. Nominal Thk.	7.000 mm	
Surface Area	13457.5 cm²		SG of Contents	0.000	
Element Volume	38485.3 cm³		Weight of Contents	0.0 kgf	
Empty Weight	71.9 kgf		Operating Weight	71.9 kgf	
			Comp. All. Ope. Stress	1018.7 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R_o) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (25.5 \cdot 70.5) / (1173 \cdot 0.85 + 0.4 \cdot 25.5)$$

$$= 1.7848 + 0.0000 = 1.7848 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1173 \cdot 0.85 \cdot 7) / (70.5 - 0.4 \cdot 7)$$

$$= 103 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1173 \cdot 0.85 \cdot 7) / (70.5 - 0.4 \cdot 7)$$

$$= 103 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R_o - 0.4 \cdot t)) / (E \cdot t)$$

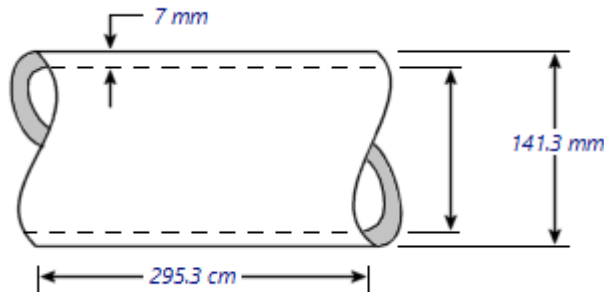
$$= (25.5 \cdot (70.5 - 0.4 \cdot 7)) / (0.85 \cdot 7)$$

$$= 290.052 \text{ kgf/cm}^2$$

SA-312 TP304L, Min Metal Temp without impact per UHA-51: -196 °C

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: shell-8

Cylindrical Shell Design Information:					
Design Pressure	P	25.492 kgf/cm²	Int. Design Temperature	85 °C	
Material	SA-312 TP304L		Ext. Design Temperature	0 °C	
Length	295.277 cm		External Pressure Chart	HA-3	
			UNS Number	S30403	
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S	1172.6 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1172.6 kgf/cm²
Outside Diameter	Do	141.300 mm	Actual Stress	290.7 kgf/cm²	
Spec. Min. Thk t _e , or s _a	7.0000 mm		Spec. Nominal Thk.	7.000 mm	
Surface Area	13107.6 cm²		SG of Contents	0.000	
Element Volume	37581.8 cm³		Weight of Contents	0.0 kgf	
Empty Weight	70.0 kgf		Operating Weight	70.0 kgf	
			Comp. All. Ope. Stress	1018.4 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R_o) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (25.5 \cdot 70.7) / (1173 \cdot 0.85 + 0.4 \cdot 25.5)$$

$$= 1.7886 + 0.0000 = 1.7886 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1173 \cdot 0.85 \cdot 7) / (70.7 - 0.4 \cdot 7)$$

$$= 103 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1173 \cdot 0.85 \cdot 7) / (70.7 - 0.4 \cdot 7)$$

$$= 103 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R_o - 0.4 \cdot t)) / (E \cdot t)$$

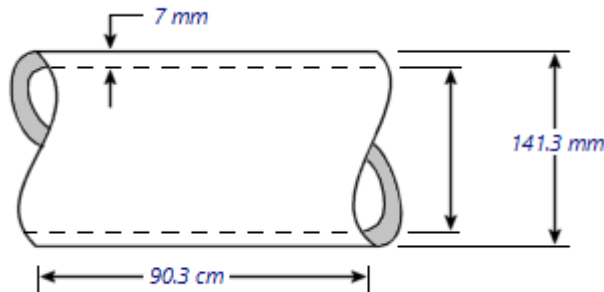
$$= (25.5 \cdot (70.7 - 0.4 \cdot 7)) / (0.85 \cdot 7)$$

$$= 290.695 \text{ kgf/cm}^2$$

SA-312 TP304L, Min Metal Temp without impact per UHA-51: -196 °C

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: SHELL9

Cylindrical Shell Design Information:					
Design Pressure	P	25.492 kgf/cm ²	Int. Design Temperature	85 °C	
Material	SA-312 TP304L		Ext. Design Temperature	0 °C	
Length		90.277 cm	External Pressure Chart	HA-3	
			UNS Number	S30403	
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S	1172.6 kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1172.6 kgf/cm ²
Outside Diameter	Do	141.300 mm	Actual Stress	290.7 kgf/cm ²	
Spec. Min. Thk t _e , or sa		7.0000 mm	Spec. Nominal Thk.	7.000 mm	
Surface Area		4007.5 cm ²	SG of Contents	0.000	
Element Volume		11490.1 cm ³	Weight of Contents	0.0 kgf	
Empty Weight		21.4 kgf	Operating Weight	21.4 kgf	
			Comp. All. Ope. Stress	1018.4 kgf/cm ²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R_o) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (25.5 \cdot 70.7) / (1173 \cdot 0.85 + 0.4 \cdot 25.5)$$

$$= 1.7886 + 0.0000 = 1.7886 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1173 \cdot 0.85 \cdot 7) / (70.7 - 0.4 \cdot 7)$$

$$= 103 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1173 \cdot 0.85 \cdot 7) / (70.7 - 0.4 \cdot 7)$$

$$= 103 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R_o - 0.4 \cdot t)) / (E \cdot t)$$

$$= (25.5 \cdot (70.7 - 0.4 \cdot 7)) / (0.85 \cdot 7)$$

$$= 290.695 \text{ kgf/cm}^2$$

SA-312 TP304L, Min Metal Temp without impact per UHA-51: -196 °C

Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	51.090 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	33.140 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	51.041 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	43.230 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	36.454 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	36.454 kgf/cm ²
App 27-4	= M.A.W.P.	39.300 kgf/cm ²

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 39.3 • 1
 = 51.090 kgf/cm²

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	1321.3	1560.1	0.847	51.14
Shell-1	1278.5	1560.1	0.820	51.14
conical-1	1525.9	1560.1	0.978	51.14
Shell-2	637.4	1560.1	0.409	51.11
SHELL-3	802.8	1560.1	0.515	51.11
SHELL-4	802.8	1560.1	0.515	51.11
Shell-5	802.8	1560.1	0.515	51.11
Shell-6	802.8	1560.1	0.515	51.11
Shell-6	802.8	1560.1	0.515	51.11
conical -2	943.0	1560.1	0.604	51.11
shell-7	581.5	1560.1	0.373	51.10
shell-8	582.8	1560.1	0.374	51.10
SHELL9	582.8	1560.1	0.374	51.10

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
B2 2''	Nozzle	1172.63	1172.63	1.000
L2 2''	Nozzle	1172.63	1172.63	1.000
L4 2''	Nozzle	1172.63	1172.63	1.000
L6 2''	Nozzle	1172.63	1172.63	1.000
L1 2''	Nozzle	1172.63	1172.63	1.000
L3 2''	Nozzle	1172.63	1172.63	1.000
L5 2''	Nozzle	1172.63	1172.63	1.000
P3-2''	Nozzle	1172.63	1172.63	1.000
A4 2''	Nozzle	1172.63	1172.63	1.000
A1 2''	Nozzle	1172.63	1172.63	1.000
A3 2''	Nozzle	1172.63	1172.63	1.000
P2 2''	Nozzle	1172.63	1172.63	1.000
A2 2''	Nozzle	1172.63	1172.63	1.000
P1 2''	Nozzle	1172.63	1172.63	1.000
B1 2''	Nozzle	1172.63	1172.63	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
Bottom head	1172.61	1172.61	1.000
Shell-1	1172.61	1172.61	1.000
conical-1	1172.61	1172.61	1.000



Shell-2	1172.61	1172.61	1.000
H6-2	1172.61	1172.61	1.000
H6-1	1172.61	1172.61	1.000
SHELL-3	1172.61	1172.61	1.000
SHELL-4	1172.61	1172.61	1.000
H5-2	1172.61	1172.61	1.000
H5-1	1172.61	1172.61	1.000
Shell-5	1172.61	1172.61	1.000
Shell-6	1172.61	1172.61	1.000
H4-2	1172.61	1172.61	1.000
H4-1	1172.61	1172.61	1.000
Shell-6	1172.61	1172.61	1.000
conical -2	1172.61	1172.61	1.000
shell-7	1172.61	1172.61	1.000
H3-2	1172.61	1172.61	1.000
H3-1	1172.61	1172.61	1.000
shell-8	1172.61	1172.61	1.000
H2-2	1172.61	1172.61	1.000
H2-1	1172.61	1172.61	1.000
SHELL9	1172.61	1172.61	1.000
H1-2	1172.61	1172.61	1.000
H1-1	1172.61	1172.61	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
B2 2''	108.17	1560.11	0.069
L2 2''	108.17	1560.11	0.069
L4 2''	108.17	1560.11	0.069
L6 2''	108.17	1560.11	0.069
L1 2''	108.17	1560.11	0.069
L3 2''	108.17	1560.11	0.069
L5 2''	108.17	1560.11	0.069
P3-2''	108.17	1560.11	0.069
A4 2''	108.17	1560.11	0.069
A1 2''	108.11	1560.11	0.069
A3 2''	108.11	1560.11	0.069
P2 2''	108.11	1560.11	0.069
A2 2''	108.09	1560.11	0.069
P1 2''	108.09	1560.11	0.069
B1 2''	108.12	1560.11	0.069

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	No Calc	508.002	9.5	0.0025973	759.252
20	30	124.075	508	10	0.0014211	674.188
30	40	60.892	508	10	0.0030266	779.748
40	50	8.71	220	10	0.042043	1075.7
50	60	No Calc	...	39.624	No Calc	No Calc
60	70	No Calc	...	39.624	No Calc	No Calc
70	80	437.737	220	8	0.0014845	680.567
80	90	437.737	220	8	0.0014845	680.567
90	100	No Calc	...	39.624	No Calc	No Calc
100	110	No Calc	...	39.624	No Calc	No Calc
110	120	407.737	220	8	0.0014867	680.787
120	130	407.737	220	8	0.0014867	680.787
130	140	No Calc	...	39.624	No Calc	No Calc
140	150	No Calc	...	39.624	No Calc	No Calc
150	160	25	220	8	0.0083617	904.808
160	170	12.7279	217.264	7	0.014042	961.171
170	180	303.806	141	7	0.0027383	766.382
180	190	No Calc	...	33.274	No Calc	No Calc
190	200	No Calc	...	33.274	No Calc	No Calc
200	210	295.277	141.3	7	0.0027293	765.94
210	220	No Calc	...	33.274	No Calc	No Calc
220	230	No Calc	...	33.274	No Calc	No Calc
230	240	90.277	141.3	7	0.0028801	773.145
240	250	No Calc	...	33.274	No Calc	No Calc
250	260	No Calc	...	33.274	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	9.5	No Calc	...	15.8
20	30	10	No Calc	...	17.7
30	40	10	No Calc	...	20.2
40	50	10	No Calc	...	65.2
50	60	39.624	37.719	...	No Calc
60	70	39.624	37.6174	...	No Calc
70	80	8	No Calc	...	33
80	90	8	No Calc	...	33
90	100	39.624	37.3126	...	No Calc
100	110	39.624	34.8488	...	No Calc
110	120	8	No Calc	...	33
120	130	8	No Calc	...	33
130	140	39.624	37.3126	...	No Calc
140	150	39.624	37.3126	...	No Calc
150	160	8	No Calc	...	43.9
160	170	7	No Calc	...	39.8
170	180	7	No Calc	...	50.7
180	190	33.274	30.6578	...	No Calc
190	200	33.274	30.6578	...	No Calc
200	210	7	No Calc	...	50.6
210	220	33.274	30.6578	...	No Calc
220	230	33.274	30.6578	...	No Calc
230	240	7	No Calc	...	51.1
240	250	33.274	30.6578	...	No Calc
250	260	33.274	28.1178	...	No Calc



Minimum

15.8

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	124.075	No Calc	No Calc	No Calc
30	40	60.892	60.892	No Calc	No Calc
40	50	8.71	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	437.737	No Calc	No Calc	No Calc
80	90	437.737	No Calc	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc
100	110	No Calc	No Calc	No Calc	No Calc
110	120	407.737	No Calc	No Calc	No Calc
120	130	407.737	No Calc	No Calc	No Calc
130	140	No Calc	No Calc	No Calc	No Calc
140	150	No Calc	No Calc	No Calc	No Calc
150	160	25	No Calc	No Calc	No Calc
160	170	12.7279	12.7279	No Calc	No Calc
170	180	303.806	No Calc	No Calc	No Calc
180	190	No Calc	No Calc	No Calc	No Calc
190	200	No Calc	No Calc	No Calc	No Calc
200	210	295.277	No Calc	No Calc	No Calc
210	220	No Calc	No Calc	No Calc	No Calc
220	230	No Calc	No Calc	No Calc	No Calc
230	240	90.277	No Calc	No Calc	No Calc
240	250	No Calc	No Calc	No Calc	No Calc
250	260	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	33.912	24696.7	33.912	24696.7	...
20	30	144.424	215094	144.424	215094	...
30	40	76.9314	83483.5	76.9314	83483.5	...
40	50	4.61264	2736.33	4.61264	2736.33	...
50	60	34.2434	3552.77	34.2434	3552.77	...
60	70	33.8151	3552.77	33.8151	3552.77	...
70	80	128.31	98055.6	128.31	98055.6	...
80	90	58.9099	45019.6	58.9099	45019.6	...
90	100	33.3903	3552.77	33.3903	3552.77	...
100	110	33.3903	3552.77	33.3903	3552.77	...
110	120	128.31	98055.6	128.31	98055.6	...
120	130	46.079	35214.1	46.079	35214.1	...
130	140	33.3903	3552.77	33.3903	3552.77	...
140	150	33.3903	3552.77	33.3903	3552.77	...
150	160	10.6925	8171.3	10.6925	8171.3	...
160	170	4.96946	3401.85	4.96946	3401.85	...
170	180	71.8641	38485.3	71.8641	38485.3	...
180	190	16.9549	1263.21	16.9549	1263.21	...
190	200	16.9549	1263.21	16.9549	1263.21	...
200	210	70.0029	37581.8	70.0029	37581.8	...
210	220	16.9549	1263.21	16.9549	1263.21	...
220	230	16.9549	1263.21	16.9549	1263.21	...
230	240	21.4024	11490.1	21.4024	11490.1	...
240	250	16.9549	1263.21	16.9549	1263.21	...
250	260	16.7334	...	16.7334	...	...
Total		1103	729118.56	1103	729118.56	0

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset Dtl. Cent. cm	Description
10	Liqd	23.4513	...	-6.35002	...	Liquid 10
10	Nozl	14.7363	...	-32.9997	...	B2 2''
10	Wght	4.99991	...	-12.7	...	Vortex breaker
20	Liqd	150.969	...	42.5	...	Liquid 20
20	Nozl	9.72789	33.1719	45	...	L2 2''
20	Nozl	9.72789	23.4561	45	-23.4561	L4 2''
20	Nozl	9.72789	...	45	-33.1719	L6 2''
20	Nozl	9.72789	33.1719	90	...	L1 2''
20	Nozl	9.72789	23.4561	90	-23.4561	L3 2''
20	Nozl	10.6119	...	90	-33.1719	L5 2''
30	Nozl	9.00165	20.5125	45	...	P3-2''
30	Nozl	9.00165	-25.5949	15	...	A4 2''
70	Pack	21.2289	...	170.25	...	BED 1
70	Nozl	10.2499	18.6709	10.5269	...	A1 2''
70	Lugs	196.998	...	71	...	Support Lugs
70	Wght	1.99996	...	40	...	Bed1 sup ring
70	Wght	4.99991	...	40.5	...	Bed1 sup grid
80	Pack	6.74908	...	41.25	...	BED 1
80	Wght	49.9991	...	100	...	Cleat wt
80	Wght	4.99991	...	90	...	Bed 1 Dist wt
80	Wght	1.99996	...	88	...	Bed1 Dist sup w
110	Pack	23.4786	...	156.5	...	BED-2
110	Wght	1.99996	...	11.9997	...	Bed2 sup ring
110	Wght	4.99991	...	12.4998	...	Bed 2 Sup grid
120	Pack	4.49939	...	27.5	...	Bed2
120	Wght	4.99991	...	60	...	A3 dist wt
120	Wght	1.99996	...	58	...	A3 dist sup wt
150	Nozl	10.2502	18.6709	14.0267	...	A3 2''
150	Nozl	9.49939	-18.6709	14.0267	...	P2 2''



150	Wght	4.99991	...	14.0267	...
170	Pack	7.51426	...	128.5	...
170	Wght	1.99996	...	8	...
170	Wght	4.99991	...	9.5	...
170	Wght	4.99991	...	260	...
170	Wght	1.99996	...	258	...
200	Pack	7.5498	...	129.7	...
200	Wght	1.99996	...	8	...
200	Wght	4.99991	...	10	...
200	Wght	4.99991	...	260	...
200	Wght	1.99996	...	258	...
200	Wght	29.9995	7	265.3	...
200	Wght	29.9995	-7	265.3	...
230	Pack	0.11468	...	77.777	...
230	Nozl	10.4884	-15.3033	10.2973	...
230	Nozl	10.5262	15.3033	30.2973	...
230	Wght	4.99991	...	10.2973	...
230	Wght	4.99991	...	70	...
250	Nozl	10.3805	...	12.0777	...
250	Forc	...	...	3	...
70	Pliq	40.27	...	170.250	...
80	Pliq	12.80	...	41.250	...
110	Pliq	44.54	...	156.500	...
120	Pliq	8.54	...	27.500	...
170	Pliq	14.25	...	128.500	...
200	Pliq	14.32	...	129.700	...
230	Pliq	0.91	...	77.777	...

A3 feed pipe
 BED 3
 Bed3 sup ring
 Bed 3 Sup grid
 Bed3 dist wt
 Bed3 dist sup w
 BED 4
 Bed1 sup ring
 Bed1 sup grid
 Bed1 dist wt
 Bed1 dist sup w
 Lifting lug1
 Lifting lug2
 Demister
 A2 2''
 P1 2''
 A2 feed pipe
 Demister sup
 B1 2''
 B1 F/M

Total Weight of Each Detail Type:

Packing	71.1
Packing Liquid	135.6
Liquid	174.4
Nozzles	153.4
Lugs	197.0
Weights	186.0

Sum of the Detail Weights 917.6 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	1103.5	1103.5	1103.5	1103.5	1103.5	1103.5
Nozzles	153.4	153.4	153.4	153.4	153.4	153.4
Lugs	197.0	197.0	197.0	197.0	197.0	197.0
Wld Weights	136.0	136.0	136.0	136.0	136.0	136.0
Packing	...	...	...	...	71.1	71.1
Empty Weights	...	...	...	50.0	50.0	...
Test Weights	...	50.0	...	...	...	...
Ope Weights	...	...	...	...	...	50.0
Ope. Liquid	...	...	...	...	...	174.4
Packing Liq	...	...	...	...	...	135.6
Test Liquid	...	728.8	...	...	...	...
Totals	1589.9	2368.7	1589.9	1639.9	1711.1	2021.1

Field Installation Options:

* Packing installed after lifting.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	1589.9 kg
Shop Test Wt.	- Fabricated weight + water (full)	2368.7 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	1589.9 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	1639.9 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	1711.1 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	2021.1 kg

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	397.9 kg
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	407.8 kg
Mass of the Upper 1/3 of the Vertical Vessel (empty)	374.2 kg

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	3637.68	3360.18
20	30	18353.2	17630.6
30	40	9858.59	9309.19
40	50	601.992	547.265
50	60	1877.27	701.116
60	70	1877.27	701.116
70	80	20734.5	19226.5
80	90	9519.7	8827.36
90	100	1868.46	701.116
100	110	1868.46	701.116
110	120	20734.5	19226.5
120	130	7446.25	6904.7
130	140	1868.46	701.116
140	150	1868.46	701.116
150	160	1727.88	1602.21
160	170	825.766	759.332
170	180	13457.5	12121.3
180	190	1140.54	393.921
190	200	1140.54	393.921
200	210	13107.6	11808.9
210	220	1140.54	393.921
220	230	1140.54	393.921
230	240	4007.46	3610.4
240	250	1140.54	393.921
250	260	911.555	...

Total	141855.3	121110.8 cm ²
Total	152.7	130.4 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg	Total. Ele. Oper. Wgt. kg	Total. Ele. Hydro. Wgt. kg	Total Dtl. Offset Mom. kgf-m.	Oper. Wgt. No Liquid kg
10	20	53.6483	77.0996	78.3338	...	53.6483
20	30	203.675	354.644	418.672	11.0174	203.675
30	40	94.9347	94.9347	178.381	0.4575	94.9347
40	50	4.61264	4.61264	7.34774	...	4.61264
50	60	34.2434	34.2434	37.7946	...	34.2434
60	70	33.8151	33.8151	37.3663	...	33.8151
70	Lugs	29.1119	41.4118	48.7142	1.91375	33.3577
Lugs	80	116.447	165.647	194.857	...	133.431
80	90	115.909	135.461	160.908	...	122.658
90	100	33.3903	33.3903	36.9415	...	33.3903
100	110	33.3903	33.3903	36.9415	...	33.3903
110	120	135.309	203.326	233.321	...	158.788
120	130	53.0788	66.1134	88.277	...	57.5782
130	140	33.3903	33.3903	36.9415	...	33.3903
140	150	33.3903	33.3903	36.9415	...	33.3903
150	160	35.442	35.442	43.6096	0.14019	35.442
160	170	4.96946	4.96946	8.36977	...	4.96946
170	180	85.8638	107.632	124.332	...	93.3781
180	190	16.9549	16.9549	18.2176	...	16.9549
190	200	16.9549	16.9549	18.2176	...	16.9549
200	210	144.002	165.873	181.566	...	151.551
210	220	16.9549	16.9549	18.2176	...	16.9549
220	230	16.9549	16.9549	18.2176	...	16.9549



230	240	52.4169	53.438	63.9018	0.0057845	52.5316
240	250	16.9549	16.9549	18.2176	...	16.9549
250	260	27.1139	27.1139	27.1139	...	27.1139

Empty Support Force + the Sum of the Y forces 1666.26 kgf
 Operating Support Force + the Sum of the Y forces 2183.07 kgf
 Hydro Support Force + the Sum of the Y forces 2530.68 kgf

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg	Cumulative Oper. Wgt. kg	Cumulative Hydro. Wgt. kg
10	20	...	...	...
20	30	-53.6483	-77.0996	-78.3338
30	40	-257.323	-431.743	-497.006
40	50	-352.258	-526.678	-675.386
50	60	-356.871	-531.291	-682.734
60	70	-391.114	-565.534	-720.529
70	Lugs	-424.929	-599.349	-757.895
Lugs	80	1055.78	1183.35	1365.11
80	90	922.347	1017.7	1170.25
90	100	799.689	882.244	1009.34
100	110	766.298	848.854	972.403
110	120	732.908	815.464	935.462
120	130	574.12	612.137	702.141
130	140	516.542	546.024	613.864
140	150	483.151	512.634	576.923
150	160	449.761	479.243	539.981
160	170	414.319	443.801	496.372
170	180	409.35	438.832	488.002
180	190	315.972	331.2	363.67
190	200	299.017	314.245	345.452
200	210	282.062	297.29	327.235
210	220	130.51	131.417	145.668
220	230	113.555	114.462	127.451
230	240	96.6004	97.5068	109.233
240	250	44.0688	44.0688	45.3315
250	260	27.1139	27.1139	27.1139

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. kgf-m.	Cumulative Oper. Mom. kgf-m.	Cumulative Hydro. Mom. kgf-m.
10	20	...	...	...
20	30	11.0174	11.0174	11.0174
30	40	11.4749	11.4749	11.4749
40	50	11.4749	11.4749	11.4749
50	60	11.4749	11.4749	11.4749
60	70	11.4749	11.4749	11.4749
70	Lugs	13.3887	13.3887	13.3887
Lugs	80	0.14597	0.14597	0.14597
80	90	0.14597	0.14597	0.14597
90	100	0.14597	0.14597	0.14597
100	110	0.14597	0.14597	0.14597
110	120	0.14597	0.14597	0.14597
120	130	0.14597	0.14597	0.14597
130	140	0.14597	0.14597	0.14597
140	150	0.14597	0.14597	0.14597
150	160	0.14597	0.14597	0.14597
160	170	0.0057845	0.0057845	0.0057845
170	180	0.0057845	0.0057845	0.0057845



180	190	0.0057845	0.0057845	0.0057845
190	200	0.0057845	0.0057845	0.0057845
200	210	0.0057845	0.0057845	0.0057845
210	220	0.0057845	0.0057845	0.0057845
220	230	0.0057845	0.0057845	0.0057845
230	240	0.0057845	0.0057845	0.0057845
240	250	...	...	...
250	260	...	...	...

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Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Rating Ope.	Flange Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	Max Pressure DNV
B2 2''	87.8	101.4	85	600	GR 2.1	64.6	67.2	87.8	67.2
L2 2''	74.65	83.92	85	600	GR 2.3	...	...	...	...
L4 2''	87.83	101.36	85	600	GR 2.1	...	...	...	...
L6 2''	87.83	101.36	85	600	GR 2.1	...	...	...	...
L1 2''	87.83	101.36	85	600	GR 2.1	...	...	...	...
L3 2''	74.65	83.92	85	600	GR 2.3	...	...	...	...
L5 2''	87.83	101.36	85	600	GR 2.1	...	...	...	...
P3-2''	87.83	101.36	85	600	GR 2.1	...	...	...	...
A4 2''	87.83	101.36	85	600	GR 2.1	...	...	...	...
A1 2''	87.83	101.36	85	600	GR 2.1	...	...	...	...
A3 2''	87.83	101.36	85	600	GR 2.1	...	...	...	...
P2 2''	74.65	83.92	85	600	GR 2.3	...	...	...	...
A2 2''	87.83	101.36	85	600	GR 2.1	...	...	...	...
P1 2''	87.83	101.36	85	600	GR 2.1	...	...	...	...
B1 2''	87.83	101.36	85	600	GR 2.1	...	...	...	...
Min Rating	67.166	83.925, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME kgf-m.	Axial Force FE kgf	MAWP PD kgf/cm ²	Maximum Pressure Pmax kgf/cm ²	Pass Fail
B2 2''	77.851	0.50	53	326	39.3	67.2	Pass

Evaluating Flange on Nozzle [B2 2'']:
 $= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$
 $= 16(53) + 4 \cdot \text{abs}(326) \cdot 77.851 \leq \pi \cdot 77.851^3 \cdot ((87.833 - 39.300) + .500 \cdot 87.833)$
 $= 0.95734E+03 \leq 0.13704E+04$ Check Passes

Maximum Pressure for Flange on Nozzle B2 2'' [Pmax]:
 $= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$
 $= \min(87.833, 0.500 \cdot 87.833 + 87.833 - 957.3 / (\pi \cdot 77.851^3))$
 $= \min(87.833, 67.166)$
 $= 67.166 \text{ kgf/cm}^2$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 0.58059 Hz.

The Natural Frequency for the Vessel (Ope...) is 0.5657 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. kgf	X,Z Moment and For Res kgf-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	...	...
130	140	...	...
140	150	...	...
150	160	...	...
160	170	...	...
170	180	...	...
180	190	...	...
190	200	...	...
200	210	...	...
210	220	...	...
220	230	...	...
230	240	...	...
240	250	...	...
250	260	448.658	72.2908

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. kgf	X,Z Moment and For Res kgf-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	...	...
130	140	...	...
140	150	...	...
150	160	...	...
160	170	...	...
170	180	...	...
180	190	...	...
190	200	...	...
200	210	...	...
210	220	...	...
220	230	...	...
230	240	...	...
240	250	...	...
250	260	448.658	72.2908

User Input Forces and Moments:

From Distance - - - - Forces - - - - - Moments - - - -



Node	From	Fx	Fy	Fz	Mx	My	Mz
250	3.00	317.	-359.	317.	42.		42.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 208.8 km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	0 cm
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 cm
Distance Upwind of Crest Lh	0 cm
Distance from Crest to the Vessel x	0 cm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 cm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Dynamic Gust-Effect Factor, Operating Case [Gf]:

$$= 0.925[(1+1.7 \cdot I_{zbar} \cdot \sqrt{(gQ^2 \cdot Q^2 + gR^2 \cdot R^2)}) / (1+1.7 \cdot gV \cdot I_{zbar})]$$

$$= 0.925[(1+1.7 \cdot 0.15 \cdot \sqrt{(3.4^2 \cdot 0.94^2 + 4.05^2 \cdot 2.07^2)}) / (1+1.7 \cdot 3.4 \cdot 0.15)]$$

$$= 1.622$$

Natural Frequency of Vessel (Operating)	0.566 Hz
Natural Frequency of Vessel (Empty)	0.581 Hz
Natural Frequency of Vessel (Test)	0.542 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.700
Structure Height to Diameter ratio	89.730
Height to top of Structure	1908.327 cm

This is classified as a flexible structure. Dynamic analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: D from Table 26.11-1
 $\alpha = 11.5 : Zg = 58979 \text{ cm}$

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 10.2 + 0 = 10.2 \text{ cm}$
 $= 0.34 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (0.34 ft.) $< 15 \text{ ft.}$
 $= 2.41(15 / Zg)^{2/\alpha}$
 $= 2.41(15 / 1935)^{2/11.5}$
 $= 1.04$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1, \text{ per Table 26.6-1}$

As there is No Hill Present: $[Kzt]$:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0 \cdot 0 \cdot 0)^2$$

$$= 1$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.04 \cdot 1 \cdot 1 \cdot 1 \cdot 130^2)$$

$$= 44.6 \text{ psf [218] kgf/m}^2$$

Force on the first element [F]:

$$= qz \cdot Gf \cdot Cf \cdot \text{WindArea}$$

$$= 44.6 \cdot 1.62 \cdot 0.7 \cdot 1.42$$

$$= 71.8 \text{ lbs. [32.6] kgf}$$

Element	Hgt (z) cm	K1	K2	K3	Kz	Kzt	qz kgf/m ²
Bottom head	10.2	0.000	0.000	0.000	1.035	1.000	217.781
Shell-1	74.7	0.000	0.000	0.000	1.035	1.000	217.781
conical-1	168.8	0.000	0.000	0.000	1.035	1.000	217.781
Shell-2	221.6	0.000	0.000	0.000	1.035	1.000	217.781
H6-2	231.7	0.000	0.000	0.000	1.035	1.000	217.781
H6-1	242.7	0.000	0.000	0.000	1.035	1.000	217.781
SHELL-3	398.2	0.000	0.000	0.000	1.035	1.000	217.781
SHELL-4	617.1	0.000	0.000	0.000	1.090	1.000	229.440
H5-2	691.7	0.000	0.000	0.000	1.112	1.000	234.044
H5-1	702.7	0.000	0.000	0.000	1.115	1.000	234.685
Shell-5	858.2	0.000	0.000	0.000	1.155	1.000	242.987
Shell-6	1062.1	0.000	0.000	0.000	1.198	1.000	252.163
H4-2	1121.7	0.000	0.000	0.000	1.210	1.000	254.571
H4-1	1132.7	0.000	0.000	0.000	1.212	1.000	255.003
Shell-6	1150.7	0.000	0.000	0.000	1.215	1.000	255.702
conical -2	1170.2	0.000	0.000	0.000	1.219	1.000	256.451
shell-7	1330.3	0.000	0.000	0.000	1.246	1.000	262.234
H3-2	1487.4	0.000	0.000	0.000	1.271	1.000	267.374
H3-1	1497.1	0.000	0.000	0.000	1.272	1.000	267.676
shell-8	1649.6	0.000	0.000	0.000	1.294	1.000	272.230
H2-2	1802.4	0.000	0.000	0.000	1.314	1.000	276.457
H2-1	1812.1	0.000	0.000	0.000	1.315	1.000	276.715
SHELL9	1862.1	0.000	0.000	0.000	1.321	1.000	278.028
H1-2	1912.4	0.000	0.000	0.000	1.328	1.000	279.320
H1-1	1918.9	0.000	0.000	0.000	1.328	1.000	279.485

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, cm
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, cm
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of (D/L²)(104)
- L - Total height of structure cm
- Lc - Total length of conical section(s) of vessel, cm
- tb - Uncorroded plate thickness at bottom of vessel, mm
- V30 - Design Wind Speed provided by user, km/hr
- Vc - Critical wind velocity, km/hr
- Vw - Maximum wind speed at top of structure, km/hr
- W - Total corroded weight of structure, kgf
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. kgf
- Z - Maximum amplitude of vibration at top of vessel, mm
- Dl - Logarithmic decrement (taken as 0.03 for Welded Structures)
- Vp - Vib. Chance, <= 0.000320369 (High); 0.000320369 < 0.000400461 (Probable)

P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5

:: 99.1/1908 = 0.052

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)

:: $[0.7/62.6^2] \cdot 10^4 = 1.78$

Compute the vibration possibility. If $V_p > 0.000400461$ no chance. $[V_p]$:

= $W / (L \cdot Dr^2)$

= $2021 / (1908 \cdot 16.3^2)$

= 0.39910E-02

Since V_p is > 0.000400461 no further vibration analysis is required !

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation cm	Pressure kgf/m ²	Force kgf
Vortex breake	0.00	-12.70	217.78	0.00
Bed1 sup ring	0.00	275.97	217.78	0.00
Bed1 sup grid	0.00	276.47	217.78	0.00
Cleat wt	0.00	635.97	230.61	0.00
Bed 1 Dist wt	0.00	625.97	229.99	0.00
Bed1 Dist sup	0.00	623.97	229.87	0.00
Bed2 sup ring	0.00	707.97	234.97	0.00
Bed 2 Sup gri	0.00	708.47	234.99	0.00
A3 dist wt	0.00	1055.97	251.89	0.00
A3 dist sup w	0.00	1053.97	251.80	0.00
A3 feed pipe	0.00	1140.00	255.29	0.00
Bed3 sup ring	0.00	1174.17	256.59	0.00
Bed 3 Sup gri	0.00	1175.67	256.65	0.00
Bed3 dist wt	0.00	1426.17	265.37	0.00
Bed3 dist sup	0.00	1424.17	265.31	0.00
Bed1 sup ring	0.00	1497.70	267.70	0.00
Bed1 sup grid	0.00	1499.70	267.75	0.00
Bed1 dist wt	0.00	1749.70	275.00	0.00
Bed1 dist sup	0.00	1747.70	274.94	0.00
Lifting lug1	0.00	1755.00	275.15	0.00
Lifting lug2	0.00	1755.00	275.15	0.00
A2 feed pipe	0.00	1815.00	276.79	0.00
Demister sup	0.00	1874.70	278.35	0.00

The Natural Frequency for the Vessel (Ope...) is 0.5657 Hz.

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm ²	Wind Pressure kgf/m ²	Element Wind Load kgf
10	20	10.2195	85.8523	1317.63	217.781	35.8496
20	30	74.7251	85.852	9872.98	217.781	268.62
30	40	168.79	61.4678	5224.77	217.781	142.153
40	50	221.58	37.18	323.838	217.781	8.81085
50	60	231.739	34.2549	375.873	217.781	10.2266
60	70	242.712	34.2549	375.873	217.781	10.2266
70	80	398.198	37.18	11154	217.781	303.474
80	90	617.067	37.18	5121.06	229.44	146.791
90	100	691.739	34.2549	375.873	234.044	10.9902
100	110	702.712	34.2549	375.873	234.685	11.0204
110	120	858.198	37.18	11154	242.987	338.597
120	130	1062.07	37.18	4005.66	252.163	126.19
130	140	1121.74	34.2549	375.873	254.571	11.9542
140	150	1132.71	34.2549	375.873	255.003	11.9744
150	160	1150.7	37.18	929.5	255.702	29.693
160	170	1170.2	30.5045	463.668	256.451	14.8553

170	180	1330.3	23.829	7239.39	262.234	237.171
180	190	1487.37	21.6776	210.334	267.374	7.02582
190	200	1497.08	21.6776	210.334	267.676	7.03377
200	210	1649.57	23.8797	7051.13	272.23	239.808
210	220	1802.37	21.6776	210.334	276.457	7.2645
220	230	1812.08	21.6776	210.334	276.715	7.27129
230	240	1862.07	23.8797	2155.79	278.028	74.8796
240	250	1912.37	21.6776	210.334	279.32	7.33973
250	260	1918.89	47.2186	157.115	279.485	5.48588

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	1.000
Site Class	D
Component Elevation Ratio	z/h
Amplification Factor	Ap
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR]
Component strength factor	[Rpo]
Overstrength factor	[Omega0]
Response Acceleration Factor	[SMS]
Response Acceleration Factor	[SM1]
Building Importance Factor	[Ie]
Site Latitude	0.000 deg
Site Longitude	0.000 deg

Seismic Analysis Results:

$$\text{SDS} = 2/3 * \text{Sms} = 2/3 * 0.33 = 0.22$$

$$\text{SD1} = 2/3 * \text{Sm1} = 2/3 * 0.14 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $\text{Ct} \cdot \text{hn}^x$ where $\text{Ct} = 0.020$, $x = 0.75$ and hn = Structural Height (ft.)
 = $0.020(62.60.75)$
 = 0.445 seconds

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/0.57)
 = 1.768

Check the Value of T which is the smaller of $\text{Cu} \cdot \text{Ta}$ and T:
 = Minimum Value of ($1.7 \cdot 0.45$, 1.77) per 12.8.2
 = 0.757

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 = $\text{Sds} / (\text{R} / \text{I})$
 = $0.22 / (1/1.5)$
 = 0.326

Maximum value of Cs per equation 12.8-3 [Cs]:
 = $\text{Sd1} / (\text{T} (\text{R} / \text{Ie}))$
 = $0.093 / (0.76 (1/1.5))$
 = 0.184

Minimum value of Cs per equation 15.4-1 [Cs]:
 = $\max((0.044 \cdot \text{SDS} \cdot \text{Ie}), 0.030)$
 = $\max((0.044 \cdot 0.22 \cdot 1.5), 0.030)$
 = 0.030

Total Base Shear [V]:
 = $\text{Cs} \cdot \text{W}$ (Equation 12.8-1):
 = $0.18 \cdot 1824$
 = 335.049 kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot \text{Sds} \cdot \text{W}$
 = $0.2 \cdot 0.22 \cdot 1824$

= 79.3 kgf

Distribute the Base shear force to each element according to the equations
 $F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor
 $C_{vx} = W_x * h_x / k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related
 to the period of Vibration.

In this case, the value of k was 1.63

The Natural Frequency for the Vessel (Ope...) is 0.5657 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height cm	Earthquake Weight kgf	Element Ope Load kgf	Element Emp Load kgf	Element Vertical Load kgf
10	20	2.5	77.0996	0.00098041	0.00070568	-3.35126
20	30	62.5	354.644	0.86732	0.47901	-18.7664
30	40	162.5	94.9347	1.10616	1.04097	-22.8929
40	50	209.355	4.61264	0.081305	0.076075	-23.0934
50	60	219.514	34.2434	0.65218	0.60958	-24.5819
60	70	230.487	33.8151	0.69745	0.65117	-26.0517
70	Lugs	295.973	41.4118	1.2852	0.83876	331.106
Lugs	80	415.973	165.647	8.96473	5.80569	-307.521
80	90	604.842	135.461	13.5143	10.5629	-314.721
90	100	679.514	33.3903	4.02906	3.67066	-320.609
100	110	690.487	33.3903	4.1359	3.76663	-322.061
110	120	845.973	203.326	35.0954	21.1724	-323.512
120	130	1049.84	66.1134	16.2386	11.7609	-332.35
130	140	1109.51	33.3903	8.97651	8.08767	-335.224
140	150	1120.49	33.3903	9.12201	8.21693	-336.675
150	160	1138.47	35.442	9.93775	8.9485	-338.127
160	170	1158.57	4.96946	1.43383	1.29059	-339.667
170	180	1318.08	107.632	38.3402	27.45	-339.883
180	190	1475.15	16.9549	7.25932	6.49843	-344.562
190	200	1484.85	16.9549	7.33749	6.56743	-345.299
200	210	1637.34	165.873	84.216	65.2939	-346.036
210	220	1790.15	16.9549	9.95925	8.87636	-353.245
220	230	1799.85	16.9549	10.0476	8.954	-353.982
230	240	1849.84	53.438	33.1174	28.9309	-354.719
240	250	1900.15	16.9549	10.9785	9.77153	-357.042
250	260	1906.66	27.1139	17.6549	15.7128	-357.779



Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support cm	Cumulative Shr Wind Cas kgf	Cumulative Shr Eqk Cas kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	298.929	...	...	...	...
20	30	233.473	...	...	...	...
30	40	133.473	...	...	...	...
40	50	86.6181	...	...	...	...
50	60	76.4592	...	...	...	...
60	70	65.4864	...	...	...	...
70	Lugs	30	...	...	...	...
Lugs	80	120	448.658	448.658	7291.33	7291.33
80	90	308.868	448.658	448.658	6214.55	6214.55
90	100	383.541	448.658	448.658	5595.16	5595.16
100	110	394.514	448.658	448.658	5545.93	5545.93
110	120	550	448.658	448.658	5496.69	5496.69
120	130	753.869	448.658	448.658	4150.72	4150.72
130	140	813.541	448.658	448.658	3665.92	3665.92
140	150	824.514	448.658	448.658	3616.69	3616.69
150	160	842.5	448.658	448.658	3567.46	3567.46
160	170	862.6	448.658	448.658	3455.3	3455.3
170	180	1022.1	448.658	448.658	3387.1	3387.1
180	190	1179.18	448.658	448.658	2022.62	2022.62
190	200	1188.88	448.658	448.658	1979.09	1979.09
200	210	1341.37	448.658	448.658	1935.56	1935.56
210	220	1494.18	448.658	448.658	609.348	609.348
220	230	1503.88	448.658	448.658	565.816	565.816
230	240	1553.87	448.658	448.658	522.284	522.284
240	250	1604.18	448.658	448.658	115.823	115.823
250	260	1610.69	448.658	448.658	72.2908	72.2908

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

Note:

Review the Vessel Design Summary for the cumulative shear force and bending moment on the support.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support cm	Cumulative Wind Shear kgf	Earthquake Shear kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	298.929	...	...	...	...
20	30	233.473	35.8496	0.00098041	2.85204	0.77997E-04
30	40	133.473	304.47	0.8683	198.536	0.49992
40	50	86.6181	446.623	1.97446	517.751	1.70809
50	60	76.4592	455.434	2.05577	557.035	1.88361
60	70	65.4864	465.661	2.70795	609.016	2.15149
70	Lugs	30	475.887	3.4054	660.674	2.4869
Lugs	80	120	2074.7	335.049	10648.7	3608.48
80	90	308.868	1295.34	321.394	8212.92	2831.3
90	100	383.541	1148.55	307.88	6526.2	2396.95
100	110	394.514	1137.56	303.851	6400.77	2363.38
110	120	550	1126.54	299.715	6276.55	2330.27
120	130	753.869	787.946	264.619	3404.81	1483.77
130	140	813.541	661.757	248.381	2621.78	1206.63
140	150	824.514	649.802	239.404	2549.82	1179.87
150	160	842.5	637.828	230.282	2479.17	1154.1
160	170	862.6	608.135	220.345	2323.43	1097.77
170	180	1022.1	593.28	218.911	2232.12	1064.39
180	190	1179.18	356.109	180.57	788.839	456.992
190	200	1188.88	349.083	173.311	754.627	439.824
200	210	1341.37	342.049	165.974	721.098	423.364
210	220	1494.18	102.241	81.7576	64.829	57.357
220	230	1503.88	94.9765	71.7984	55.2612	49.9074
230	240	1553.87	87.7053	61.7508	46.3986	43.4284
240	250	1604.18	12.8256	28.6334	0.97962	2.53933
250	260	1610.69	5.48588	17.6549	0.091264	0.29371

Note:
 The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

Note:
 Review the Vessel Design Summary for the cumulative shear force and bending moment on the support.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear kgf	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	...	17.426	17.6129	...
20	30	...	12.9416	17.2392	...
30	40	...	5.46764	8.64409	...
40	50	...	1.9657	2.2912	...
50	60	...	1.23016	1.64021	...
60	70	...	0.41005	0.82011	...
70	Lugs	...	-1.68547	...	...
Lugs	80	448.658	15.3615	49.383	0.019878
80	90	448.658	75.8124	106.968	0.033602
90	100	448.658	109.633	112.302	0.033738
100	110	448.658	114.975	117.652	0.033873
110	120	448.658	200.894	302.452	0.058295
120	130	448.658	342.903	385.268	0.0654
130	140	448.658	389.678	394.09	0.065489
140	150	448.658	398.504	402.921	0.065577
150	160	448.658	413.033	423.238	0.067058
160	170	448.658	429.505	435.842	0.068888
170	180	448.658	590.987	793.728	0.13155
180	190	448.658	800.837	807.951	0.13172
190	200	448.658	815.069	822.19	0.13189
200	210	448.658	1053.13	1305.07	0.16034
210	220	448.658	1313.58	1322.08	0.16039
220	230	448.658	1330.59	1339.1	0.16044
230	240	448.658	1418.62	1498.63	0.16262
240	250	448.658	1507.24	1515.86	0.16263
250	260	448.658	1518.81	1521.76	0.16263

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)

Allowable deflection : 95.416 Actual deflection : 1521.765 mm

**** Warning: Allowable Deflection Criteria Exceeded ****

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear kgf	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	...	27.2585	27.5531	0.00031646
20	30	35.8496	20.1945	26.964	0.00031644
30	40	304.47	8.46184	13.4365	0.00027709
40	50	446.623	3.02112	3.52416	0.00010391
50	60	455.434	1.88905	2.51962	0.32869E-04
60	70	465.661	0.62922	1.25891	0.17084E-04
70	Lugs	475.887	-2.60072	...	...
Lugs	80	2074.7	24.4751	77.2922	0.030133
80	90	1295.34	117.063	162.827	0.047544
90	100	1148.55	166.695	170.568	0.047705
100	110	1137.56	174.446	178.328	0.047863
110	120	1126.54	295.609	431.211	0.072399
120	130	787.946	483.377	537.039	0.077959
130	140	661.757	542.575	548.113	0.078023
140	150	649.802	553.652	559.194	0.078086
150	160	637.828	571.853	584.578	0.079118
160	170	608.135	592.357	600.184	0.080359
170	180	593.28	774.527	974.607	0.11459



180	190	356.109	981.28	987.954	0.11466
190	200	349.083	994.63	1001.31	0.11473
200	210	342.049	1209.31	1422.69	0.12234
210	220	102.241	1429.73	1436.78	0.12234
220	230	94.9765	1443.83	1450.88	0.12235
230	240	87.7053	1516.48	1582.11	0.12247
240	250	12.8256	1589.16	1596.21	0.12247
250	260	5.48588	1598.63	1601.05	0.12247

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	8.7185	54.4906
20	30	8.71846	54.4904
30	40	6.2422	39.0137
40	50	3.77571	23.5982
50	60	3.47867	21.7417
60	70	3.47867	21.7417
70	80	3.77571	23.5982
80	90	3.77571	23.5982
90	100	3.47867	21.7417
100	110	3.47867	21.7417
110	120	3.77571	23.5982
120	130	3.77571	23.5982
130	140	3.47867	21.7417
140	150	3.47867	21.7417
150	160	3.77571	23.5982
160	170	3.0978	19.3612
170	180	2.41989	15.1243
180	190	2.20141	13.7588
190	200	2.20141	13.7588
200	210	2.42504	15.1565
210	220	2.20141	13.7588
220	230	2.20141	13.7588
230	240	2.42504	15.1565
240	250	2.20141	13.7588
250	260	4.79515	29.9697

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)

Allowable deflection : 95.416 Actual deflection : 1601.052 mm

**** Warning: Allowable Deflection Criteria Exceeded ****

Total Deflection in the Operating Condition + Applied Forces:

Allowable deflection : 95.416 Actual deflection : 3122.817 mm

**** Warning: Allowable Deflection Criteria Exceeded ****

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	148.779	148.779	1820147	1820147
20	30	156.451	156.451	1910244	1910244
30	40	158.635	158.635	1935823	1935823
40	50	65.9734	65.9734	331367	331367
50	60	65.9734	65.9734	331367	331367
60	70	53.2814	53.2814	272510	272510
70	80	53.2814	53.2814	272510	272510
80	90	53.2814	53.2814	272510	272510
90	100	53.2814	53.2814	272510	272510
100	110	53.2814	53.2814	272510	272510
110	120	53.2814	53.2814	272510	272510
120	130	53.2814	53.2814	272510	272510
130	140	53.2814	53.2814	272510	272510
140	150	53.2814	53.2814	272510	272510
150	160	53.2814	53.2814	272510	272510
160	170	53.7841	53.7841	277763	277763
170	180	29.4682	29.4682	94073.4	94073.4
180	190	29.4682	29.4682	94073.4	94073.4
190	200	29.5341	29.5341	94504.4	94504.4
200	210	29.5341	29.5341	94504.4	94504.4
210	220	29.5341	29.5341	94504.4	94504.4
220	230	29.5341	29.5341	94504.4	94504.4
230	240	29.5341	29.5341	94504.4	94504.4
240	250	29.5341	29.5341	94504.4	94504.4
250	260	29.5341	29.5341	94504.4	94504.4

Longitudinal Allowable Stresses:

From	To	Tensile kgf/cm ²	Hydrotest Tensile kgf/cm ²	Compressive kgf/cm ²	Hydrotest Compressive kgf/cm ²
10	20	1407.14	1872.1	-890.068	-1002.75
20	30	1407.13	1872.1	-897.09	-1010.38
30	40	1196.07	1591.28	-899.021	-1012.47
40	50	1196.07	1591.28	-1007.49	-1126.52
50	60	1407.13	1872.1	-1007.49	-1126.52
60	70	1407.13	1872.1	-978.82	-1097.09
70	Lugs	1196.06	1591.29	-978.82	-1097.09
Lugs	80	1196.06	1591.29	-978.82	-1097.09
80	90	1196.06	1591.29	-978.82	-1097.09
90	100	1407.13	1872.1	-978.82	-1097.09
100	110	1407.13	1872.1	-978.82	-1097.09
110	120	1196.06	1591.29	-978.82	-1097.09
120	130	1196.06	1591.29	-978.82	-1097.09
130	140	1407.13	1872.1	-978.82	-1097.09
140	150	1407.13	1872.1	-978.82	-1097.09
150	160	1196.06	1591.29	-978.82	-1097.09
160	170	1196.06	1591.29	-977.646	-1095.87
170	180	1196.06	1591.29	-1018.67	-1137.84
180	190	1407.13	1872.1	-1018.67	-1137.84
190	200	1407.13	1872.1	-1018.4	-1137.57
200	210	1196.06	1591.29	-1018.4	-1137.57
210	220	1407.13	1872.1	-1018.4	-1137.57
220	230	1407.13	1872.1	-1018.4	-1137.57
230	240	1196.06	1591.29	-1018.4	-1137.57
240	250	1407.13	1872.1	-1018.4	-1137.57
250	260	1407.13	1872.1	-1018.4	-1137.57

Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have been computed in the new and cold condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure kgf/cm ²	Longitudinal Stress External Pressure kgf/cm ²	Longitudinal Stress Hydrotest Pressure kgf/cm ²
10	20	322.945	...	647.231
20	30	305.904	...	613.078
30	40	305.65	...	612.569
40	50	122.362	...	245.231
50	60	...	...	...
60	70	...	...	...
70	80	157.413	...	315.48
80	90	157.413	...	315.48
90	100	...	...	...
100	110	...	...	...
110	120	157.413	...	315.48
120	130	157.413	...	315.48
130	140	...	...	...
140	150	...	...	...
150	160	157.413	...	315.48
160	170	164.786	...	330.257
170	180	110.526	...	221.511
180	190	...	...	...
190	200	...	...	...
200	210	110.799	...	222.058
210	220	...	...	...
220	230	...	...	...
230	240	110.799	...	222.058
240	250	...	...	...
250	260	...	...	...

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty kgf/cm ²	Wght. Str. Operating kgf/cm ²	Wght. Str. Hydrotest kgf/cm ²	Wght. Str. Emp. Mom. kgf/cm ²	Wght. Str. Opr. Mom. kgf/cm ²
10	20	...	...	...	...	...
20	30	0.34291	0.4928	...	0.57675	0.57675
30	40	1.62211	2.72162	...	0.59277	0.59277
40	50	5.3394	7.98319	...	3.46291	3.46291
50	60	5.40932	8.05311	...	3.46291	3.46291
60	70	7.34055	10.6141	...	4.21082	4.21082
70	Lugs	7.9752	11.2488	...	4.91309	4.91309
Lugs	80	-19.8152	-22.2095	...	0.053565	0.053565
80	90	-17.3109	-19.1006	...	0.053565	0.053565
90	100	-15.0088	-16.5582	...	0.053565	0.053565
100	110	-14.3821	-15.9316	...	0.053565	0.053565
110	120	-13.7554	-15.3049	...	0.053565	0.053565
120	130	-10.7753	-11.4888	...	0.053565	0.053565
130	140	-9.69462	-10.248	...	0.053565	0.053565
140	150	-9.06794	-9.62127	...	0.053565	0.053565
150	160	-8.44126	-8.99459	...	0.053565	0.053565
160	170	-7.70339	-8.25155	...	0.0020825	0.0020825
170	180	-13.8913	-14.8918	...	0.006149	0.006149
180	190	-10.7225	-11.2393	...	0.006149	0.006149
190	200	-10.1245	-10.6401	...	0.0061209	0.0061209
200	210	-9.55038	-10.066	...	0.0061209	0.0061209
210	220	-4.41897	-4.44966	...	0.0061209	0.0061209



220	230	-3.84489	-3.87558	...	0.0061209	0.0061209
230	240	-3.27081	-3.3015	...	0.0061209	0.0061209
240	250	-1.49214	-1.49214	...	...	...
250	260	-0.91805	-0.91805	...	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. kgf/cm ²	Bend. Str. Oper. Wind kgf/cm ²	Bend. Str. Oper. Equ. kgf/cm ²	Bend. Str. Hyd. Wind kgf/cm ²	Bend. Str. Hyd. Equ. kgf/cm ²
10	20	...	...	...	...	...
20	30	...	0.1493	0.40831E-05	...	...
30	40	...	10.2559	0.025824	...	...
40	50	...	156.247	0.51547	...	...
50	60	...	168.102	0.56844	...	...
60	70	...	223.484	0.78951	...	...
70	Lugs	...	242.44	0.91259	...	...
Lugs	80	...	3907.63	1324.17	...	...
80	90	...	3013.8	1038.97	...	...
90	100	...	2394.85	879.581	...	...
100	110	...	2348.82	867.265	...	...
110	120	...	2303.24	855.113	...	...
120	130	...	1249.43	544.482	...	...
130	140	...	962.084	442.785	...	...
140	150	...	935.679	432.964	...	...
150	160	...	909.755	423.508	...	...
160	170	...	836.479	395.219	...	...
170	180	...	2372.74	1131.45	...	...
180	190	...	838.536	485.782	...	...
190	200	...	798.51	465.4	...	...
200	210	...	763.031	447.983	...	...
210	220	...	68.5989	60.6924	...	...
220	230	...	58.4747	52.8096	...	...
230	240	...	49.0967	45.9538	...	...
240	250	...	1.03659	2.687	...	...
250	260	...	0.096571	0.31079	...	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case kgf/cm ²	Vortex Shedding Empty Case kgf/cm ²	Vortex Shedding Test Case kgf/cm ²	Earthquake Empty Case kgf/cm ²
10	20	...	...	...	...
20	30	...	...	...	0.29389E-05
30	40	...	...	...	0.014273
40	50	...	...	...	0.33995
50	60	...	...	...	0.38092
60	70	...	...	...	0.54162
70	Lugs	...	...	...	0.64357
Lugs	80	...	...	...	1074.37
80	90	...	...	...	848.132
90	100	...	...	...	721.36
100	110	...	...	...	711.57
110	120	...	...	...	701.93
120	130	...	...	...	452.09
130	140	...	...	...	368.638
140	150	...	...	...	360.562
150	160	...	...	...	352.815
160	170	...	...	...	329.598
170	180	...	...	...	944.304
180	190	...	...	...	413.097
190	200	...	...	...	396.084
200	210	...	...	...	381.626
210	220	...	...	...	53.6482
220	230	...	...	...	46.6864
230	240	...	...	...	40.6399



240	250	...	...	...	2.39146
250	260	...	...	...	0.2766

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind kgf/cm ²	Longitudinal Stress Y Forces Seismic kgf/cm ²
10	20	...	-0.022525
20	30	...	-0.11995
30	40	...	-0.14431
40	50	...	-0.35004
50	60	...	-0.3726
60	70	...	-0.48895
70	Lugs	6.73703	6.2143
Lugs	80	-6.73703	-5.77166
80	90	-6.73703	-5.90679
90	100	-6.73703	-6.0173
100	110	-6.73703	-6.04454
110	120	-6.73703	-6.07178
120	130	-6.73703	-6.23765
130	140	-6.73703	-6.29159
140	150	-6.73703	-6.31883
150	160	-6.73703	-6.34607
160	170	-6.67406	-6.3154
170	180	-12.1812	-11.5339
180	190	-12.1812	-11.6927
190	200	-12.154	-11.6915
200	210	-12.154	-11.7165
210	220	-12.154	-11.9606
220	230	-12.154	-11.9856
230	240	-12.154	-12.0105
240	250	-12.154	-12.0892
250	260	-12.154	-12.1141

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded kgf/cm ²	Earthquake For/Mom Corroded kgf/cm ²	Wind For/Mom No Corrosion kgf/cm ²	Earthquake For/Mom No Corrosion kgf/cm ²
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...
50	60	...	...	...	...
60	70	...	...	...	...
70	Lugs	...	...	...	...
Lugs	80	2675.62	2675.62	2675.62	2675.62
80	90	2280.49	2280.49	2280.49	2280.49
90	100	2053.19	2053.19	2053.19	2053.19
100	110	2035.13	2035.13	2035.13	2035.13
110	120	2017.06	2017.06	2017.06	2017.06
120	130	1523.14	1523.14	1523.14	1523.14
130	140	1345.24	1345.24	1345.24	1345.24
140	150	1327.18	1327.18	1327.18	1327.18
150	160	1309.11	1309.11	1309.11	1309.11
160	170	1243.97	1243.97	1243.97	1243.97
170	180	3600.49	3600.49	3600.49	3600.49
180	190	2150.05	2150.05	2150.05	2150.05
190	200	2094.18	2094.18	2094.18	2094.18
200	210	2048.11	2048.11	2048.11	2048.11
210	220	644.783	644.783	644.783	644.783
220	230	598.719	598.719	598.719	598.719
230	240	552.655	552.655	552.655	552.655
240	250	122.558	122.558	122.558	122.558
250	260	76.4946	76.4946	76.4946	76.4946

Bending Stresses due to Blast Load:

From	To	Bending Stress kgf/cm ²
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	Lugs	...
Lugs	80	...
80	90	...
90	100	...
100	110	...
110	120	...
120	130	...
130	140	...
140	150	...
150	160	...
160	170	...
170	180	...
180	190	...
190	200	...
200	210	...
210	220	...
220	230	...
230	240	...
240	250	...
250	260	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Un-Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	1407.14	0.00	890.07	0.0000	-0.0000
20	1.07	1407.13	-0.38	897.09	0.0008	0.0004
30	12.47	1196.07	-9.23	899.02	0.0104	0.0103
40	165.05	1196.07	-154.37	1007.49	0.1380	0.1532
70	262.07	1196.06	-232.64	978.82	0.2191	0.2377
70	6556.75	1196.06	-6609.85	978.82	5.4819	6.7529 *
80	5270.30	1196.06	-5318.39	978.82	4.4064	5.4335 *
110	4299.86	1196.06	-4340.84	978.82	3.5950	4.4348 *
120	2755.11	1196.06	-2790.14	978.82	2.3035	2.8505 *
150	2203.74	1196.06	-2234.10	978.82	1.8425	2.2824 *
160	2066.08	1196.06	-2094.83	977.65	1.7274	2.1427 *
170	5947.17	1196.06	-5999.31	1018.67	4.9723	5.8893 *
200	2789.45	1196.06	-2832.86	1018.40	2.3322	2.7817 *
230	586.33	1196.06	-617.18	1018.40	0.4902	0.6060

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1407.14	-0.02	890.07		0.0000
20	0.80	1407.13	-0.35	897.09	0.0006	0.0004
30	2.08	1196.07		899.02	0.0017	
40	8.79	1196.07		1007.49	0.0074	
70	19.75	1196.06		978.82	0.0165	



70	3724.46	1196.06	-3775.63	978.82	3.1139	3.8573 *
80	3105.45	1196.06	-3151.89	978.82	2.5964	3.2201 *
110	2699.22	1196.06	-2738.87	978.82	2.2567	2.7981 *
120	1958.27	1196.06	-1992.30	978.82	1.6373	2.0354 *
150	1647.19	1196.06	-1676.77	978.82	1.3772	1.7131 *
160	1559.55	1196.06	-1587.59	977.65	1.3039	1.6239 *
170	4519.37	1196.06	-4570.22	1018.67	3.7785	4.4864 *
200	2408.48	1196.06	-2451.01	1018.40	2.0137	2.4067 *
230	578.02	1196.06	-608.58	1018.40	0.4833	0.5976

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	1407.14	0.00	890.07	0.0000	-0.0000
20	1.22	1407.13	-0.23	897.09	0.0009	0.0003
30	13.57	1196.07	-8.13	899.02	0.0113	0.0090
40	167.69	1196.07	-151.73	1007.49	0.1402	0.1506
70	265.34	1196.06	-229.37	978.82	0.2218	0.2343
70	6554.35	1196.06	-6612.25	978.82	5.4799	6.7553 *
80	5268.51	1196.06	-5320.18	978.82	4.4049	5.4353 *
110	4298.31	1196.06	-4342.39	978.82	3.5937	4.4364 *
120	2754.40	1196.06	-2790.85	978.82	2.3029	2.8512 *
150	2203.19	1196.06	-2234.65	978.82	1.8420	2.2830 *
160	2065.53	1196.06	-2095.38	977.65	1.7269	2.1433 *
170	5946.16	1196.06	-6000.31	1018.67	4.9714	5.8903 *
200	2788.93	1196.06	-2833.37	1018.40	2.3318	2.7822 *
230	586.30	1196.06	-617.21	1018.40	0.4902	0.6061

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1407.14	-0.02	890.07		0.0000
20	0.95	1407.13	-0.20	897.09	0.0007	0.0002
30	3.20	1196.07		899.02	0.0027	
40	11.61	1196.07		1007.49	0.0097	
70	23.29	1196.06		978.82	0.0195	
70	3971.86	1196.06	-4027.82	978.82	3.3208	4.1150 *
80	3294.50	1196.06	-3344.52	978.82	2.7545	3.4169 *
110	2850.85	1196.06	-2893.60	978.82	2.3835	2.9562 *
120	2049.95	1196.06	-2085.41	978.82	1.7139	2.1305 *
150	1717.33	1196.06	-1748.01	978.82	1.4358	1.7858 *
160	1624.63	1196.06	-1653.76	977.65	1.3583	1.6916 *
170	4705.51	1196.06	-4758.37	1018.67	3.9342	4.6711 *
200	2474.32	1196.06	-2517.89	1018.40	2.0687	2.4724 *
230	583.30	1196.06	-613.93	1018.40	0.4877	0.6028

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	1872.10	0.00	1002.75	0.0000	-0.0000
20	0.00	1872.10	0.00	1010.38	0.0000	-0.0000
30	0.00	1591.28	0.00	1012.47	0.0000	-0.0000
40	0.00	1591.28	0.00	1126.52	0.0000	-0.0000
70	0.00	1591.29	0.00	1097.09	0.0000	-0.0000
70	0.00	1591.29	0.00	1097.09	0.0000	-0.0000
80	0.00	1591.29	0.00	1097.09	0.0000	-0.0000
110	0.00	1591.29	0.00	1097.09	0.0000	-0.0000
120	0.00	1591.29	0.00	1097.09	0.0000	-0.0000
150	0.00	1591.29	0.00	1097.09	0.0000	-0.0000
160	0.00	1591.29	0.00	1095.87	0.0000	-0.0000
170	0.00	1591.29	0.00	1137.84	0.0000	-0.0000
200	0.00	1591.29	0.00	1137.57	0.0000	-0.0000
230	0.00	1591.29	0.00	1137.57	0.0000	-0.0000

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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10	0.00	1872.10	0.00	1002.75	0.0000	-0.0000
20	0.00	1872.10	0.00	1010.38	0.0000	-0.0000
30	0.00	1591.28	0.00	1012.47	0.0000	-0.0000
40	0.00	1591.28	0.00	1126.52	0.0000	-0.0000
70	0.00	1591.29	0.00	1097.09	0.0000	-0.0000
70	0.00	1591.29	0.00	1097.09	0.0000	-0.0000
80	0.00	1591.29	0.00	1097.09	0.0000	-0.0000
110	0.00	1591.29	0.00	1097.09	0.0000	-0.0000
120	0.00	1591.29	0.00	1097.09	0.0000	-0.0000
150	0.00	1591.29	0.00	1097.09	0.0000	-0.0000
160	0.00	1591.29	0.00	1095.87	0.0000	-0.0000
170	0.00	1591.29	0.00	1137.84	0.0000	-0.0000
200	0.00	1591.29	0.00	1137.57	0.0000	-0.0000
230	0.00	1591.29	0.00	1137.57	0.0000	-0.0000

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	322.95	1407.14		890.07	0.2295	
20	307.12	1407.13		897.09	0.2183	
30	319.22	1196.07		897.17	0.2669	
40	290.05	1196.07	-29.37	1007.49	0.2425	0.0291
70	422.75	1196.06	-71.95	978.82	0.3535	0.0735
70	6711.77	1196.06	-6454.83	978.82	5.6115	6.5945 *
80	5425.92	1196.06	-5162.77	978.82	4.5365	5.2745 *
110	4455.72	1196.06	-4184.98	978.82	3.7253	4.2755 *
120	2911.81	1196.06	-2633.44	978.82	2.4345	2.6904 *
150	2360.60	1196.06	-2077.24	978.82	1.9736	2.1222 *
160	2230.31	1196.06	-1930.59	961.40	1.8647	2.0081 *
170	6056.69	1196.06	-5889.78	1018.67	5.0639	5.7818 *
200	2899.73	1196.06	-2722.57	1018.40	2.4244	2.6734 *
230	697.10	1196.06	-506.41	1018.40	0.5828	0.4973

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	322.92	1407.14		890.07	0.2295	
20	306.85	1407.13		897.09	0.2181	
30	308.85	1196.07		897.17	0.2582	
40	133.97	1196.07		1007.49	0.1120	
70	180.70	1196.06		978.82	0.1511	
70	4129.27	1196.06	-3870.41	978.82	3.4524	3.9542 *
80	3451.91	1196.06	-3187.10	978.82	2.8861	3.2561 *
110	3008.26	1196.06	-2736.19	978.82	2.5151	2.7954 *
120	2207.37	1196.06	-1927.99	978.82	1.8455	1.9697 *
150	1874.75	1196.06	-1590.60	978.82	1.5674	1.6250 *
160	1789.41	1196.06	-1488.98	961.40	1.4961	1.5488 *
170	4816.04	1196.06	-4647.84	1018.67	4.0266	4.5626 *
200	2585.12	1196.06	-2407.09	1018.40	2.1614	2.3636 *
230	694.10	1196.06	-503.13	1018.40	0.5803	0.4940

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	1407.14	0.00	1002.75	0.0000	-0.0000
20	1.22	1407.13	-0.23	1010.38	0.0009	0.0002
30	13.57	1196.07	-8.13	1010.46	0.0113	0.0080
40	167.69	1196.07	-151.73	1126.52	0.1402	0.1347
70	265.34	1196.06	-229.37	1097.09	0.2218	0.2091
70	6554.35	1196.06	-6612.25	1097.09	5.4799	6.0271 *
80	5268.51	1196.06	-5320.18	1097.09	4.4049	4.8494 *
110	4298.31	1196.06	-4342.39	1097.09	3.5937	3.9581 *
120	2754.40	1196.06	-2790.85	1097.09	2.3029	2.5439 *
150	2203.19	1196.06	-2234.65	1097.09	1.8420	2.0369 *
160	2065.53	1196.06	-2095.38	1078.94	1.7269	1.9421 *
170	5946.16	1196.06	-6000.31	1137.84	4.9714	5.2734 *



200	2788.93	1196.06	-2833.37	1137.57	2.3318	2.4907 *
230	586.30	1196.06	-617.21	1137.57	0.4902	0.5426

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1407.14	-0.02	1002.75		0.0000
20	0.95	1407.13	-0.20	1010.38	0.0007	0.0002
30	3.20	1196.07		1010.46	0.0027	
40	11.61	1196.07		1126.52	0.0097	
70	23.29	1196.06		1097.09	0.0195	
70	3971.86	1196.06	-4027.82	1097.09	3.3208	3.6714 *
80	3294.50	1196.06	-3344.52	1097.09	2.7545	3.0485 *
110	2850.85	1196.06	-2893.60	1097.09	2.3835	2.6375 *
120	2049.95	1196.06	-2085.41	1097.09	1.7139	1.9009 *
150	1717.33	1196.06	-1748.01	1097.09	1.4358	1.5933 *
160	1624.63	1196.06	-1653.76	1078.94	1.3583	1.5328 *
170	4705.51	1196.06	-4758.37	1137.84	3.9342	4.1819 *
200	2474.32	1196.06	-2517.89	1137.57	2.0687	2.2134 *
230	583.30	1196.06	-613.93	1137.57	0.4877	0.5397

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	647.23	1872.10		1002.75	0.3457	
20	613.08	1872.10		1010.38	0.3275	
30	612.57	1591.28		1012.47	0.3850	
40	245.23	1591.28		1126.52	0.1541	
70	315.48	1591.29		1097.09	0.1983	
70	315.48	1591.29		1097.09	0.1983	
80	315.48	1591.29		1097.09	0.1983	
110	315.48	1591.29		1097.09	0.1983	
120	315.48	1591.29		1097.09	0.1983	
150	315.48	1591.29		1097.09	0.1983	
160	330.26	1591.29		1095.87	0.2075	
170	221.51	1591.29		1137.84	0.1392	
200	222.06	1591.29		1137.57	0.1395	
230	222.06	1591.29		1137.57	0.1395	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	647.23	1872.10		1002.75	0.3457	
20	613.08	1872.10		1010.38	0.3275	
30	612.57	1591.28		1012.47	0.3850	
40	245.23	1591.28		1126.52	0.1541	
70	315.48	1591.29		1097.09	0.1983	
70	315.48	1591.29		1097.09	0.1983	
80	315.48	1591.29		1097.09	0.1983	
110	315.48	1591.29		1097.09	0.1983	
120	315.48	1591.29		1097.09	0.1983	
150	315.48	1591.29		1097.09	0.1983	
160	330.26	1591.29		1095.87	0.2075	
170	221.51	1591.29		1137.84	0.1392	
200	222.06	1591.29		1137.57	0.1395	
230	222.06	1591.29		1137.57	0.1395	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	322.95	1407.14		890.07	0.2295	
20	306.82	1407.13		897.09	0.2180	
30	307.86	1196.07		897.17	0.2574	
40	131.16	1196.07		1007.49	0.1097	
70	170.30	1196.06		978.82	0.1424	
70	137.65	1196.06		978.82	0.1151	



80	140.16	1196.06	978.82	0.1172
110	143.71	1196.06	978.82	0.1202
120	146.69	1196.06	978.82	0.1226
150	149.03	1196.06	978.82	0.1246
160	157.08	1196.06	961.40	0.1313
170	96.64	1196.06	1018.67	0.0808
200	101.25	1196.06	1018.40	0.0847
230	107.53	1196.06	1018.40	0.0899

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	322.95	1407.14		890.07	0.2295	
20	306.40	1407.13		897.09	0.2177	
30	308.37	1196.07		897.17	0.2578	
40	130.34	1196.07		1007.49	0.1090	
70	168.66	1196.06		978.82	0.1410	
70	135.20	1196.06		978.82	0.1130	
80	138.31	1196.06		978.82	0.1156	
110	142.11	1196.06		978.82	0.1188	
120	145.92	1196.06		978.82	0.1220	
150	148.42	1196.06		978.82	0.1241	
160	156.53	1196.06		961.40	0.1309	
170	95.63	1196.06		1018.67	0.0800	
200	100.73	1196.06		1018.40	0.0842	
230	107.50	1196.06		1018.40	0.0899	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	322.95	1407.14		890.07	0.2295	
20	306.97	1407.13		897.09	0.2182	
30	308.96	1196.07		897.17	0.2583	
40	133.81	1196.07		1007.49	0.1119	
70	173.58	1196.06		978.82	0.1451	
70	135.26	1196.06		978.82	0.1131	
80	138.37	1196.06		978.82	0.1157	
110	142.16	1196.06		978.82	0.1189	
120	145.98	1196.06		978.82	0.1220	
150	148.47	1196.06		978.82	0.1241	
160	156.54	1196.06		961.40	0.1309	
170	95.64	1196.06		1018.67	0.0800	
200	100.74	1196.06		1018.40	0.0842	
230	107.50	1196.06		1018.40	0.0899	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	322.95	1407.14		890.07	0.2295	
20	306.82	1407.13		897.09	0.2180	
30	307.86	1196.07		897.17	0.2574	
40	131.16	1196.07		1007.49	0.1097	
70	170.30	1196.06		978.82	0.1424	
70	137.65	1196.06		978.82	0.1151	
80	140.16	1196.06		978.82	0.1172	
110	143.71	1196.06		978.82	0.1202	
120	146.69	1196.06		978.82	0.1226	
150	149.03	1196.06		978.82	0.1246	
160	157.08	1196.06		961.40	0.1313	
170	96.64	1196.06		1018.67	0.0808	
200	101.25	1196.06		1018.40	0.0847	
230	107.53	1196.06		1018.40	0.0899	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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10	0.00	1407.14	0.00	890.07	0.0000	-0.0000
20	1.07	1407.13	-0.08	897.09	0.0008	0.0001
30	3.31	1196.07		899.02	0.0028	
40	11.45	1196.07		1007.49	0.0096	
70	16.16	1196.06		978.82	0.0135	
70		1196.06	-22.26	978.82		0.0227
80		1196.06	-19.15	978.82		0.0196
110		1196.06	-15.36	978.82		0.0157
120		1196.06	-11.54	978.82		0.0118
150		1196.06	-9.05	978.82		0.0092
160		1196.06	-8.25	977.65		0.0084
170		1196.06	-14.90	1018.67		0.0146
200		1196.06	-10.07	1018.40		0.0099
230		1196.06	-3.31	1018.40		0.0032

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	322.92	1172.61		741.72	0.2754	
20	306.85	1172.61		747.57	0.2617	
30	308.82	996.72		747.64	0.3098	
40	133.46	996.72		839.58	0.1339	
70	179.79	996.72		815.68	0.1804	
70	2805.11	996.72	-2546.24	815.68	2.8143	3.1216 *
80	2412.94	996.72	-2148.13	815.68	2.4209	2.6335 *
110	2153.15	996.72	-1881.08	815.68	2.1602	2.3061 *
120	1662.88	996.72	-1383.51	815.68	1.6684	1.6961 *
150	1451.24	996.72	-1167.09	815.68	1.4560	1.4308 *
160	1394.19	996.72	-1093.76	801.16	1.3988	1.3652 *
170	3684.59	996.72	-3516.39	848.89	3.6967	4.1423 *
200	2137.14	996.72	-1959.10	848.67	2.1442	2.3084 *
230	648.15	996.72	-457.17	848.67	0.6503	0.5387

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1172.61	-0.02	835.62		0.0000
20	0.95	1172.61	-0.20	841.98	0.0008	0.0002
30	3.17	996.72		842.05	0.0032	
40	11.10	996.72		938.77	0.0111	
70	22.38	996.72		914.24	0.0224	
70	2647.69	996.72	-2703.65	914.24	2.6564	2.9573 *
80	2255.53	996.72	-2305.55	914.24	2.2630	2.5218 *
110	1995.74	996.72	-2038.49	914.24	2.0023	2.2297 *
120	1505.47	996.72	-1540.92	914.24	1.5104	1.6855 *
150	1293.82	996.72	-1324.51	914.24	1.2981	1.4488 *
160	1229.41	996.72	-1258.54	899.12	1.2335	1.3998 *
170	3574.07	996.72	-3626.92	948.20	3.5858	3.8250 *
200	2026.34	996.72	-2069.90	947.98	2.0330	2.1835 *
230	537.35	996.72	-567.97	947.98	0.5391	0.5991

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	322.95	1407.14		890.07	0.2295	
20	306.97	1407.13		897.09	0.2182	
30	308.96	1196.07		897.17	0.2583	
40	133.81	1196.07		1007.49	0.1119	
70	173.58	1196.06		978.82	0.1451	
70	135.26	1196.06		978.82	0.1131	
80	138.37	1196.06		978.82	0.1157	
110	142.16	1196.06		978.82	0.1189	
120	145.98	1196.06		978.82	0.1220	
150	148.47	1196.06		978.82	0.1241	
160	156.54	1196.06		961.40	0.1309	
170	95.64	1196.06		1018.67	0.0800	
200	100.74	1196.06		1018.40	0.0842	

230 107.50 1196.06 1018.40 0.0899

*Absolute Maximum of the all of the Stress Ratio's 6.7553 **

*** Warning: One or more elements were Overstressed due to combined loads !*

Governing Element: SHELL-3
 Governing Load Case 3 : NP+OW+WI+FW+BW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Bottom head	...	...	...	...	...	+ Tens
Bottom head	...	...	...	...	...	- Comp
Shell-1	...	0.5	0.6	0.1	1.2	+ Tens
Shell-1	...	0.5	-0.6	-0.1	-0.2	- Comp
conical-1	...	2.7	0.6	10.3	13.6	+ Tens
conical-1	...	2.7	-0.6	-10.3	-8.1	- Comp
Shell-2	...	8.0	3.5	156.2	167.7	+ Tens
Shell-2	...	8.0	-3.5	-156.2	-151.7	- Comp
SHELL-3	...	18.0	4.9	242.4	265.3	+ Tens
SHELL-3	...	18.0	-4.9	-242.4	-229.4	- Comp
SHELL-3	...	-28.9	0.1	6583.2	6554.4	+ Tens
SHELL-3	...	-28.9	-0.1	-6583.2	-6612.2	- Comp
SHELL-4	...	-25.8	0.1	5294.3	5268.5	+ Tens
SHELL-4	...	-25.8	-0.1	-5294.3	-5320.2	- Comp
Shell-5	...	-22.0	0.1	4320.3	4298.3	+ Tens
Shell-5	...	-22.0	-0.1	-4320.3	-4342.4	- Comp
Shell-6	...	-18.2	0.1	2772.6	2754.4	+ Tens
Shell-6	...	-18.2	-0.1	-2772.6	-2790.8	- Comp
Shell-6	...	-15.7	0.1	2218.9	2203.2	+ Tens
Shell-6	...	-15.7	-0.1	-2218.9	-2234.7	- Comp
conical -2	...	-14.9	...	2080.5	2065.5	+ Tens
conical -2	...	-14.9	...	-2080.5	-2095.4	- Comp
shell-7	...	-27.1	...	5973.2	5946.2	+ Tens
shell-7	...	-27.1	...	-5973.2	-6000.3	- Comp
shell-8	...	-22.2	...	2811.1	2788.9	+ Tens
shell-8	...	-22.2	...	-2811.1	-2833.4	- Comp
SHELL9	...	-15.5	...	601.8	586.3	+ Tens
SHELL9	...	-15.5	...	-601.8	-617.2	- Comp



Shop/Field Installation Options :

Packing is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Packing	833.768 cm
Center of Gravity of the Liquid	40.260 cm
Center of Gravity of the Nozzles	585.604 cm
Center of Gravity of the Lugs	306.973 cm
Center of Gravity of the Added Weights (Operating)	1187.021 cm
Center of Gravity of the Added Weights (Empty)	1187.021 cm
Center of Gravity of Bare Shell New and Cold	762.587 cm
Center of Gravity of Bare Shell Corroded	762.587 cm
Vessel CG in the Operating Condition	678.320 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	739.440 cm
Vessel CG in the Test Condition	673.899 cm

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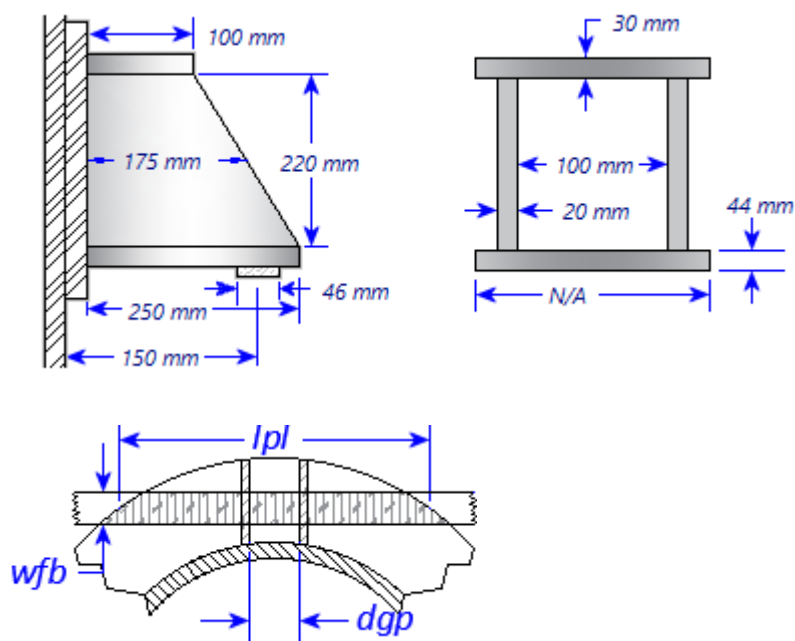
Support Lug Calculations: Operating Case

Input Data: Support Lugs

Type of Geometry : Gussets with Continuous Top Ring

Number of Support Lugs	Nlug	4
Distance from Vessel OD to Lug Contact Point	Dlug	150.0000 mm
Lug Support Force Bearing Width	Wfb	46.0000 mm
Lug Material		SA-516 70 +IT
Lug Yield Stress		2467.60 kgf/cm ²
Radial Width of bottom Support Lug Plate	Wpl	250.0000 mm
Effective Force Bearing Length	Lpl	154.0000 mm
Thickness of bottom Support Lug Plate	Tpl	44.0000 mm
Distance between Gussets	Dgp	100.0000 mm
Mean Width of Gusset Plate	Wgp	175.0000 mm
Height of Gusset Plate	Hgp	220.0000 mm
Thickness of Gusset Plate	Tgp	20.0000 mm
Radial Width of Reinforcing Ring	Wtp	100.0000 mm
Thickness of Top Plate	Ttp	30.0000 mm

Bolt Material	SA-36 +IT
Bolt Allowable Stress at Design Temperature	1529.52 kgf/cm ²
Thread Series	TEMA Metric
Bolt Diameter	72.00 mm



Results for Support Lugs:

Description: Support Lugs

Overturning Moment at Support Lug	17940.0 kgf-m.
Weight Load at the top of one Lug	545.8 kgf

Force on one Lug, Operating Condition [Flug]:

$$= (W/Nlug + Mlug/(Rlug \cdot Nlug/2))$$

$$= (2183/4 + 17940/(26 \cdot 4/2))$$

$$= 35045.73 \text{ kgf}$$

Force on one lug in the Empty Condition [Flugemp]:

$$= (-Wemp/Nlug + Mlugemp/(Rlug \cdot Nlug/2))$$

$$= (-1666/4 + 17940/(26 \cdot 4/2))$$

$$= 34083.40 \text{ kgf}$$

Ring Girder Calculations Per Bednar pp.157-159:



Force P acting in the plane of the ring [P]:

$$= F_{lug} \cdot D_{lug} / (H_{gp} + T_{pl})$$

$$= 35046 \cdot 150 / (220 + 44)$$

$$= 19912.35 \text{ kgf}$$

Angle [α]:

$$= \pi / N_{lug}$$

$$= \pi / 4$$

$$= 45.0 \text{ deg}$$

Maximum bending moment (at the load points) [M1]:

$$= 0.5 \cdot P(OD/2 + W_{tp}/2) \cdot (1/\alpha - \cos(\alpha)/\sin(\alpha))$$

$$= 0.5 \cdot 19912 \cdot 160 (1/0.79 - \cos(45)/\sin(45))$$

$$= 435.27 \text{ kgf-m.}$$

Tangential (axial) thrust induced in the ring [T1]:

(Top ring in compression, bottom ring in tension)

$$= 0.5 \cdot P \cdot \cos(\alpha)/\sin(\alpha)$$

$$= 0.5 \cdot 19912 \cdot \cos(45)/\sin(45)$$

$$= 9956.17 \text{ kgf}$$

Resulting combined stress at the load point [S1]:

$$= T1/(T_{tp} \cdot W_{tp}) + M1/(T_{tp} \cdot W_{tp}^2/6)$$

$$= 9956/(30 \cdot 100) + 435/(30 \cdot 100^2/6)$$

$$= 1202.41 \text{ kgf/cm}^2$$

Bending moment at the midpoints between the loads [M2]:

$$= 0.5 \cdot P(OD/2 + W_{tp}/2) \cdot (1/\sin(\alpha) - 1/\alpha)$$

$$= 0.5 \cdot 19912 \cdot 160 (1/\sin(45) - 1/0.79)$$

$$= 224.57 \text{ kgf-m.}$$

Axial thrust [T2]:

$$= 0.5 \cdot P / \sin(\alpha)$$

$$= 0.5 \cdot 19912/\sin(45)$$

$$= 14080.16 \text{ kgf}$$

Combined maximum stress [S2]:

$$= T2/(W_{tp} \cdot T_{tp}) + M2/(T_{tp} \cdot W_{tp}^2/6)$$

$$= 14080/(100 \cdot 30) + 225/(30 \cdot 100^2/6)$$

$$= 918.48 \text{ kgf/cm}^2$$

Ring Allowable Stress [Sa]:

$$= 2/3 \cdot Y_{lug}$$

$$= 2/3 \cdot 2468$$

$$= 1645.06 \text{ kgf/cm}^2$$

Bottom Support Plate Bending Stress (Point Load) [Spl]:

$$= (F_{lugemp}/2 \cdot D_{gp}/2) / ((W_{pl}/6) \cdot T_{pl}^2) \text{ (Uplift Condition)}$$

$$= (34083/2 \cdot 100/2) / ((250/6) \cdot 44^2)$$

$$= 1056.31 \text{ kgf/cm}^2$$

Bottom Plate Required Thickness (Point Load) 35.2579 mm

Bottom Plate required thickness for Uplift per ADM S 3/4 [trUpl]:

$$= 0.72 \cdot \sqrt{(F_{lugemp} \cdot D_{gp} / ((W_{pl} - dt) \cdot S_{pa}))}$$

$$= 0.72 \cdot \sqrt{(34083 \cdot 100 / ((250 - 75.2) \cdot 1645))}$$

$$= 24.79 \text{ mm}$$

Bearing Area [Ba]:

$$= L_{pl} \cdot W_{fb}$$

$$= 154 \cdot 46$$

$$= 70.84 \text{ cm}^2$$

Bending Stress in bottom Plate (Unif. Load) Per Bednar p.156 [Spl2]:

$$= \beta_1 \cdot F_{lug}/Ba \cdot W_{fb}^2 / T_{pl}^2 \text{ per Roark \& Young 5th Ed.}$$

$$= 2.11 \cdot 35046/70.8 \cdot 46^2/44^2$$

$$= 1138.20 \text{ kgf/cm}^2$$

Bottom Plate Required Thickness (Uniform Load) 36.5991 mm



Bottom Plate Required Thickness based on ADM S 3/4 [trAD]:

$$= 0.71 \cdot D_{gp} \cdot \sqrt{(F_{lug} / (L_{pl} \cdot W_{fb}) / S_{pa}})$$

$$= 0.71 \cdot 100 \cdot \sqrt{(35046 / (154 \cdot 46) / 1645)}$$

$$= 38.935 \text{ mm}$$

If using the AD Code recommendations, the force bearing width (Wfb) must be greater than or equal to 1/3 of the bottom plate radial width (Wpl) plus the pad thickness (Padthk), if there is a pad.

Bottom Support Plate Allowable Stress [Spa]:

$$= 2/3 \cdot Y_{lug}$$

$$= 2/3 \cdot 2468$$

$$= 1645.06 \text{ kgf/cm}^2$$

Gusset Plate Axial Stress (Force / Gusset Plate Area) [Sgp]:

$$= (F_{lug} / 2) / (W_{gp} \cdot T_{gp})$$

$$= (35046 / 2) / (175 \cdot 20)$$

$$= 500.65 \text{ kgf/cm}^2$$

Check required Gusset Plate Thickness:

Required Thickness of Gussets per AISC: 10.1441 mm

Gusset Plate Allowable Stress [Sga]:

$$= (1 - (K_{lr}^2 / (2 \cdot C_c^2))) F_y / (5/3 + 3 \cdot (K_{lr} / (8 \cdot C_c)) - (K_{lr}^3 / (8 \cdot C_c^3)))$$

$$= (1 - (44^2 / (2 \cdot 125^2))) 2468 /$$

$$((5/3 + 3 \cdot (44 / (8 \cdot 125)) - (44^3 / (8 \cdot 125^3)))$$

$$= 1289.95 \text{ kgf/cm}^2$$

Dimension [Gtp]:

$$= 2 \cdot W_{gp} - W_{pl}$$

$$= 2 \cdot 175 - 250$$

$$= 100.0 \text{ mm}$$

Angle [αG]:

$$= \pi / 2 - \text{atan}((W_{pl} - G_{tp}) / H_{gp})$$

$$= \pi / 2 - \text{atan}((250 - 100) / 220)$$

$$= 55.7 \text{ deg}$$

Maximum Compressive Gusset Plate Stress per Bednar [SgpB]:

$$= F_{lug} (3 \cdot D_{lug} - W_{pl}) / (T_{gp} \cdot W_{pl}^2 \cdot (\sin(\alpha G))^2)$$

$$= 35046 (3 \cdot 150 - 250) / (20 \cdot 250^2 \cdot (\sin(55.7))^2)$$

$$= 821.40 \text{ kgf/cm}^2$$

Gusset Plate Allowable Compressive Stress [SgaB]:

$$= 18000 / (1 + (1/18000) \cdot (H_{gp} / \sin(\alpha G) / (0.289 \cdot T_{gp}))^2)$$

$$= 18000 / (1 + (1/18000) \cdot (220 / \sin(55.7) / (0.289 \cdot 20))^2)$$

$$= 1132.06 \text{ kgf/cm}^2$$

Determine the required bolt area due to Uplift [Abr]:

$$= F_{lugemp} / S_{bolt}$$

$$= 34083 / 1530$$

$$= 22.28 \text{ cm}^2$$

The Area available in a single bolt : 32.218 cm² [OK]

Review notes about nozzle loadings and supports in the Warnings report.

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Conical Reinforcement Calculations, ASME VIII Div. 1, App. 1

Input Values:

conical-1

Design Internal Pressure, Small End	25.492	kgf/cm ²
Design Internal Pressure, Large End	25.492	kgf/cm ²
Temperature for Internal Pressure	85.0	°C
Design External Pressure	0.000	kgf/cm ²
Temperature for External Pressure	0.0	°C

CONE Properties:

Cone Material	SA-312 TP304L	+IT
Cone Allowable Stress at Temperature	Ss	1172.61 kgf/cm ²
Cone Allowable Stress At Ambient		1172.61 kgf/cm ²
Longitudinal Joint Efficiency of Cone	Ec,E2	0.85
Circumferential Joint Efficiency of Cone		0.85
Actual Thickness of Cone	Tc	10.0000 mm
Corrosion Allowance for Cone	Cac	0.0000 mm

Diameter Basis for Cone and Cylinders	ID	
Diameter of Small End of Cone	Ds	200.000 mm
Diameter of Large End of Cone	Dl	488.000 mm
Half Apex Angle for Cone		9.62 degrees
Axial Length of Cone	Lc	850.000 mm

SMALL-END Properties:

Attached Small End Material Specification	SA-312 TP304L	+IT
Operating Allowable Stress	Ss	1172.61 kgf/cm ²
Ambient Allowable Stress		1172.61 kgf/cm ²
Longitudinal Joint Efficiency	Es,E1	0.85
Actual Thickness	Ts	10.0000 mm
Corrosion Allowance	Cas	0.0000 mm
Axial Length of Attached Element	Ls	87.100 mm

LARGE-END Properties:

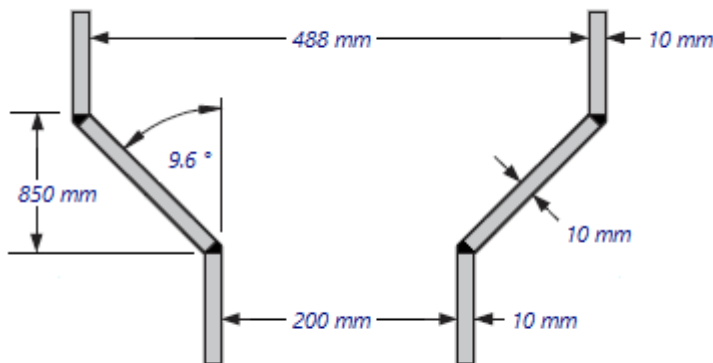
Attached Large End Material Specification	SA-312 TP304L	
Operating Allowable Stress	Sl	1172.61 kgf/cm ²
Ambient Allowable Stress		1172.61 kgf/cm ²
Longitudinal Joint Efficiency	El,E1	1.00
Actual Thickness	Tl	10.0000 mm
Corrosion Allowance	Cal	0.0000 mm
Axial Length of Attached Element	Ll	1240.750 mm

Type of Reinforcement at Large End of Cone: None

Type of Reinforcement at Small End of Cone: None

Force Arl & Ars calculation, cone -> flange No

Force Arl & Ars calculation, cone -> cylinder No



Cone Geometry

Calculated Values:

Elastic Modulus Data from ASME Section II Part D at 0 °C

Elastic Modulus of Cone Material	0.200E+07 kgf/cm ² at 0 °C
Elastic Modulus of Small Cylinder Material	0.200E+07 kgf/cm ² at 0 °C
Elastic Modulus of Large Cylinder Material	0.200E+07 kgf/cm ² at 0 °C
Elastic Modulus of Large End Reinforcement	0.200E+07 kgf/cm ² at 0 °C
Elastic Modulus of Small End Reinforcement	0.200E+07 kgf/cm ² at 0 °C

Elastic Modulus Data from ASME Section II Part D at 85 °C

Elastic Modulus of Cone Material	0.194E+07 kgf/cm ² at 85 °C
Elastic Modulus of Small Cylinder Material	0.194E+07 kgf/cm ² at 85 °C
Elastic Modulus of Large Cylinder Material	0.194E+07 kgf/cm ² at 85 °C
Elastic Modulus of Large End Reinforcement	0.194E+07 kgf/cm ² at 85 °C
Elastic Modulus of Small End Reinforcement	0.194E+07 kgf/cm ² at 85 °C

Forces and Moments on the ends of the Cone:

Axial Force on Small End of Cone	-526.68 kgf
Axial Force on Large End of Cone	-431.74 kgf
Moment on Small End of Cone	529.23 kgf-m.
Moment on Large End of Cone	210.01 kgf-m.

The moments above are computed from the Wind & Seismic loads as well as any user defined moments and include effects due to eccentric masses. The vertical forces include the weight of the elements and details above and below the cone as well as any user defined vertical forces. Vertical forces also include loads cause from vertical accelerations due to earthquake loading.

Note: Both ends of the Cone are Lines of Support

Maximum Centroid Reinforcement Distance Large End	12.5996 mm
Maximum Centroid Reinforcement Distance Small End	8.2916 mm

Note: No ring was found close enough to the large end to be considered.

Note: No ring was found close enough to the small end to be considered.

Reinforcement Calculations for Cone / Large Cylinder:

Required Area of Reinforcement for Large End Under Internal Pressure

Note: Large end is in compression, 1.0 will be used for E1 and E2.

Large end ratio of pressure to allowable stress [P/Ss E1]: 0.02174

Large End Maximum half apex angle w/o reinforcement [δ]:

$$= 326.6 \cdot \sqrt{\frac{P}{(S_s \cdot E_1)}} \text{ per 1-5 equation 1}$$

$$= 326.6 \cdot \sqrt{\frac{25.5}{(1173 \cdot 1)}}$$

$$= 48.155 \text{ degrees}$$

Large end actual half apex angle [alpha]: 9.615 degrees

Because delta is greater than alpha, reinforcement calculations are not required for the large end under internal pressure.

Reinforcement Calculations for Cone / Small Cylinder:

Required Area of Reinforcement for Small End under Internal Pressure

Small end ratio of pressure to allowable stress [P/Ss E1]: 0.02558

Small End Maximum half apex angle w/o reinforcement [δ]:

$$= 89 \cdot \sqrt{\frac{P}{(S_s \cdot E_1)}} \text{ per 1-5 equation 6}$$

$$= 89 \cdot \sqrt{\frac{25.5}{(1173 \cdot 0.85)}}$$

$$= 14.233 \text{ degrees}$$

Small end actual half apex angle [alpha]: 9.615 degrees

Because delta is greater than alpha, reinforcement calculations are not required for the small end under internal pressure.



Because the 1/2 apex angle is less than 30 degrees, the following discontinuity stress evaluation results are for information only.

The following calculations are only required per 1-5(g) and do include external loads due to wind or seismic. These discontinuity stresses are computed at the shell/cone junction and do not include effects of local stiffening from a junction ring.

Results for Discontinuity Stresses per Bednar p. 236 2nd Edition:

Stress Type	Stress	Allowable	Location
Tensile Stress	482.72	3517.84	Small Cyl. Long.
Compres. Stress	-17.99	-3517.84	Small Cyl. Long.
Membrane Stress	371.21	1758.92	Small End Tang.
Tensile Stress	486.92	3517.84	Cone Longitudinal
Compres. Stress	-18.15	-3517.84	Cone Longitudinal
Tensile Stress	375.02	1758.92	Cone Tangential
Tensile Stress	656.29	3517.84	Large Cyl. Long.
Compres. Stress	0.00	-3517.84	Large Cyl. Long. *
Membrane Stress	467.32	-1758.92	Large End Tang.
Tensile Stress	661.01	3517.84	Cone Longitudinal
Compres. Stress	0.00	-3517.84	Cone Longitudinal *
Compres Stress	476.37	-1758.92	Cone Tangential

An asterisk (*) denotes that this stress was not applicable for this combination of loads.

Maximum Allowable Pressure Calculations for Cone to Shell Junction:

Pressure Case	Pressure kgf/cm ²	Reason for Failure at this Pressure
MAWP	39.348	Thickness due to internal pressure, Cone Large End
MAPnc	39.348	Thickness due to internal pressure, Cone Large End

These pressures were determined by iteration.

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Conical Reinforcement Calculations, ASME VIII Div. 1, App. 1

Input Values:

conical -2

Design Internal Pressure, Small End	25.492	kgf/cm ²
Design Internal Pressure, Large End	25.492	kgf/cm ²
Temperature for Internal Pressure	85.0	°C
Design External Pressure	0.000	kgf/cm ²
Temperature for External Pressure	0.0	°C

CONE Properties:

Cone Material	SA-312 TP304L	
Cone Allowable Stress at Temperature	Ss	1172.61 kgf/cm ²
Cone Allowable Stress At Ambient		1172.61 kgf/cm ²
Longitudinal Joint Efficiency of Cone	Ec,E2	0.85
Circumferential Joint Efficiency of Cone		0.85
Actual Thickness of Cone	Tc	7.0000 mm
Corrosion Allowance for Cone	Cac	0.0000 mm
Diameter Basis for Cone and Cylinders	ID	
Diameter of Small End of Cone	Ds	127.000 mm
Diameter of Large End of Cone	Dl	204.000 mm
Half Apex Angle for Cone		15.22 degrees
Axial Length of Cone	Lc	152.000 mm

SMALL-END Properties:

Attached Small End Material Specification	SA-312 TP304L	
Operating Allowable Stress	Ss	1172.61 kgf/cm ²
Ambient Allowable Stress		1172.61 kgf/cm ²
Longitudinal Joint Efficiency	Es,E1	0.85
Actual Thickness	Ts	7.0000 mm
Corrosion Allowance	Cas	0.0000 mm
Axial Length of Attached Element	Ls	3038.060 mm

LARGE-END Properties:

Attached Large End Material Specification	SA-312 TP304L	
Operating Allowable Stress	Sl	1172.61 kgf/cm ²
Ambient Allowable Stress		1172.61 kgf/cm ²
Longitudinal Joint Efficiency	El,E1	0.85
Actual Thickness	Tl	8.0000 mm
Corrosion Allowance	Cal	0.0000 mm
Axial Length of Attached Element	Ll	250.000 mm

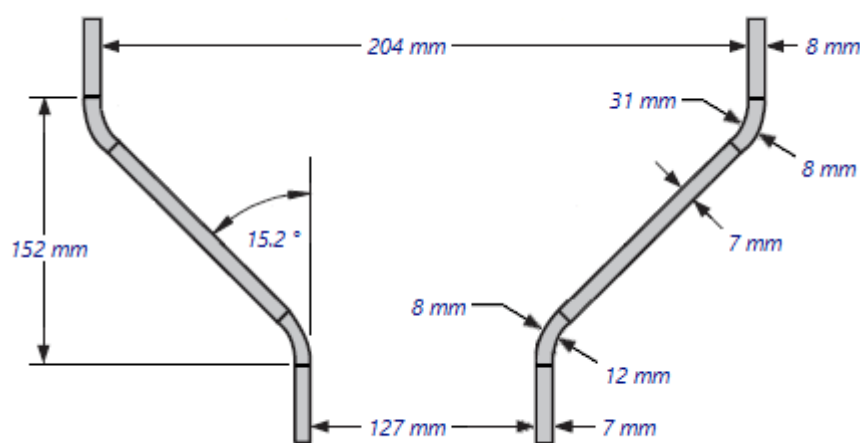
Type of Reinforcement at Large End of Cone: Knuckle

Large End Reinforcing/Knuckle Material	None	
Large Reinforcing/Knuckle Allowable, Operating		1172.61 kgf/cm ²
Large Reinforcing/Knuckle Allowable, Ambient		1172.61 kgf/cm ²
Bend Radius for Knuckle at Large End	Rblk	31.0000 mm
Thickness for Knuckle at Large End	Rtlk	8.0000 mm

Type of Reinforcement at Small End of Cone: Knuckle

Small End Reinforcing/Knuckle Material	None	
Small Reinforcing/Knuckle Allowable, Operating		1172.61 kgf/cm ²
Small Reinforcing/Knuckle Allowable, Ambient		1172.61 kgf/cm ²
Bend Radius for Knuckle at Small End	Rbsk	12.0000 mm
Thickness for Knuckle at Small End	Rtsk	8.0000 mm

Force Arl & Ars calculation, cone -> flange No
 Force Arl & Ars calculation, cone -> cylinder No



Cone Geometry

Calculated Values:

Elastic Modulus Data from ASME Section II Part D at 0 °C

Elastic Modulus of Cone Material	0.200E+07 kgf/cm ² at 0 °C
Elastic Modulus of Small Cylinder Material	0.200E+07 kgf/cm ² at 0 °C
Elastic Modulus of Large Cylinder Material	0.200E+07 kgf/cm ² at 0 °C

Elastic Modulus Data from ASME Section II Part D at 85 °C

Elastic Modulus of Cone Material	0.194E+07 kgf/cm ² at 85 °C
Elastic Modulus of Small Cylinder Material	0.194E+07 kgf/cm ² at 85 °C
Elastic Modulus of Large Cylinder Material	0.194E+07 kgf/cm ² at 85 °C

Forces and Moments on the ends of the Cone:

Axial Force on Small End of Cone	768.31 kgf
Axial Force on Large End of Cone	773.28 kgf
Moment on Small End of Cone	5619.23 kgf-m.
Moment on Large End of Cone	5778.73 kgf-m.

The moments above are computed from the Wind & Seismic loads as well as any user defined moments and include effects due to eccentric masses. The vertical forces include the weight of the elements and details above and below the cone as well as any user defined vertical forces. Vertical forces also include loads cause from vertical accelerations due to earthquake loading.

Note: Both ends of the Cone are Lines of Support

Maximum Centroid Reinforcement Distance Large End	7.4162 mm
Maximum Centroid Reinforcement Distance Small End	5.5537 mm

Note: No ring was found close enough to the large end to be considered.

Note: No ring was found close enough to the small end to be considered.

Thickness and Maximum Pressure for Large Knuckle:

Large End Knuckle Thickness	8.000 mm
Large End Knuckle Radius	31.000 mm
Corrosion Allowance	0.000 mm

% Elongation per Table UG-79-1 $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$: 17.143 %

Thickness Due to Internal Pressure [Tr]:
 $= (P \cdot (L + ca) \cdot M) / (2 \cdot S \cdot E \cdot 0.2 \cdot P) + ca$ per App. 1-4(d)
 $= (25.5(105) \cdot 1.21) / (2 \cdot 1173 \cdot 0.85 \cdot 0.2 \cdot 25.5)$
 $= 1.6213$ mm



Equivalent Crown Radius [L]:
 $= Di/(2(\cos(\alpha)) + ca \text{ per UG-32(g)})$
 $= 202/(2(\cos(15.2)) + 0)$
 $= 104.5808 \text{ mm}$

Stress Concentration Factor [M]:
 $= (3 + \sqrt{(L/(r + ca))})/4$
 $= (3 + \sqrt{(105/(31 + 0))})/4$
 $= 1.2092$

Maximum Allowable Working Pressure at Given Thickness:
 $= (2 \cdot S \cdot E(t - ca))/(M(L + ca) + 0.2(t - ca)) \text{ per Appendix 1-4 (d)}$
 $= (2 \cdot 1173 \cdot 0.85(8 - 0))/(1.21(105 + 0) + 0.2(8 - 0))$
 $= 124.5344 \text{ kgf/cm}^2$

Required Thickness for Small Knuckle:

Small End Knuckle Thickness	8.000 mm
Small End Knuckle Radius	12.000 mm
Corrosion Allowance	0.000 mm
Minimum Required Small End Thickness	1.587 mm

% Elongation per Table UG-79-1 $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$: 37.500 %

Maximum Allowable Pressure Calculations for Cone to Shell Junction:

Pressure Case	Pressure kgf/cm ²	Reason for Failure at this Pressure
MAWP	64.144	Thickness due to internal pressure, Cone Large End
MAPnc	64.144	Thickness due to internal pressure, Cone Large End

These pressures were determined by iteration.

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Input, Nozzle Desc: B2 2"

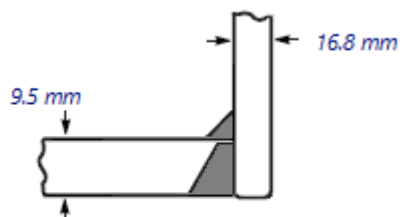
From: 10

Pressure for Reinforcement Calculations	P	39.397	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-403 304L	+IT
Parent Allowable Stress at Temperature	Sv	1172.61	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1172.61	kgf/cm ²
Inside Diameter of Elliptical Head	D	489.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	9.5000	mm
Head Internal Corrosion Allowance	cas	0.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F304L	+IT
Material UNS Number		S30403	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1172.63	kgf/cm ²
Allowable Stress At Ambient	Sna	1172.63	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-182 F304	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	325.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 2.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: B2 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.



Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 39.4 \cdot 489 \cdot 0.9 / (2 \cdot 1173 \cdot 1 - 0.2 \cdot 39.4)$
 $= 7.4182 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (Sn \cdot E)] - 1) - 1)$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([39.4 / (1173 \cdot 1)] - 1) - 1)$
 $= 0.8679 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	103.3280	mm
Parallel to Vessel Wall	Rn+tn+t	51.6640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	23.7500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: B2 2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.8679	mm
Wall Thickness per UG16(b),	tr16b	= 1.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.2424	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.2424	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 1.5000	mm
Wall Thickness per table UG-45	tb3	= 4.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(8.24, 1.5)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.87, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	117.9,	Allowable	:	1172.6	kgf/cm ²	Passed
Expansion	:	0.0,	Allowable	:	2813.7	kgf/cm ²	Passed
Occasional	:	22.4,	Allowable	:	1559.6	kgf/cm ²	Passed
Shear	:	102.5,	Allowable	:	820.8	kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -22 °C

Weld Size Calculations, Description: B2 2"

Intermediate Calc. for nozzle/shell Welds Tmin 11.1765 mm



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.9899 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 336.4899 mm

Input Echo, WRC107/537 Item 1, Description: B2 2" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	0.0000	mm
Vessel Diameter	Dv	880.204	mm
Vessel Thickness	Tv	9.500	mm
Design Temperature	T1	85.0	C
Vessel Material		SA-403 304L	
Vessel UNS Number		S30403	
Vessel Cold S.I. Allowable	Smc	1172.61	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1172.61	kgf/cm ²
Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	0.0000	mm
Nozzle Diameter	Dn	50.800	mm
Nozzle Thickness	Tn	16.764	mm
Nozzle Material		SA-182 F304L	
Nozzle UNS Number		S30403	
Nozzle Cold S.I. Allowable	SNmc	1172.63	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1172.63	kgf/cm ²
Design Internal Pressure	Dp	39.397	kgf/cm ²
Include Pressure Thrust		No	

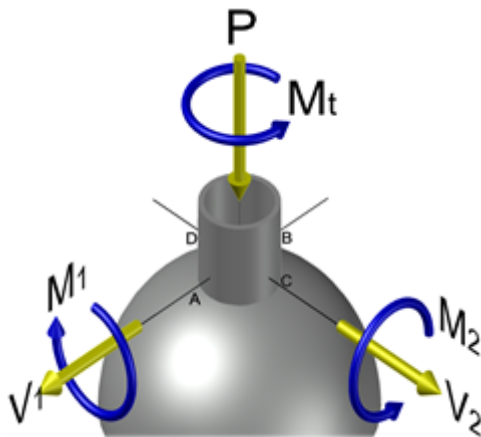
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-326.3	kgf
Longitudinal Shear (SUS)	(V1) V1	288.4	kgf
Circumferential Shear (SUS)	(Vc) V2	288.4	kgf
Circumferential Moment (SUS)	(Mc) M1	37.8	kgf-m.
Longitudinal Moment (SUS)	(Ml) M2	37.8	kgf-m.
Torsional Moment (SUS)	Mt	48.9	kgf-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 84.328 + 2 \cdot 1.65 \cdot \sqrt{(444.852 (9.5 - 0))}$
 $= 298.856 \text{ mm}$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	-326.3	kgf
Circumferential Shear	(VC) V2	288.4	kgf
Longitudinal Shear	(VL) V1	288.4	kgf
Circumferential Moment	(MC) M1	37.8	kgf-m.
Longitudinal Moment	(ML) M2	37.8	kgf-m.
Torsional Moment	MT	48.9	kgf-m.

Dimensionless Param: $U = 0.65 \text{ TAU} = 5.00 (2.02) \text{ RHO} = 0.57$

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.06064	(A,B,C,D)
$M(x) / P$	SP 1	0.07035	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.08355	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.15860	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.08355	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.15860	(A,B,C,D)
$N(y) * T / P$	SP 1	0.10209	(A,B,C,D)
$M(y) / P$	SP 1	0.02201	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.09372	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.05795	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.09372	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.05795	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	21.9	21.9	21.9	21.9	21.9	21.9	21.9	21.9
Rad. Bend.	P	152.6	-152.6	152.6	-152.6	152.6	-152.6	152.6	-152.6
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-53.9	-53.9	53.9	53.9
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-613.4	613.4	613.4	-613.4
Rad. Memb.	ML	-53.9	-53.9	53.9	53.9	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-613.4	613.4	613.4	-613.4	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-492.7	428.8	841.8	-690.2	-492.7	428.8	841.8	-690.2



Tang. Memb. P	36.9	36.9	36.9	36.9	36.9	36.9	36.9	36.9
Tang. Bend. P	47.7	-47.7	47.7	-47.7	47.7	-47.7	47.7	-47.7
Tang. Memb. MC	0.0	0.0	0.0	0.0	-60.4	-60.4	60.4	60.4
Tang. Bend. MC	0.0	0.0	0.0	0.0	-224.1	224.1	224.1	-224.1
Tang. Memb. ML	-60.4	-60.4	60.4	60.4	0.0	0.0	0.0	0.0
Tang. Bend. ML	-224.1	224.1	224.1	-224.1	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-199.9	152.9	369.2	-174.5	-199.9	152.9	369.2	-174.5
Shear VC	22.9	22.9	-22.9	-22.9	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-22.9	-22.9	22.9	22.9
Shear MT	46.1	46.1	46.1	46.1	46.1	46.1	46.1	46.1
Tot. Shear	69.0	69.0	23.2	23.2	23.2	23.2	69.0	69.0
Str. Int.	508.1	445.1	842.9	691.3	494.5	430.8	851.7	699.3

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		912.9	912.9	912.9	912.9	912.9	912.9	912.9	912.9
Rad. Pl (SUS)		-31.9	-31.9	75.8	75.8	-31.9	-31.9	75.8	75.8
Rad. Q (SUS)		-460.8	460.8	766.0	-766.0	-460.8	460.8	766.0	-766.0
Long. Pm (SUS)		912.9	912.9	912.9	912.9	912.9	912.9	912.9	912.9
Long. Pl (SUS)		-23.5	-23.5	97.3	97.3	-23.5	-23.5	97.3	97.3
Long. Q (SUS)		-176.4	176.4	271.9	-271.9	-176.4	176.4	271.9	-271.9
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		22.9	22.9	-22.9	-22.9	-22.9	-22.9	22.9	22.9
Shear Q (SUS)		46.1	46.1	46.1	46.1	46.1	46.1	46.1	46.1
Pm (SUS)		912.9	912.9	912.9	912.9	912.9	912.9	912.9	912.9
Pm+Pl (SUS)		908.4	908.4	1024.7	1024.7	908.4	908.4	1024.7	1024.7
Pm+Pl+Q (Total)		728.4	1358.0	1755.8	739.4	714.8	1343.6	1764.5	747.4

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	912.85	1172.61	Passed
Pm+Pl (SUS)	1024.73	1758.92	Passed
Pm+Pl+Q (TOTAL)	1764.51	3517.84	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: L2 2''

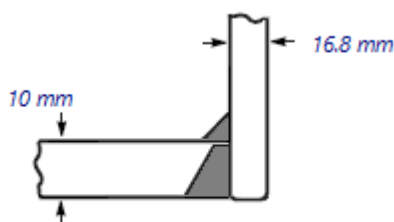
From: 20

Pressure for Reinforcement Calculations	P	39.338	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material	SA-312 TP304L		
Parent Allowable Stress at Temperature	Sv	1172.61	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1172.61	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	488.00	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	cas	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		50.00	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material	SA-182 F304L +IT		
Material UNS Number	S30403		
Material Specification/Type	Forgings		
Allowable Stress at Temperature	Sn	1172.63	kgf/cm ²
Allowable Stress At Ambient	Sna	1172.63	kgf/cm ²
Diameter Basis (for tr calc only)	Inside		
Layout Angle	0.00	deg	
Diameter	2.0000	in.	
Size and Thickness Basis	Actual		
Actual Thickness	tn	16.7640	mm
Flange Material	SA-182 F304L +IT		
Flange Type	Long Weld Neck		
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	175.4260	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class	600		
Flange Grade	GR 2.3		

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L2 2''

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 39.3 \cdot 244 / (1173 \cdot 1 - 0.6 \cdot 39.3)$
 $= 8.3537 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([39.3 / (1173 \cdot 1)] - 1))$
 $= 0.8665 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	104.3280	mm
Parallel to Vessel Wall	Rn+tn+tt	52.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	25.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L2 2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.8665	mm
Wall Thickness per UG16(b),	tr16b	= 1.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.3537	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.3537	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 1.5000	mm
Wall Thickness per table UG-45	tb3	= 4.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(8.35, 1.5)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.87, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -19 °C

Weld Size Calculations, Description: L2 2"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.0000	mm
---	------	---------	----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 3.6706 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 189.0966 mm

*This nozzle or its diameter limit intersects weld seam # [1] on this element. The
 distance between the nozzle and seam is, 0.00 mm , while the distance from the center*

PV Elite 28 U2 Licensee: SL User

FileName : =A150NRL31HN003_Ope-Lug loc as per MDS

Nozzle Calcs.: L2 2''

Nozl: 32

4:02pm

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of the nozzle to the edge of the diameter limit is: 52.16 mm

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Input, Nozzle Desc: L4 2''

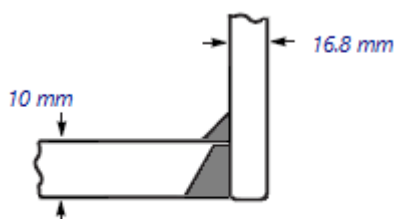
From: 20

Pressure for Reinforcement Calculations	P	39.338	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material	SA-312 TP304L		
Parent Allowable Stress at Temperature	Sv	1172.61	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1172.61	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	488.00	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	cas	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		50.00	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material	SA-182 F304L +IT		
Material UNS Number	S30403		
Material Specification/Type	Forgings		
Allowable Stress at Temperature	Sn	1172.63	kgf/cm ²
Allowable Stress At Ambient	Sna	1172.63	kgf/cm ²
Diameter Basis (for tr calc only)	Inside		
Layout Angle	45.00	deg	
Diameter	2.0000	in.	
Size and Thickness Basis	Actual		
Actual Thickness	tn	16.7640	mm
Flange Material	SA-182 F304 +IT		
Flange Type	Long Weld Neck		
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	175.4260	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class	600		
Flange Grade	GR 2.1		

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L4 2''

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 39.3 \cdot 244 / (1173 \cdot 1 - 0.6 \cdot 39.3)$
 $= 8.3537 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([39.3 / (1173 \cdot 1)] - 1))$
 $= 0.8665 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	104.3280	mm
Parallel to Vessel Wall	Rn+tn+tt	52.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	25.0000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: L4 2''.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.8665	mm
Wall Thickness per UG16(b),	tr16b	= 1.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.3537	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.3537	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 1.5000	mm
Wall Thickness per table UG-45	tb3	= 4.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(8.35, 1.5)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.87, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -19 °C

Weld Size Calculations, Description: L4 2''

Intermediate Calc. for nozzle/shell Welds	Tmin	10.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.6706 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 189.0966 mm

Input, Nozzle Desc: L6 2''

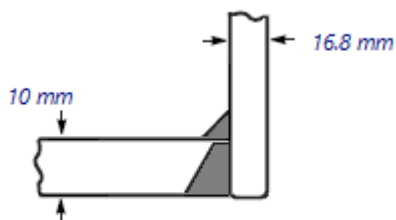
From: 20

Pressure for Reinforcement Calculations	P	39.338	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material	SA-312 TP304L		
Parent Allowable Stress at Temperature	Sv	1172.61	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1172.61	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	488.00	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	cas	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		50.00	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material	SA-182 F304L +IT		
Material UNS Number	S30403		
Material Specification/Type	Forgings		
Allowable Stress at Temperature	Sn	1172.63	kgf/cm ²
Allowable Stress At Ambient	Sna	1172.63	kgf/cm ²
Diameter Basis (for tr calc only)	Inside		
Layout Angle	90.00	deg	
Diameter	2.0000	in.	
Size and Thickness Basis	Actual		
Actual Thickness	tn	16.7640	mm
Flange Material	SA-182 F304 +IT		
Flange Type	Long Weld Neck		
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	175.4260	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class	600		
Flange Grade	GR 2.1		

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L6 2''

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 39.3 \cdot 244 / (1173 \cdot 1 - 0.6 \cdot 39.3)$
 $= 8.3537 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([39.3 / (1173 \cdot 1)] - 1))$
 $= 0.8665 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	104.3280	mm
Parallel to Vessel Wall	Rn+tn+tt	52.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	25.0000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: L6 2''.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.8665	mm
Wall Thickness per UG16(b),	tr16b	= 1.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.3537	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.3537	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 1.5000	mm
Wall Thickness per table UG-45	tb3	= 4.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(8.35, 1.5)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.87, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -19 °C

Weld Size Calculations, Description: L6 2''

Intermediate Calc. for nozzle/shell Welds	Tmin	10.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.6706 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 189.0966 mm



Input, Nozzle Desc: L1 2''

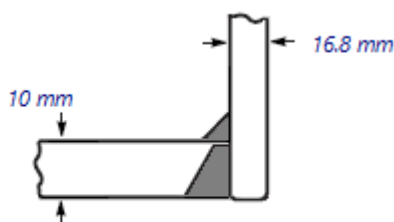
From: 20

Pressure for Reinforcement Calculations	P	39.300	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material	SA-312 TP304L		
Parent Allowable Stress at Temperature	Sv	1172.61	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1172.61	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	488.00	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	cas	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		95.00	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material	SA-182 F304L +IT		
Material UNS Number	S30403		
Material Specification/Type	Forgings		
Allowable Stress at Temperature	Sn	1172.63	kgf/cm ²
Allowable Stress At Ambient	Sna	1172.63	kgf/cm ²
Diameter Basis (for tr calc only)	Inside		
Layout Angle	0.00	deg	
Diameter	2.0000	in.	
Size and Thickness Basis	Actual		
Actual Thickness	tn	16.7640	mm
Flange Material	SA-182 F304 +IT		
Flange Type	Long Weld Neck		
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	175.4260	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class	600		
Flange Grade	GR 2.1		

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L1 2''

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 39.3 \cdot 244 / (1173 \cdot 1 - 0.6 \cdot 39.3)$
 $= 8.3454 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([39.3 / (1173 \cdot 1)] - 1))$
 $= 0.8657 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	104.3280	mm
Parallel to Vessel Wall	Rn+tn+tt	52.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	25.0000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: L1 2''.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.8657	mm
Wall Thickness per UG16(b),	tr16b	= 1.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.3454	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.3454	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 1.5000	mm
Wall Thickness per table UG-45	tb3	= 4.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(8.35, 1.5)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.87, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -19 °C

Weld Size Calculations, Description: L1 2''

Intermediate Calc. for nozzle/shell Welds	Tmin	10.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.6706 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 189.0966 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center



of the nozzle to the edge of the diameter limit is: 52.16 mm

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Input, Nozzle Desc: L3 2"

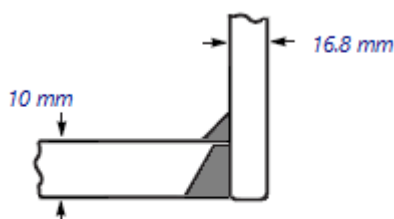
From: 20

Pressure for Reinforcement Calculations	P	39.300	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material	SA-312 TP304L		
Parent Allowable Stress at Temperature	Sv	1172.61	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1172.61	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	488.00	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	cas	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		95.00	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material	SA-182 F304L +IT		
Material UNS Number	S30403		
Material Specification/Type	Forgings		
Allowable Stress at Temperature	Sn	1172.63	kgf/cm ²
Allowable Stress At Ambient	Sna	1172.63	kgf/cm ²
Diameter Basis (for tr calc only)	Inside		
Layout Angle	45.00	deg	
Diameter	2.0000	in.	
Size and Thickness Basis	Actual		
Actual Thickness	tn	16.7640	mm
Flange Material	SA-182 F304L +IT		
Flange Type	Long Weld Neck		
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	175.4260	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class	600		
Flange Grade	GR 2.3		

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L3 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 39.3 \cdot 244 / (1173 \cdot 1 - 0.6 \cdot 39.3)$
 $= 8.3454 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([39.3 / (1173 \cdot 1)] - 1))$
 $= 0.8657 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	104.3280	mm
Parallel to Vessel Wall	Rn+tn+tt	52.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	25.0000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: L3 2''.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.8657	mm
Wall Thickness per UG16(b),	tr16b	= 1.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.3454	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.3454	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 1.5000	mm
Wall Thickness per table UG-45	tb3	= 4.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(8.35, 1.5)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.87, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -19 °C

Weld Size Calculations, Description: L3 2''

Intermediate Calc. for nozzle/shell Welds	Tmin	10.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.6706 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 189.0966 mm

Input, Nozzle Desc: L5 2''

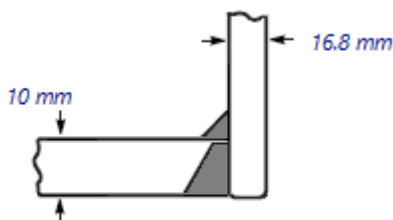
From: 20

Pressure for Reinforcement Calculations	P	39.300	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material	SA-312 TP304L		
Parent Allowable Stress at Temperature	Sv	1172.61	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1172.61	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	488.00	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	cas	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		95.00	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material	SA-182 F304L +IT		
Material UNS Number	S30403		
Material Specification/Type	Forgings		
Allowable Stress at Temperature	Sn	1172.63	kgf/cm ²
Allowable Stress At Ambient	Sna	1172.63	kgf/cm ²
Diameter Basis (for tr calc only)	Inside		
Layout Angle	90.00	deg	
Diameter	2.0000	in.	
Size and Thickness Basis	Actual		
Actual Thickness	tn	16.7640	mm
Flange Material	SA-182 F304 +IT		
Flange Type	Long Weld Neck		
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	175.4260	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class	600		
Flange Grade	GR 2.1		

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L5 2''

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 39.3 \cdot 244 / (1173 \cdot 1 - 0.6 \cdot 39.3)$
 $= 8.3454 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([39.3 / (1173 \cdot 1)] - 1))$
 $= 0.8657 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	104.3280	mm
Parallel to Vessel Wall	Rn+tn+tt	52.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	25.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L5 2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.8657	mm
Wall Thickness per UG16(b),	tr16b	= 1.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.3454	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.3454	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 1.5000	mm
Wall Thickness per table UG-45	tb3	= 4.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(8.35, 1.5)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.87, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -19 °C

Weld Size Calculations, Description: L5 2"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 3.6706 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 189.0966 mm



Input, Nozzle Desc: P3-2''

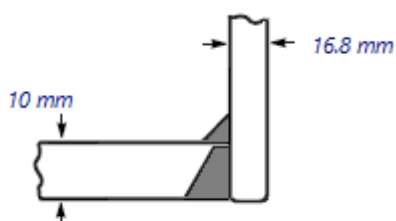
From: 30

Pressure for Reinforcement Calculations	P	39.300	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material	SA-312 TP304L	+IT	
Parent Allowable Stress at Temperature	Sv	1172.61	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1172.61	kgf/cm ²
Inside Diameter of Cone at Nozzle Location	D	335.24	mm
Cone Half Apex Angle	Alpha	9.62	Degrees
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	cas	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		165.00	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material	SA-182 F304L	+IT	
Material UNS Number	S30403		
Material Specification/Type	Forgings		
Allowable Stress at Temperature	Sn	1172.63	kgf/cm ²
Allowable Stress At Ambient	Sna	1172.63	kgf/cm ²
Diameter Basis (for tr calc only)	Inside		
Layout Angle	0.00	deg	
Diameter	2.0000	in.	
Size and Thickness Basis	Actual		
Actual Thickness	tn	16.7640	mm
Flange Material	SA-182 F304	+IT	
Flange Type	Long Weld Neck		
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 2.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: P3-2''

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Conical Transition, tr [Int. Press]
 $= P \cdot D / (2 \cdot \cos(a) \cdot (S_v \cdot E - 0.6 \cdot P))$ Appendix 1-4(e)
 $= 39.3 \cdot 335 / (2 \cdot \cos(9.62) \cdot (1173 \cdot 1 - 0.6 \cdot 39.3))$
 $= 5.8148 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([39.3 / (1173 \cdot 1)] - 1))$
 $= 0.8657 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	105.0518	mm
Parallel to Vessel Wall	Rn+tn+t	52.5259	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	25.0000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: P3-2''.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.8657	mm
Wall Thickness per UG16(b),	tr16b	= 1.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 5.8148	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 5.8148	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 1.5000	mm
Wall Thickness per table UG-45	tb3	= 4.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(5.81, 1.5)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.87, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -34 °C

Weld Size Calculations, Description: P3-2''

Intermediate Calc. for nozzle/shell Welds Tmin 10.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 5.3897 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 165.3896 mm

PV Elite 28 U2 Licensee: SL User

FileName : =A150NRL31HN003_Ope-Lug loc as per MDS

Nozzle Calcs.: P3-2''

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Nozl: 38 4:02pm Sep 18,2026



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Input, Nozzle Desc: A4 2"

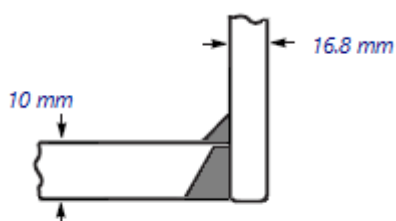
From: 30

Pressure for Reinforcement Calculations	P	39.300	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material	SA-312 TP304L	+IT	
Parent Allowable Stress at Temperature	Sv	1172.61	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1172.61	kgf/cm ²
Inside Diameter of Cone at Nozzle Location	D	436.89	mm
Cone Half Apex Angle	Alpha	9.62	Degrees
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	cas	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		135.00	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material	SA-182 F304L	+IT	
Material UNS Number	S30403		
Material Specification/Type	Forgings		
Allowable Stress at Temperature	Sn	1172.63	kgf/cm ²
Allowable Stress At Ambient	Sna	1172.63	kgf/cm ²
Diameter Basis (for tr calc only)	Inside		
Layout Angle	180.00	deg	
Diameter	2.0000	in.	
Size and Thickness Basis	Actual		
Actual Thickness	tn	16.7640	mm
Flange Material	SA-182 F304	+IT	
Flange Type	Long Weld Neck		
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 2.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: A4 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Conical Transition, tr [Int. Press]
 $= P \cdot D / (2 \cdot \cos(a) \cdot (S_v \cdot E - 0.6 \cdot P))$ Appendix 1-4(e)
 $= 39.3 \cdot 437 / (2 \cdot \cos(9.62) \cdot (1173 \cdot 1 - 0.6 \cdot 39.3))$
 $= 7.5779 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([39.3 / (1173 \cdot 1)] - 1))$
 $= 0.8657 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	105.0518	mm
Parallel to Vessel Wall	Rn+tn+t	52.5259	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	25.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: A4 2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.8657	mm
Wall Thickness per UG16(b),	tr16b	= 1.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 7.5779	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.5779	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 1.5000	mm
Wall Thickness per table UG-45	tb3	= 4.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.58, 1.5)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.87, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -23 °C

Weld Size Calculations, Description: A4 2"

Intermediate Calc. for nozzle/shell Welds Tmin 10.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 4.1078 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 164.1078 mm

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FileName : =A150NRL31HN003_Ope-Lug loc as per MDS

Nozzle Calcs.: A4 2''

Nozl: 39 4:02pm Sep 18,2026

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Input, Nozzle Desc: A1 2"

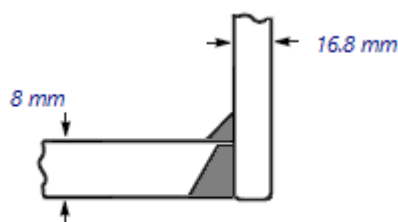
From: 70

Pressure for Reinforcement Calculations	P	39.300	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material	SA-312 TP304L		
Parent Allowable Stress at Temperature	Sv	1172.61	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1172.61	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	204.00	mm
Shell Finished (Minimum) Thickness	t	8.0000	mm
Shell Internal Corrosion Allowance	cas	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		246.50	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material	SA-182 F304L		
Material UNS Number	S30403		
Material Specification/Type	Forgings		
Allowable Stress at Temperature	Sn	1172.63	kgf/cm ²
Allowable Stress At Ambient	Sna	1172.63	kgf/cm ²
Diameter Basis (for tr calc only)	Inside		
Layout Angle	0.00	deg	
Diameter	2.0000	in.	
Size and Thickness Basis	Actual		
Actual Thickness	tn	16.7640	mm
Flange Material	SA-182 F304		
Flange Type	Long Weld Neck		
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	169.4260	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class	600		
Flange Grade	GR 2.1		

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: A1 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 39.3 \cdot 102 / (1173 \cdot 1 - 0.6 \cdot 39.3)$
 $= 3.4887 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([39.3 / (1173 \cdot 1)] - 1))$
 $= 0.8657 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	101.6000	mm
Parallel to Vessel Wall, opening length	d	50.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	20.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: A1 2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.8657	mm
Wall Thickness per UG16(b),	tr16b	= 1.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 3.4887	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 3.4887	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 1.5000	mm
Wall Thickness per table UG-45	tb3	= 4.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(3.49, 1.5)]$
 $= 3.4887 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.87, 3.49)$
 $= 3.4887 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 117.9,	Allowable	: 1172.6	kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2813.7	kgf/cm ²	Passed
Occasional	: 22.4,	Allowable	: 1559.6	kgf/cm ²	Passed
Shear	: 102.5,	Allowable	: 820.8	kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

SA-312 TP304L, Min Metal Temp without impact per UHA-51: -196 °C

SA-182 F304L, Min Metal Temp without impact per UHA-51: -196 °C

Weld Size Calculations, Description: A1 2"

Intermediate Calc. for nozzle/shell Welds	Tmin	8.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.6000 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.



The Drop for this Nozzle is : 9.1227 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 186.5487 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 50.80 mm

Input Echo, WRC297 Item 1, Description: A1 2'' :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	0.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	204.000	mm	
Shell Thickness	8.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-312 TP304L		
Vessel Cold S.I. Allowable	Smc	1172.61	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1172.61	kgf/cm ²
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	0.0000	mm	
Nozzle Diameter	50.800	mm	
Nozzle Thickness	16.7640	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-182 F304L		
Nozzle Cold S.I. Allowable	SNmc	1172.63	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1172.63	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	39.30	kgf/cm ²
Radial Load	P	-326.32	kgf
Circumferential Shear	Vc	244.74	kgf
Longitudinal Shear	Vl	326.32	kgf
Circumferential Moment	Mc	32.62	kgf-m.
Longitudinal Moment	Ml	42.41	kgf-m.
Torsional Moment	Mt	48.93	kgf-m.
Include Axial Pressure Thrust		No	
Include Pressure Stress Indices per Div. 2		No	
Local Loads applied at end of Nozzle/Attachment		No	

*Warning - The ratio, Dn/Tn (5.030) is < 10
 check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 84.328 + 2 \cdot 1.65 \cdot \sqrt{(106.0(8.0 - 0.0))}$
 $= 180.425 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	212.000 mm
Nozzle Outside Diameter	(d) =	84.328 mm
Vessel Thickness used	(T) =	8.000 mm
Nozzle Thickness used	(t) =	16.764 mm
T / t	=	0.477
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.048$$

$$Nr/P = 0.068$$



Mr/P = 0.139
 M0/P = 0.041
 N0/P = 0.095
 MrD/Mc = 0.283
 NrDL/Mc = 0.099
 M0d/Mc = 0.087
 N0DL/Mc = 0.071
 MrD/Ml = 0.110
 NrDL/Ml = 0.065
 M0D/Ml = 0.036
 N0DL/Ml = 0.120

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	48	48	48	48
Outplane Bending (P)	124	-124	124	-124
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-94	-94	94	94
Outplane Bending (ML)	-167	167	167	-167
Normal Pressure Stress	482	521	482	521
Outplane Stress Summary	392	518	917	372

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	34	34	34	34
Inplane Bending (P)	426	-426	426	-426
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-51	-51	51	51
Inplane Bending (ML)	-519	519	519	-519
Inplane Pressure Stress	241	241	241	241
Inplane Stress Summary	130	317	1273	-619

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	23	23	-23	-23
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	54	54	54	54
Shear Stress Summary	77	77	31	31

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	414	545	1276	993

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	48	48	48	48
Outplane Bending (P)	124	-124	124	-124
Outplane Membrane (Mc)	-42	-42	42	42
Outplane Bending (Mc)	-317	317	317	-317
Outplane Membrane (ML)	0	0	0	0



Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	241	241	241	241

Outplane Stress Summary	53	439	774	-109
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	34	34	34	34
Inplane Bending (P)	426	-426	426	-426
Inplane Membrane (Mc)	-60	-60	60	60
Inplane Bending (Mc)	-1025	1025	1025	-1025
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	482	521	482	521

Inplane Stress Summary	-142	1095	2029	-836
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-30	-30	30	30
Torsional Shear (Mt)	54	54	54	54

Shear Stress Summary	23	23	85	85
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	201	1095	2035	846
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	48	48	48	48
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-94	-94	94	94
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	53	100	53	100

Hoop Stress Summary	7	54	196	243
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	9	9	9	9
Axial Bending (P)	73	-73	73	-73
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-82	-82	82	82
Axial Bending (ML)	-83	83	83	-83
Axial Pressure Stress	26	26	26	26

Axial Stress Summary	-56	-37	275	-37
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	10	10	-10	-10
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	26	26	26	26
Shear Stress Summary	37	37	15	15

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	98	118	278	283

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	48	48	48	48
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-42	-42	42	42
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	53	100	53	100
Hoop Stress Summary	59	106	145	192

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	9	9	9	9
Axial Bending (P)	73	-73	73	-73
Axial Membrane (Mc)	-63	-63	63	63
Axial Bending (Mc)	-192	192	192	-192
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	26	26	26	26
Axial Stress Summary	-146	91	365	-166

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-14	-14	14	14
Shear due to Torsion	26	26	26	26
Shear Stress Summary	11	11	40	40

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	207	112	373	367

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	482	521	482	521	482	521	482	521
Circ. Pl (SUS)	-46	-46	142	142	-26	-26	94	94



Circ. Q (SUS)	-43	43	291	-291	-599	599	1451	-1451
Long. Pm (SUS)	241	241	241	241	241	241	241	241
Long. Pl (SUS)	-17	-17	85	85	6	6	90	90
Long. Q (SUS)	-93	93	945	-945	-193	193	441	-441
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	23	23	-23	-23	-30	-30	30	30
Shear Q (SUS)	54	54	54	54	54	54	54	54
Pm (SUS)	482.0	521.0	482.0	521.0	482.0	521.0	482.0	521.0
Pm+Pl (SUS)	438.5	477.1	625.8	664.6	460.2	498.6	579.6	618.1
Pm+Pl+Q (Total)	413	544	1273	992	202	1094	2032	845

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	521.00	1172.61	Passed
Pm+Pl (SUS)	664.56	1758.92	Passed
Pm+Pl+Q (TOTAL)	2032.60	3517.84	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	53	100	53	100	53	100	53	100
Circ. Pl (SUS)	-46	-46	142	142	6	6	90	90
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	26	26	26	26	26	26	26	26
Long. Pl (SUS)	-73	-73	91	91	-54	-54	72	72
Long. Q (SUS)	-10	10	156	-156	-119	119	265	-265
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	10	10	-10	-10	-14	-14	14	14
Shear Q (SUS)	26	26	26	26	26	26	26	26
Pm (SUS)	53.0	100.0	53.0	100.0	53.0	100.0	53.0	100.0
Pm+Pl (SUS)	57.6	103.0	196.3	242.8	91.4	136.9	147.0	192.1
Pm+Pl+Q (Total)	96.3	116.0	276.2	282.8	207.4	112.7	370.0	365.9

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	100.00	1172.63	Passed
Pm+Pl (SUS)	242.79	1758.95	Passed
Pm+Pl+Q (TOTAL)	370.05	3517.90	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: A3 2"

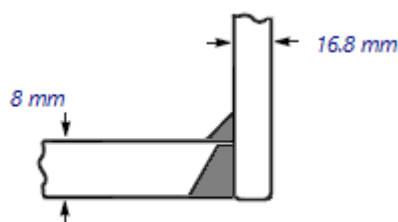
From: 150

Pressure for Reinforcement Calculations	P	39.300	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material	SA-312 TP304L		
Parent Allowable Stress at Temperature	Sv	1172.61	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1172.61	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	204.00	mm
Shell Finished (Minimum) Thickness	t	8.0000	mm
Shell Internal Corrosion Allowance	cas	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1140.00	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material	SA-182 F304L +IT		
Material UNS Number	S30403		
Material Specification/Type	Forgings		
Allowable Stress at Temperature	Sn	1172.63	kgf/cm ²
Allowable Stress At Ambient	Sna	1172.63	kgf/cm ²
Diameter Basis (for tr calc only)	Inside		
Layout Angle	0.00	deg	
Diameter	2.0000	in.	
Size and Thickness Basis	Actual		
Actual Thickness	tn	16.7640	mm
Flange Material	SA-182 F304 +IT		
Flange Type	Long Weld Neck		
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	169.4260	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class	600		
Flange Grade	GR 2.1		

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: A3 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 39.3 \cdot 102 / (1173 \cdot 1 - 0.6 \cdot 39.3)$
 $= 3.4887 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([39.3 / (1173 \cdot 1)] - 1))$
 $= 0.8657 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	101.6000	mm
Parallel to Vessel Wall, opening length	d	50.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	20.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: A3 2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.8657	mm
Wall Thickness per UG16(b),	tr16b	= 1.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 3.4887	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 3.4887	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 1.5000	mm
Wall Thickness per table UG-45	tb3	= 4.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(3.49, 1.5)]$
 $= 3.4887 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.87, 3.49)$
 $= 3.4887 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 117.9,	Allowable	: 1172.6	kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2813.7	kgf/cm ²	Passed
Occasional	: 22.4,	Allowable	: 1559.6	kgf/cm ²	Passed
Shear	: 102.5,	Allowable	: 820.8	kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -53 °C

Weld Size Calculations, Description: A3 2"

Intermediate Calc. for nozzle/shell Welds Tmin 8.0000 mm

Results Per UW-16.1:



	Required Thickness	Actual Thickness
Nozzle Weld	$5.6000 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 9.1227 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 186.5487 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 50.80 mm

Input Echo, WRC297 Item 1, Description: A3 2" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	204.000	mm
Shell Thickness	8.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-312 TP304L	
Vessel Cold S.I. Allowable	Smc 1172.61	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1172.61	kgf/cm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	50.800	mm
Nozzle Thickness	16.7640	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-182 F304L	
Nozzle Cold S.I. Allowable	SNmc 1172.63	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1172.63	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	39.30	kgf/cm ²
Radial Load	P	-326.32	kgf
Circumferential Shear	Vc	244.74	kgf
Longitudinal Shear	Vl	326.32	kgf
Circumferential Moment	Mc	32.62	kgf-m.
Longitudinal Moment	Ml	42.41	kgf-m.
Torsional Moment	Mt	48.93	kgf-m.
Include Axial Pressure Thrust		No	
Include Pressure Stress Indices per Div. 2		No	
Local Loads applied at end of Nozzle/Attachment		No	

*Warning - The ratio, Dn/Tn (5.030) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 84.328 + 2 \cdot 1.65 \cdot \sqrt{(106.0(8.0 - 0.0))}$
 $= 180.425 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	212.000 mm
Nozzle Outside Diameter	(d) =	84.328 mm
Vessel Thickness used	(T) =	8.000 mm
Nozzle Thickness used	(t) =	16.764 mm
T / t	=	0.477



$$d / t = 10.000$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.048$$

Nr/P	=	0.068
Mr/P	=	0.139
M0/P	=	0.041
N0/P	=	0.095
MrD/Mc	=	0.283
NrDL/Mc	=	0.099
M0d/Mc	=	0.087
N0DL/Mc	=	0.071
MrD/Ml	=	0.110
NrDL/Ml	=	0.065
M0D/Ml	=	0.036
N0DL/Ml	=	0.120

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	48	48	48	48
Outplane Bending (P)	124	-124	124	-124
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-94	-94	94	94
Outplane Bending (ML)	-167	167	167	-167
Normal Pressure Stress	482	521	482	521
Outplane Stress Summary	392	518	917	372

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	34	34	34	34
Inplane Bending (P)	426	-426	426	-426
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-51	-51	51	51
Inplane Bending (ML)	-519	519	519	-519
Inplane Pressure Stress	241	241	241	241
Inplane Stress Summary	130	317	1273	-619

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	23	23	-23	-23
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	54	54	54	54
Shear Stress Summary	77	77	31	31

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	414	545	1276	993

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	48	48	48	48
Outplane Bending (P)	124	-124	124	-124
Outplane Membrane (Mc)	-42	-42	42	42
Outplane Bending (Mc)	-317	317	317	-317
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	241	241	241	241

Outplane Stress Summary	53	439	774	-109
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	34	34	34	34
Inplane Bending (P)	426	-426	426	-426
Inplane Membrane (Mc)	-60	-60	60	60
Inplane Bending (Mc)	-1025	1025	1025	-1025
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	482	521	482	521

Inplane Stress Summary	-142	1095	2029	-836
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-30	-30	30	30
Torsional Shear (Mt)	54	54	54	54

Shear Stress Summary	23	23	85	85
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	201	1095	2035	846
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	48	48	48	48
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-94	-94	94	94
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	53	100	53	100

Hoop Stress Summary	7	54	196	243
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	9	9	9	9
Axial Bending (P)	73	-73	73	-73
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-82	-82	82	82
Axial Bending (ML)	-83	83	83	-83
Axial Pressure Stress	26	26	26	26



Axial Stress Summary	-56	-37	275	-37
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	10	10	-10	-10
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	26	26	26	26
Shear Stress Summary	37	37	15	15

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	98	118	278	283

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	48	48	48	48
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-42	-42	42	42
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	53	100	53	100
Hoop Stress Summary	59	106	145	192

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	9	9	9	9
Axial Bending (P)	73	-73	73	-73
Axial Membrane (Mc)	-63	-63	63	63
Axial Bending (Mc)	-192	192	192	-192
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	26	26	26	26
Axial Stress Summary	-146	91	365	-166

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-14	-14	14	14
Shear due to Torsion	26	26	26	26
Shear Stress Summary	11	11	40	40

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	207	112	373	367

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):**

Type of

Stress Values at

Stress Int. Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	482	521	482	521	482	521	482	521
Circ. Pl (SUS)	-46	-46	142	142	-26	-26	94	94
Circ. Q (SUS)	-43	43	291	-291	-599	599	1451	-1451
Long. Pm (SUS)	241	241	241	241	241	241	241	241
Long. Pl (SUS)	-17	-17	85	85	6	6	90	90
Long. Q (SUS)	-93	93	945	-945	-193	193	441	-441
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	23	23	-23	-23	-30	-30	30	30
Shear Q (SUS)	54	54	54	54	54	54	54	54
Pm (SUS)	482.0	521.0	482.0	521.0	482.0	521.0	482.0	521.0
Pm+Pl (SUS)	438.5	477.1	625.8	664.6	460.2	498.6	579.6	618.1
Pm+Pl+Q (Total)	413	544	1273	992	202	1094	2032	845

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	521.00	1172.61	Passed
Pm+Pl (SUS)	664.56	1758.92	Passed
Pm+Pl+Q (TOTAL)	2032.60	3517.84	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	53	100	53	100	53	100	53	100
Circ. Pl (SUS)	-46	-46	142	142	6	6	90	90
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	26	26	26	26	26	26	26	26
Long. Pl (SUS)	-73	-73	91	91	-54	-54	72	72
Long. Q (SUS)	-10	10	156	-156	-119	119	265	-265
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	10	10	-10	-10	-14	-14	14	14
Shear Q (SUS)	26	26	26	26	26	26	26	26
Pm (SUS)	53.0	100.0	53.0	100.0	53.0	100.0	53.0	100.0
Pm+Pl (SUS)	57.6	103.0	196.3	242.8	91.4	136.9	147.0	192.1
Pm+Pl+Q (Total)	96.3	116.0	276.2	282.8	207.4	112.7	370.0	365.9

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	100.00	1172.63	Passed
Pm+Pl (SUS)	242.79	1758.95	Passed
Pm+Pl+Q (TOTAL)	370.05	3517.90	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

PV Elite 28 U2 Licensee: SL User

FileName : =A150NRL31HN003_Ope-Lug loc as per MDS

Nozzle Calcs.: A3 2'' Nozl: 41 4:02pm Sep 18,2026

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Input, Nozzle Desc: P2 2"

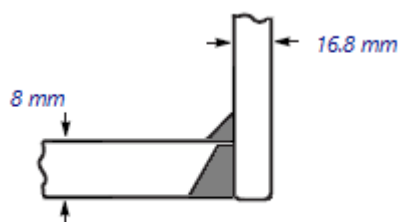
From: 150

Pressure for Reinforcement Calculations	P	39.300	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material	SA-312 TP304L		
Parent Allowable Stress at Temperature	Sv	1172.61	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1172.61	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	204.00	mm
Shell Finished (Minimum) Thickness	t	8.0000	mm
Shell Internal Corrosion Allowance	cas	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1140.00	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material	SA-182 F304L +IT		
Material UNS Number	S30403		
Material Specification/Type	Forgings		
Allowable Stress at Temperature	Sn	1172.63	kgf/cm ²
Allowable Stress At Ambient	Sna	1172.63	kgf/cm ²
Diameter Basis (for tr calc only)	Inside		
Layout Angle	180.00	deg	
Diameter	2.0000	in.	
Size and Thickness Basis	Actual		
Actual Thickness	tn	16.7640	mm
Flange Material	SA-182 F304L +IT		
Flange Type	Long Weld Neck		
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	169.4260	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class	600		
Flange Grade	GR 2.3		

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: P2 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 39.3 \cdot 102 / (1173 \cdot 1 - 0.6 \cdot 39.3)$
 $= 3.4887 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([39.3 / (1173 \cdot 1)] - 1))$
 $= 0.8657 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	101.6000	mm
Parallel to Vessel Wall, opening length	d	50.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	20.0000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: P2 2''.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.8657	mm
Wall Thickness per UG16(b),	tr16b	= 1.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 3.4887	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 3.4887	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 1.5000	mm
Wall Thickness per table UG-45	tb3	= 4.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(3.49, 1.5)]$
 $= 3.4887 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.87, 3.49)$
 $= 3.4887 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -53 °C

Weld Size Calculations, Description: P2 2''

Intermediate Calc. for nozzle/shell Welds	Tmin	8.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6000 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 9.1227 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 186.5487 mm

Input, Nozzle Desc: A2 2"

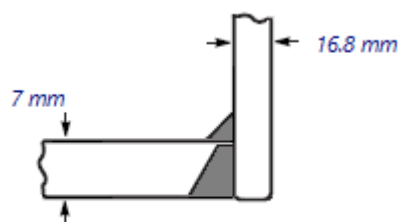
From: 230

Pressure for Reinforcement Calculations	P	39.300	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material	SA-312 TP304L		
Parent Allowable Stress at Temperature	Sv	1172.61	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1172.61	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	127.30	mm
Shell Finished (Minimum) Thickness	t	7.0000	mm
Shell Internal Corrosion Allowance	cas	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1815.00	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material	SA-182 F304L +IT		
Material UNS Number	S30403		
Material Specification/Type	Forgings		
Allowable Stress at Temperature	Sn	1172.63	kgf/cm ²
Allowable Stress At Ambient	Sna	1172.63	kgf/cm ²
Diameter Basis (for tr calc only)	Inside		
Layout Angle	180.00	deg	
Diameter	2.0000	in.	
Size and Thickness Basis	Actual		
Actual Thickness	tn	16.7640	mm
Flange Material	SA-182 F304 +IT		
Flange Type	Long Weld Neck		
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	178.7760	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	7.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class	600		
Flange Grade	GR 2.1		

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: A2 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 39.3 \cdot 63.6 / (1173 \cdot 1 - 0.6 \cdot 39.3)$
 $= 2.1770 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([39.3 / (1173 \cdot 1)] - 1))$
 $= 0.8657 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	101.6000	mm
Parallel to Vessel Wall, opening length	d	50.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: A2 2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.8657	mm
Wall Thickness per UG16(b),	tr16b	= 1.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 2.1770	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 2.1770	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 1.5000	mm
Wall Thickness per table UG-45	tb3	= 4.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(2.18, 1.5)]$
 $= 2.1770 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.87, 2.18)$
 $= 2.1770 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 117.9,	Allowable	: 1172.6	kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2813.7	kgf/cm ²	Passed
Occasional	: 22.4,	Allowable	: 1559.6	kgf/cm ²	Passed
Shear	: 102.5,	Allowable	: 820.8	kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: A2 2"

Intermediate Calc. for nozzle/shell Welds	Tmin	7.0000	mm
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Results Per UW-16.1:



	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 15.9686 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 201.7446 mm

Input Echo, WRC297 Item 1, Description: A2 2" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	127.300	mm
Shell Thickness	7.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-312 TP304L	
Vessel Cold S.I. Allowable	Smc	1172.61 kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1172.61 kgf/cm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	50.800	mm
Nozzle Thickness	16.7640	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-182 F304L	
Nozzle Cold S.I. Allowable	SNmc	1172.63 kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1172.63 kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	39.30	kgf/cm ²
Radial Load	P	-326.32	kgf
Circumferential Shear	Vc	244.74	kgf
Longitudinal Shear	Vl	326.32	kgf
Circumferential Moment	Mc	32.62	kgf-m.
Longitudinal Moment	Ml	42.41	kgf-m.
Torsional Moment	Mt	48.93	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_n/D_v (.597) is > 0.5
check the limitations of WRC bulletin 297.

Warning - The ratio, D_n/T_n (5.030) is < 10
check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{mean}(t - ca))}$$

$$= 84.328 + 2 \cdot 1.65 \cdot \sqrt{(67.15(7.0 - 0.0))}$$

$$= 155.874 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	134.300 mm
Nozzle Outside Diameter	(d) =	84.328 mm
Vessel Thickness used	(T) =	7.000 mm
Nozzle Thickness used	(t) =	16.764 mm
T / t	=	0.418
d / t	=	10.000



$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.750$$

Nr/P	=	0.061
Mr/P	=	0.112
M0/P	=	0.035
N0/P	=	0.081
MrD/Mc	=	0.269
NrDL/Mc	=	0.098
M0d/Mc	=	0.081
N0DL/Mc	=	0.083
MrD/Ml	=	0.095
NrDL/Ml	=	0.056
M0D/Ml	=	0.027
N0DL/Ml	=	0.095

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	54	54	54	54
Outplane Bending (P)	140	-140	140	-140
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-97	-97	97	97
Outplane Bending (ML)	-164	164	164	-164
Normal Pressure Stress	338	377	338	377
Outplane Stress Summary	271	358	795	224

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	40	40	40	40
Inplane Bending (P)	446	-446	446	-446
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-57	-57	57	57
Inplane Bending (ML)	-587	587	587	-587
Inplane Pressure Stress	169	169	169	169
Inplane Stress Summary	11	294	1301	-767

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	26	26	-26	-26
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	62	62	62	62
Shear Stress Summary	88	88	36	36

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	314	420	1304	994

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	54	54	54	54
Outplane Bending (P)	140	-140	140	-140



Outplane Membrane (Mc)	-65	-65	65	65
Outplane Bending (Mc)	-384	384	384	-384
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	169	169	169	169

Outplane Stress Summary	-85	401	814	-235
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	40	40	40	40
Inplane Bending (P)	446	-446	446	-446
Inplane Membrane (Mc)	-77	-77	77	77
Inplane Bending (Mc)	-1272	1272	1272	-1272
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	338	377	338	377

Inplane Stress Summary	-523	1167	2176	-1222
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-35	-35	35	35
Torsional Shear (Mt)	62	62	62	62

Shear Stress Summary	27	27	97	97
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	525	1168	2183	1232
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	54	54	54	54
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-97	-97	97	97
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	53	100	53	100

Hoop Stress Summary	10	57	205	252
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	9	9	9	9
Axial Bending (P)	56	-56	56	-56
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-82	-82	82	82
Axial Bending (ML)	-72	72	72	-72
Axial Pressure Stress	26	26	26	26

Axial Stress Summary	-63	-30	247	-10
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**Nozzle Stresses (kgf/cm²):**

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	10	10	-10	-10
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	26	26	26	26
Shear Stress Summary	37	37	15	15

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	104	115	252	264

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	54	54	54	54
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-65	-65	65	65
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	53	100	53	100
Hoop Stress Summary	42	89	173	220

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	9	9	9	9
Axial Bending (P)	56	-56	56	-56
Axial Membrane (Mc)	-63	-63	63	63
Axial Bending (Mc)	-181	181	181	-181
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	26	26	26	26
Axial Stress Summary	-152	97	337	-138

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-14	-14	14	14
Shear due to Torsion	26	26	26	26
Shear Stress Summary	11	11	40	40

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	196	105	347	367

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl

Circ. Pm (SUS)	338	377	338	377	338	377	338	377
Circ. Pl (SUS)	-43	-43	151	151	-37	-37	117	117
Circ. Q (SUS)	-24	24	304	-304	-826	826	1718	-1718
Long. Pm (SUS)	169	169	169	169	169	169	169	169
Long. Pl (SUS)	-17	-17	97	97	-11	-11	119	119
Long. Q (SUS)	-141	141	1033	-1033	-244	244	524	-524
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	26	26	-26	-26	-35	-35	35	35
Shear Q (SUS)	62	62	62	62	62	62	62	62
Pm (SUS)	338.0	377.0	338.0	377.0	338.0	377.0	338.0	377.0
Pm+Pl (SUS)	299.6	337.6	492.0	530.6	309.1	346.5	462.0	499.8
Pm+Pl+Q (Total)	313	419	1301	993	526	1166	2179	1233

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	377.00	1172.61	Passed
Pm+Pl (SUS)	530.56	1758.92	Passed
Pm+Pl+Q (TOTAL)	2179.88	3517.84	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	53	100	53	100	53	100	53	100
Circ. Pl (SUS)	-43	-43	151	151	-11	-11	119	119
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	26	26	26	26	26	26	26	26
Long. Pl (SUS)	-73	-73	91	91	-54	-54	72	72
Long. Q (SUS)	-16	16	128	-128	-125	125	237	-237
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	10	10	-10	-10	-14	-14	14	14
Shear Q (SUS)	26	26	26	26	26	26	26	26
Pm (SUS)	53.0	100.0	53.0	100.0	53.0	100.0	53.0	100.0
Pm+Pl (SUS)	60.4	105.9	205.1	251.7	75.4	120.3	174.6	220.6
Pm+Pl+Q (Total)	102.5	113.7	250.5	263.9	196.5	105.6	344.3	366.8

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	100.00	1172.63	Passed
Pm+Pl (SUS)	251.74	1758.95	Passed
Pm+Pl+Q (TOTAL)	366.83	3517.90	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: P1 2"

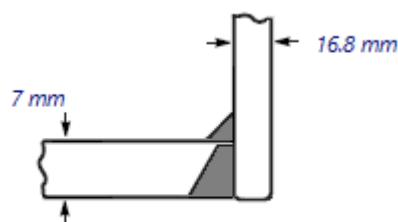
From: 230

Pressure for Reinforcement Calculations	P	39.300	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material	SA-312 TP304L		
Parent Allowable Stress at Temperature	Sv	1172.61	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1172.61	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	127.30	mm
Shell Finished (Minimum) Thickness	t	7.0000	mm
Shell Internal Corrosion Allowance	cas	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1835.00	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material	SA-182 F304L +IT		
Material UNS Number	S30403		
Material Specification/Type	Forgings		
Allowable Stress at Temperature	Sn	1172.63	kgf/cm ²
Allowable Stress At Ambient	Sna	1172.63	kgf/cm ²
Diameter Basis (for tr calc only)	Inside		
Layout Angle	0.00	deg	
Diameter	2.0000	in.	
Size and Thickness Basis	Actual		
Actual Thickness	tn	16.7640	mm
Flange Material	SA-182 F304 +IT		
Flange Type	Long Weld Neck		
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	178.7760	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	7.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class	600		
Flange Grade	GR 2.1		

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: P1 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 39.3 \cdot 63.6 / (1173 \cdot 1 - 0.6 \cdot 39.3)$
 $= 2.1770$ mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([39.3 / (1173 \cdot 1)] - 1))$
 $= 0.8657$ mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	101.6000	mm
Parallel to Vessel Wall, opening length	d	50.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: P1 2''.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.8657	mm
Wall Thickness per UG16(b),	tr16b	= 1.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 2.1770	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 2.1770	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 1.5000	mm
Wall Thickness per table UG-45	tb3	= 4.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(2.18, 1.5)]$
 $= 2.1770$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.87, 2.18)$
 $= 2.1770$ mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: P1 2''

Intermediate Calc. for nozzle/shell Welds	Tmin	7.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 15.9686 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 201.7446 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center



of the nozzle to the edge of the diameter limit is: 50.80 mm

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Input, Nozzle Desc: B1 2"

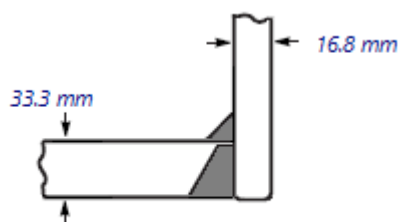
From: 250

Pressure for Reinforcement Calculations	P	39.300	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-182 F304L	
Parent Allowable Stress at Temperature	Sv	1172.61	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1172.61	kgf/cm ²
Outside Diameter of Bolted Blind Flange	D	279.40	mm
Head Finished (Minimum) Thickness	t	33.2740	mm
Head Internal Corrosion Allowance	cas	0.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F304L +IT	
Material UNS Number		S30403	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1172.63	kgf/cm ²
Allowable Stress At Ambient	Sna	1172.63	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-182 F304 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	175.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	33.2740	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 2.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: B1 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R(\exp([P/(Sn \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4(\exp([39.3/(1173 \cdot 1)] - 1))$
 $= 0.8657 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	150.8760	mm
Parallel to Vessel Wall	Rn+tn+t	75.4380	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.9100	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: B1 2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.8657	mm
Wall Thickness per UG16(b),	tr16b	= 1.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 28.1082	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 28.1082	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 1.5000	mm
Wall Thickness per table UG-45	tb3	= 4.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(28.1, 1.5)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.87, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: B1 2"

Intermediate Calc. for nozzle/shell Welds Tmin 16.7640 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Cut Length for this Nozzle is, Drop + Ho + H + T : 208.3248 mm

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: Element Overstress due to Combined Loads!**

**** Warning: Wind/Earthquake Deflection Exceeded the Specified Allowable!**

Please review all reports carefully!

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